Scaling Online Exploration for Robot Agents

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Project Page: https://robot-agent-exploration.com

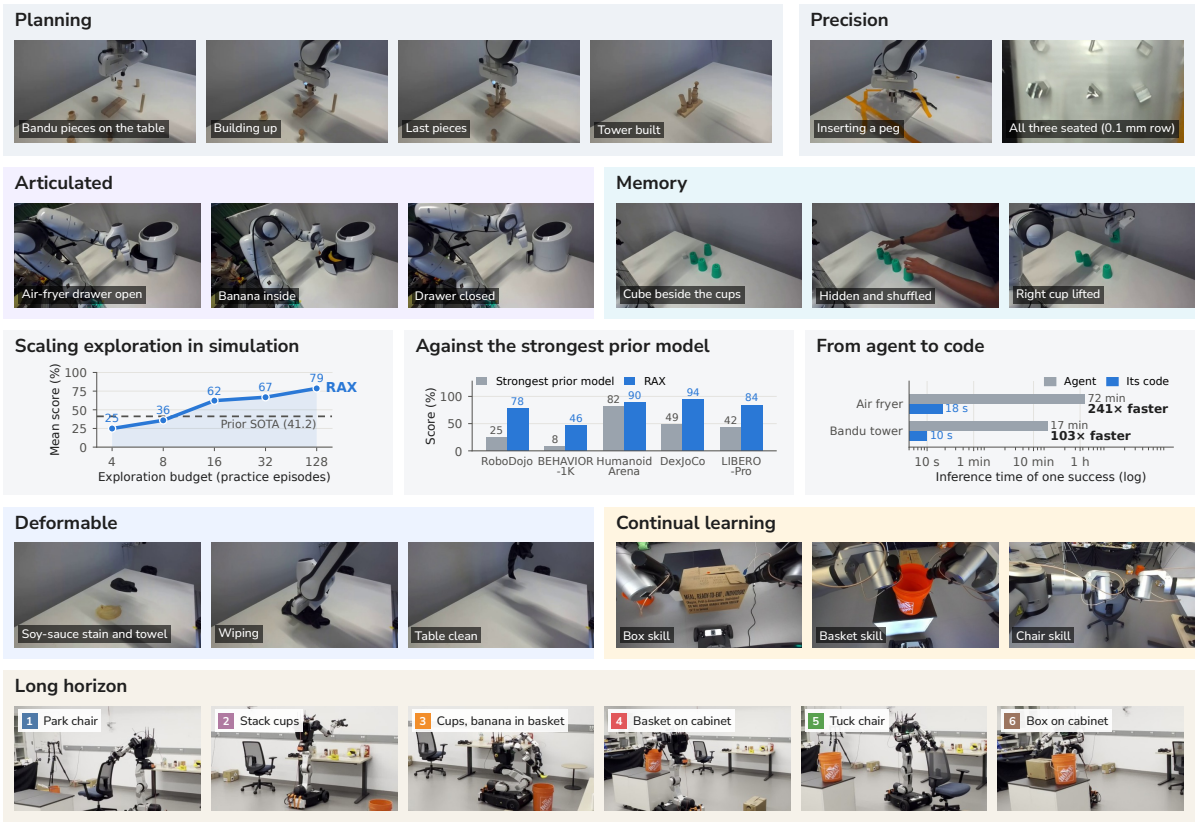


Figure 1 | **Overview of RAX.** By scaling online exploration, a coding agent masters robot tasks across embodiments without any action demonstrations, from precise insertion and articulated, deformable, and memory tasks to long-horizon mobile manipulation, and surpasses the state of the art on unseen scenes.

Abstract

Robot learning has advanced significantly by scaling demonstration datasets and model capacity; however, deploying a new robot embodiment or task typically necessitates a dedicated teleoperation corpus. Furthermore, resulting policies remain bounded by their imitation distributions, limiting generalization and continual self-improvement. We introduce **RAX (Robot Agent eXploration)**, an open-source framework for robot agents centered on **scaling**

exploration: enabling agents to improve by acquiring online experience and synthesizing executable programs. In RAX, a general-purpose coding agent generates robot control code directly, operating without action demonstration data, hand-engineered action primitives, or value-function guidance. Through online exploration, the agent identifies decision sequences that achieve task goals and compiles them into modular programs encapsulating the full perception-planning-control loop for fast inference. Using RAX, we present a systematic multi-embodiment study on scaling online exploration across fine-grained precision and long-horizon manipulation tasks using single-arm, bimanual, dexterous, mobile, and humanoid systems in simulation and on real robots. Generalization performance on unseen environments improves consistently as exploration scales. In simulation, RAX outperforms top baseline models by $+37.4\%$ on average. On real robots, RAX autonomously learns memory-dependent, articulated, high-precision, deformable, and long-horizon mobile manipulation tasks without human demonstrations, continually improving performance by composing verified skills from prior exploration sessions. As exploration scales, we observe emergent system capabilities including active perception, error recovery, safety guardrails, and human-robot agent collaboration. We open-source RAX as a generalist robot-agent platform that allows new robot embodiments to be integrated solely via API interfaces, bypassing teleoperated data collection and policy training entirely. More details are available on our project page at `robot-agent-exploration.com`.

Contents

1 Introduction **5**

2 Related Work **7**

3 The RAX Cognitive Architecture **9**
- 3.1 Multimodal Perception . . . 9
- 3.2 Planning and Control . . . 10
- 3.3 From Exploration to Autonomous Robot Program . . . 11
- 3.4 Continual Learning with Memory . . . 11
- 3.5 Parallel Multi-Agent and Multi-Environment Exploration . . . 12

4 The RAX System **12**
- 4.1 Frontend–Backend Separation . . . 13
- 4.2 Multi-Embodiment Control Interface . . . 13

5 Experiments **16**
- 5.1 Setup . . . 16
- 5.2 Simulation Results . . . 16
  - 5.2.1 RoboDojo . . . 16
  - 5.2.2 BEHAVIOR-1K . . . 17
  - 5.2.3 HumanoidArena . . . 18
  - 5.2.4 DexJoCo . . . 18
  - 5.2.5 LIBERO and LIBERO-Pro . . . 20
- 5.3 Ablations . . . 21
  - 5.3.1 Parallelization . . . 21
  - 5.3.2 Depth versus RGB-Only Perception . . . 22
  - 5.3.3 Inference Time and Tokens . . . 23
  - 5.3.4 Demonstrations Without Actions . . . 24
- 5.4 Intriguing Emerging Agentic Behaviors with Scaling Exploration . . . 25
  - 5.4.1 Active Perception . . . 25
  - 5.4.2 Error Recovery and Safety . . . 26
  - 5.4.3 Human–Robot Interaction . . . 27
- 5.5 Continual Learning . . . 27
  - 5.5.1 Reusing Skills . . . 27
  - 5.5.2 Rapid Adaptation of Programs . . . 29

6 Discussion **30**

A Interface Contracts **35**

B Agent Instructions and Skills **35**

C Benchmarks and Tasks **37**
- C.1 Sessions . . . 37
- C.2 Rehearsal and Evaluation . . . 37
- C.3 RoboDojo . . . 38
- C.4 BEHAVIOR-1K . . . 39
- C.5 HumanoidArena . . . 39
- C.6 DexJoCo . . . 40
- C.7 LIBERO and LIBERO-Pro . . . 41

D Simulation Case Studies **43**
D.1 Re-Perception Under Perturbation: Two Cans into a Basket . . . 43
D.2 Self-Calibration and Grasp Design: Tower of Hanoi with Two Dexterous Hands 46
D.3 From Exploration to Code: Carrying a Hammer over a Bar on a Humanoid . . . 50
D.4 Adaptation Under Perturbation on LIBERO-Pro . . . 54

E Exploration Budget of ASPIRE on LIBERO-Pro **56**

F Real-Robot Tasks **57**
F.1 Find the Cube . . . 57
F.2 Air Fryer . . . 61
F.2.1 Agent-Written 3D Perception . . . 61
F.3 Wiping a Stain . . . 67
F.4 Peg Insertion: the ManipulationNet Peg-in-Hole Challenge . . . 70
F.5 Bandu Tower . . . 72
F.6 Long-Horizon Tidying on the RB-Y1 . . . 74

G Compute and Cost **75**

1 Introduction

"We want AI agents that can discover like we can, not which contain what we have discovered. Building in our discoveries only makes it harder to see how the discovering process can be done."

— Richard S. Sutton, *The Bitter Lesson* (2019)

Robot learning has made substantial progress by scaling demonstration data and model capacity with imitation learning, aiming to replicate the training recipe that transformed language modeling (Kaplan et al., 2020). Vision-language-action (VLA) models and World-Action models (WAMs) fine-tune pretrained VLMs and video generation models, respectively, using large teleoperation corpora (Brohan et al., 2023; Kim et al., 2024; Black et al., 2024). However, deploying a new robot or task still requires a dedicated teleoperation corpus of its own (O'Neill et al., 2024; Team et al., 2024; Black et al., 2024; Intelligence et al., 2025; Bjorck et al., 2025; Liu et al., 2026; Wang et al., 2026d; Yuan et al., 2026; Kim et al., 2026; Bi et al., 2026). Furthermore, the knowledge of such policy models is inherently bounded by the demonstrations they imitate. Consider a mobile robot tasked with loading a car in an unfamiliar house: while opening doors, grasping bags, and placing objects may each be familiar skills, the robot must adapt to novel scenes, visual appearances, and task compositions while discovering new approaches. This adaptation and discovery process is absent during imitation learning on static teleoperation data. Consequently, these policies struggle to generalize and self-improve, even when provided with environment interactions and test-time compute. This limitation motivates a central question: *can robots self-improve and learn to discover new skills through autonomous online experience and active exploration?*

Humans learn physical skills by attempting a task, observing the outcome, diagnosing failures, and practicing until their behavior becomes reliable. LLM agents exhibit a similar pattern in software domains: they explore unfamiliar systems by executing programs, inspecting outputs, and revising their work (Yang et al., 2024). Instead of training policies exclusively on teleoperated demonstration data, we let *agents program robots directly* (Lozano-Perez, 1983) without hand-designed action primitives. The agent explores online to discover decisions that drive task success, then automates these decision-making principles into modular robot programs that integrate perception, planning, and control. It continuously debugs and tests these programs through online execution. These programs encode the agent's underlying principles for *how to* generate successful actions based on environment perception, rather than merely mimicking the actions themselves, thereby enabling more robust generalization. Furthermore, the agent can compose, modify, and update these programs when confronting increasingly complex scenes and tasks, continually improving through online experience.

We introduce **RAX (Robot Agent eXploration)**, a cognitive architecture for robot agents. Its core organizing principle is **scaling exploration**: robot agents should improve by gathering their own online experience and optimizing their programs. In RAX, a general-purpose coding agent receives a natural-language instruction, at most a few goal images or visual demonstrations, and a budget of exploration episodes. It interacts directly through the robot's perception and control interfaces without requiring human teleoperation or value function guidance. The underlying language model parameters remain fixed and no policies are trained; instead, learning updates a persistent workspace of programs, notes, and verified skills. The agent converts successful interactions into autonomous code programs and stores reusable knowledge for subsequent episodes and tasks.

As exploration scales, we observe several emergent system capabilities: the agent constructs task-relevant 3D perception models, plans actions via pose search in 3D joint space, generates

fully automated programs with closed-loop visual and force feedback, and adapts its programs when encountering novel task conditions. Furthermore, we observe agentic behaviors on real robots, including active perception, error recovery, safety guardrails, and active human-robot interaction. For example, the robot executes end-effector contact probing for precise localization when depth measurements are inaccurate. In mobile manipulation experiments, following an initial collision with a cabinet, the agent automatically instantiates a safety check to verify workspace clearance before every base rotation. To test code robustness, the agent actively requests the human operator to vary object layouts, systematically testing the spatial generalization of the autonomous program.

In RAX, we equip the agent with rich multimodal sensing by exposing metric depth and force signals through the robot interface. A unified architecture supports tabletop bimanual arms, dexterous hands, mobile household robots, and humanoids through interfaces to each robot's sensors, controllers, and kinematic models (Section 4). Unlike standard VLAs or WAMs operating on RGB images alone, the agent autonomously leverages these rich sensing signals to construct 3D point clouds and fit object models directly from perception. It then searches for actions to manipulate these 3D models while accounting for their unique shapes and affordances. Force feedback further enables the agent to resolve spatial uncertainty through contact and enable close-loop control on force signals in insertion tasks. Such multimodal world modeling have been largely absent from generalist policy models, yet naturally align with human cognitive processes.

To enable high-frequency closed-loop control, the agent compiles its decision-making process into executable Python programs that generate actions directly from sensor signals. Unlike prior approaches that rely on human-designed action primitives or policies trained on demonstrations (Lu et al., 2026; Xiao et al., 2026), our approach directly generates robot control programs that encapsulate the full perception-planning-control loop, whereas standard policies discard the underlying decision-making process essential for generalization. In practice, we observe the agent writing programs that automatically segment and identify objects in 3D using point-cloud fitting and hue-based identification, then plan actions grounded in this 3D world model, with retry loops and exception handling naturally incorporated into the code. Because actions are planned directly on perceived 3D objects rather than learned through imitation, this approach generalizes to far more diverse object layouts and domain variations than standard VLAs or WAMs. For semantically rich perception in open-world environments, the agent can program direct API calls to LLMs (such as Gemini) within the code to assist with object identification, while incorporating checkpoints and exception handling to step in and verify progress. Over the course of exploration, both action inference latency and token counts decrease as an increasing fraction of decisions becomes automated through code.

As experience scales with exploration, RAX maintains an intra-task memory file system across episodes to record failure lessons and successful strategies, allowing the agent to continually refine its programs. Across multiple tasks and sessions, we maintain a repository of modular programs as verified skills for various tasks and manipulation primitives. Each skill includes the executable code for a specific task, accompanied by execution logs from successful runs, clear preconditions, and verification checkpoints, allowing skills to be composed and reused for complex downstream tasks. In real-world mobile manipulation experiments in household settings, the agent practices individual tasks, each in its own exploration session, such as picking and placing boxes or tucking in office chairs, and commits them as verified skills. When subsequently evaluated on a composite, long-horizon task involving more than six sequential object manipulations, the agent achieves substantial automation and efficiency gains by reusing these stored skills. We envision agents practicing skills in a distributed, parallel

fashion to build a rich repository of diverse capabilities, thereby enabling continual lifelong learning.

Using RAX, we conduct **the first large-scale, multi-embodiment, systematic study of scaling exploration for robot agents**, spanning fine-grained precision and long-horizon tasks across LIBERO, LIBERO-Pro, RoboDojo, DexJoCo, BEHAVIOR-1K, and HumanoidArena in simulation and on two real robots. We evaluate the resulting policies on unseen scenes, separating exploration from evaluation. Performance grows steadily with exploration, and on fine-grained and long-horizon tasks on which state-of-the-art methods rarely succeed, RAX achieves high success rates, surpassing the strongest prior models by +37.4 on average over five benchmarks (Figure 1). Without any expert action trajectories, RAX sets a new state of the art on RoboDojo (+53.4), BEHAVIOR-1K (+38.3), DexJoCo (+31.4), and LIBERO-Pro (+21.2). On HumanoidArena, it sets a new state of the art (+7.9) with three teleoperated demonstrations per task, where every baseline uses 100. On real Franka and Rainbow RB-Y1 robots, RAX masters five complex tabletop manipulation tasks and a long-horizon mobile manipulation task with zero demonstrations, and verified skills accumulated across sessions let later sessions build on earlier ones.

We summarize our contributions below:

1. We introduce **RAX** and establish **scaling online exploration** as a foundational principle for agentic robotics. In RAX, the agent operates directly through perception and robot control interfaces without policy training, compiling successful decision sequences into executable code programs that run autonomous execution loops.
2. Using RAX, we conduct the first systematic, multi-embodiment study of scaling online exploration across simulation and real robot environments. RAX acquires diverse manipulation capabilities solely through autonomous exploration, outperforming state-of-the-art baselines by $+37.4$ points on average across five simulation benchmarks, while solving high-precision, deformable, articulated, and long-horizon manipulation tasks on real robots without human demonstrations.
3. We demonstrate rapid adaptation and continual learning on real robots, showing that generated programs can be selectively updated in response to environmental and task shifts while retaining verified skills in a persistent repository for composition and reuse in complex tasks.

We release the full architecture of RAX, together with its agent instructions, skills, benchmark integrations, and evaluation protocols, as a generalist robot-agent platform, on which a new robot can be brought up through an interface alone, without data collection or policy training.

2 Related Work

Language Models for Robot Control. A substantial body of work uses pretrained language models as high-level planners or program generators for robots. SayCan grounds language-model plans in learned affordances (Ahn et al., 2022), Inner Monologue folds environment feedback into iterative replanning (Huang et al., 2022), and Code as Policies, ProgPrompt, and VoxPoser synthesize executable programs, situated task plans, or 3D value maps directly from instructions (Liang et al., 2023; Singh et al., 2022; Huang et al., 2023). These methods typically invoke the model zero-shot or with a few in-context examples, producing a plan or a program that is executed with little opportunity to improve from experience. **RAX** instead treats the language model as a *learner*: it interacts with the robot over many episodes and revises its

perception and control programs according to what it observes.

Agentic Robot Learning. More recently, several works let coding agents improve robot behavior through interaction. ENPIRE runs agents in a closed loop of automatic reset, rollout, and policy training on a fleet of bimanual arms, and accelerates discovery by scaling the fleet (Xiao et al., 2026); FAEA and Encore refine policy programs from execution feedback, the latter starting from a few teleoperated demonstrations (Tsui et al., 2026; Kang et al., 2026); Baton explores long-horizon tasks subtask by subtask with transition-aware memory (Xu et al., 2026); SimEX develops robot skills in a simulator reconstructed from the physical workspace before a few trials on the real robot (Hu et al., 2026); RPG practices in simulation on tasks derived from an offline trajectory dataset and refines a library of reusable skills (Wang et al., 2026e); EmbodiedSWE turns coding-agent solutions into training data for vision-language-action models (You et al., 2026); and CaP-X benchmarks coding agents on manipulation (Fu et al., 2026). These works establish that agents can improve robot behavior through interaction, and have so far demonstrated it mainly in tabletop manipulation with parallel-jaw grippers (Xiao et al., 2026; Tsui et al., 2026; Kang et al., 2026; Xu et al., 2026; Hu et al., 2026; Wang et al., 2026e), often on a single platform (Xiao et al., 2026; Hu et al., 2026; Wang et al., 2026e), with humanoid and mobile settings explored through pretrained locomotion (You et al., 2026) or short pick-up tasks (Fu et al., 2026). Several also draw on demonstrations (Kang et al., 2026), offline datasets (Wang et al., 2026e), reconstructed simulators (Hu et al., 2026), or neural policy training (Xiao et al., 2026). **RAX** complements these efforts: it learns each task on the robot itself, from its own exploration, and does so across single-arm and bimanual manipulators, dexterous hands, mobile manipulators performing long-horizon household activities, and legged humanoids whose locomotion the agent constructs itself, on established benchmarks and on real robots. On this basis, it studies systematically how performance scales with exploration, with 3D perception, and with accumulated memory.

Scaling Robot Learning. Vision-language-action models (Brohan et al., 2023; Kim et al., 2024; Black et al., 2024) and imitation learning methods (Zhao et al., 2023; Chi et al., 2025) improve robot capability by scaling demonstration data and model size, and world action models extend this recipe by learning to predict how the scene evolves together with the actions that produce it (Wang et al., 2026d; Yuan et al., 2026; Kim et al., 2026; Bi et al., 2026). To cover multiple embodiments, generalist policies pool demonstrations across robots into a single training mixture (O'Neill et al., 2024; Team et al., 2024; Doshi et al., 2024; Intelligence et al., 2025; Bjorck et al., 2025), and a robot outside this mixture requires data of its own and further fine-tuning. In language modeling, computation spent at inference has emerged as a further scaling axis (Guo et al., 2025; OpenAI, 2024). **RAX** instead studies how robot capability grows with online experience acquired and organized by the agent itself, without additional demonstrations or parameter updates. Because this experience is gathered on the robot at hand, RAX extends to a new embodiment without pooled data, and adapts to it within a few episodes of exploration.

Memory and Cognitive Architectures. Language agents can retain experience as explicit memory: Voyager grows a library of code skills through open-ended exploration in a game (Wang et al., 2023), and Reflexion and ExpeL distill feedback into textual lessons that improve later attempts (Shinn et al., 2023; Zhao et al., 2024). At a broader level, classical cognitive architectures such as Soar and ACT-R structure intelligent behavior around perception, memory, and deliberation (Laird et al., 1987; Anderson et al., 2004), CoALA revisits this structure for language agents (Sumers et al., 2023), and Sutton's common model of the intelligent agent (Sutton, 2022) and OaK architecture (Sutton et al., 2022) hold that all task-specific knowledge should be acquired at runtime from the agent's own experience, a view developed further as the era of experience (Silver & Sutton). **RAX** brings this view to physical robots, where every skill is

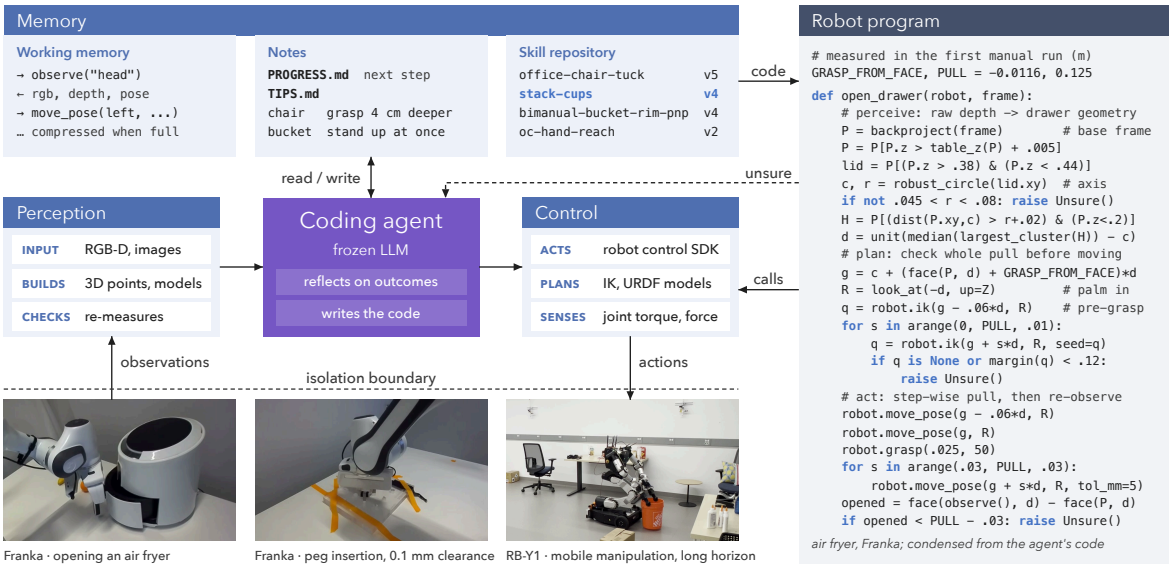


Figure 2 | **System architecture and self-improving loop of RAX.** The coding agent, powered by a frozen LLM, interacts with physical hardware across an isolation boundary using multimodal perception and control SDK interfaces. By reflecting on execution outcomes, the agent continually updates a persistent memory system containing working context, task notes, and a versioned repository of verified skills. During exploration, the agent compiles its perception and planning logic into standalone Python control programs that execute high-frequency loops directly on the robot. When encountering perception or planning failures due to environment variations, the program raises an exception (`raise Unsure()`), delegating control back to the agent to make adjustments and refine the codebase.

executable control code that enters memory only after it succeeds without supervision.

3 The RAX Cognitive Architecture

Inspired by Sutton's OaK architecture (Sutton et al., 2022), RAX acquires all task-specific knowledge at runtime from the agent's own experience. Building on task-agnostic robot control interfaces and low-level controllers, the agent autonomously develops task-specific perception, manipulation strategies, and executable skills through exploration. The following modules surround the language-model core: **perception and control**, **autonomous robot program**, and **memory for continual learning** (Figure 2).

3.1 Multimodal Perception

To build a physically accurate model of the world, we equip the agent with multimodal sensory inputs, including RGB images, depth maps, camera parameters, and force signals. The agent can thus construct 3D world models using depth data while regulating contact via force feedback. Notably, the 3D perception and object model fitting behaviors completely emerges from the agent exploration process, while we provide no skills, instructions or examples regarding specific usage of these multimodal sensing signals.

A real-robot task illustrates this process: opening an air fryer's drawer, placing a banana in its basket, and closing the drawer (Figure 3). The agent lifts RGB-D observations into a point cloud in the robot's base frame and constructs a geometric model of the fryer from its own measurements, localizing it in each new scene layout by fitting the fryer's axis and handle direction with robust estimators. Because the drawer moves, the model is articulated,

consisting of a fixed body and a drawer that slides along a prismatic joint. The agent plans every action, including grasping, pulling, and releasing, on this model and checks each motion for collisions. Appendix F.2.1 details the model, the estimators, and how the program uses them. On a humanoid, the same process yields a gravity-aligned frame from the floor plane, landmarks for self-localization, and tracking updates after every gait cycle.

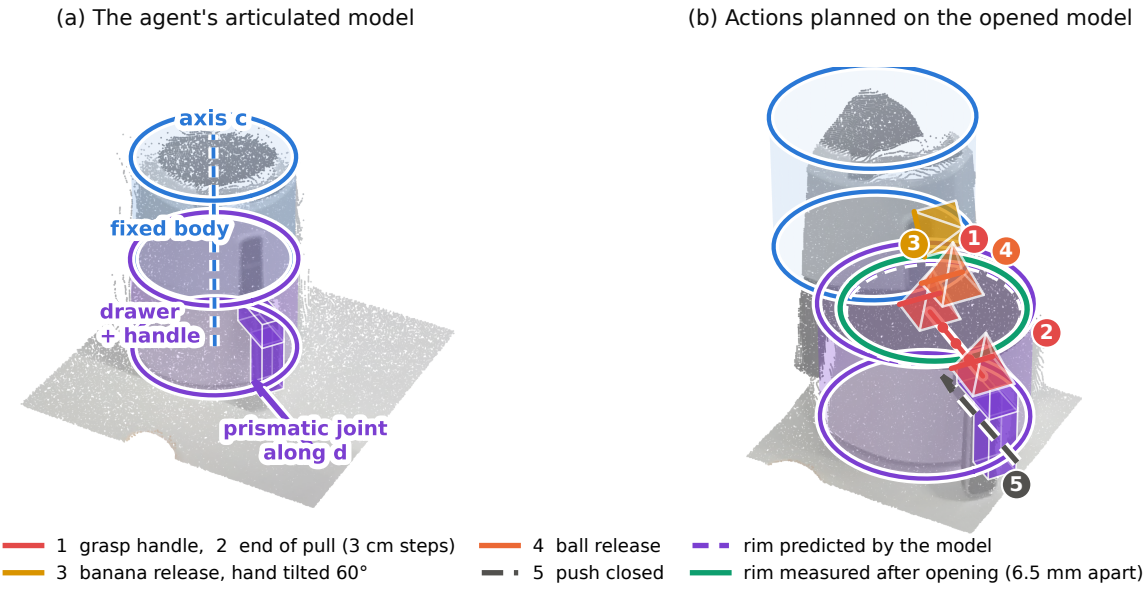


Figure 3 | **Agentic 3D perception and motion planning on a real robot. (a)** The articulated air fryer model constructed programmatically by the agent from point-cloud measurements and fit to RGB-D observations, consisting of a fixed chassis and a drawer articulated via a prismatic joint along pull direction d relative to central axis c. **(b)** Action trajectories planned on the instantiated model with the drawer opened by 13.1 cm. End-effector poses are visualized as pyramids whose bases face the approach vector and whose thick edges indicate the finger closure axis. The underlying geometric estimator maintains high fidelity, yielding only a 6.5 mm displacement between the predicted basket rim and the post-opening physical measurement. Figure 26 illustrates the corresponding execution sequence on the physical robot.

3.2 Planning and Control

Given the accurate world model, the agent then directly plans the robot actions in 3D world using robot control SDK and kinematic solvers. As the interface between the agent and the physical robot, we expose the robot's control SDK as MCP tools, together with helpers such as an inverse kinematics (IK) solver and the robot's URDF model. With these tools, the agent plans each motion and searches for feasible trajectories, accounting for the robot's posture, collisions, and joint limits. In the air fryer task (Appendix F.2), the drawer's handle has a short horizontal arm on top and a grip hanging down from its end. Grasping the grip from the front, as a person would, leaves the arm almost no joint margin near its base for the pull, so the agent grasps the short arm from above and pulls the drawer open in 3 cm steps (actions 1 and 2 in Figure 3(b)). The 18 cm banana barely fits the 19 cm basket: released from a level hand, it touched the top of the fryer, and the opening fingers would hit the fryer's body. The agent therefore tilts the hand by 60° about the banana's axis, with the banana's concave side toward the fryer, and releases it with its middle just beyond the basket's center (action 3). The agent found these motions itself, by searching through the control tools during exploration, without human advice or demonstrations; Figure 26 shows them on the real robot.

3.3 From Exploration to Autonomous Robot Program

Generating each action with LLM-reasoning is expensive and slow. To enable high-frequency close-loop planning, the agent automates the decision process into a autonomous robot program with code that can directly execute to complete the task. Unlike prior work on coding agents for robotics (Lu et al., 2026; Xiao et al., 2026), we do not rely on hand-designed primitive skills or train policies by imitating human demonstrations, which inherently restricts scaling beyond human capability (Sutton, 2019). Instead, RAX agents directly control the robot without pre-trained policies or hand-coded primitives, progressively transferring control from high-level deliberation to autonomous code programs, resulting in **autonomous decision-making through agent-native coding**. Initially, the agent controls the robot through tool calls while perceiving the scene via the core LLM. Upon achieving task success, it compiles these decisions into code programs that run independently of main-agent deliberation. For example, after identifying object shapes and geometries, the agent generates a 3D object model and automatically estimates its position and pose using depth-derived point clouds when encountering new layouts. This eliminates the need for the main agent to repeatedly perceive and process raw images, which is slow and costly.

The agent decomposes tasks into stages and implements modular skills with programs that check preconditions, execute motions, and verify outcomes, operating much like a software development process through testing, debugging, and code versioning. Empirical measurements from exploration, such as gait-cycle distance or reliable grasp orientation, become program parameters. The agent tests successive code versions during exploration and uses stage-level failures to identify which specific skill requires revision. The hammer-delivery program on the HumanoidArena DoubleDesk task (Appendix D.3), for instance, decouples approach, grasp, transit, obstacle crossing, and placement so that every failure traces directly to a specific stage (Figure 19).

Once reliable, the program is tested unattended in a new scene with exception handling and agent monitoring the generated logs in case of emergency situations. If perception or planning fails due to significant open-world variations, the program throws an exception, enabling the agent to step in, perceive the environment, and make necessary adjustments. Through scaled program testing and debugging during exploration, the result is a **fully automated program** that perceives, decides, and acts in unseen scenes with minimal or no agent intervention. Automated perception and planning allow it to adapt to new layouts in ways that replayed trajectories cannot. Deployment thus spans a continuum from the agent acting alone to a fully automated program, along which the agent advances as it accumulates experience.

3.4 Continual Learning with Memory

As online experience scales and decision-making becomes automated, RAX continually expands its capabilities by constructing persistent memory systems for reuse across episodes and tasks. Persistent memory carries experience at multiple timescales (Figure 2, top). *Within a session*, notes retain the current strategy, confirmed results, failed approaches, and useful parameters through context compression, which lives in the agent scratch pad in workspace. *Across sessions and tasks*, a skill *repository* stores each skill with its code, the conditions under which it applies, and evidence of its execution. It is continually contributed and merged by agents across different sessions and tasks, evolving with more experience.

A skill enters the repository only after it executes successfully without supervision, together with the evidence of that execution. Later sessions retrieve and compose skills that are known to work instead of rediscovering the same behavior. The benefit is largest on real robots, where

every episode costs operator time: a grasp or a drawer-opening skill becomes a building block for longer tasks, and each session leaves the library more capable than it found it.

Learning via Practice Much like humans, who repeatedly practice individual skills to gain proficiency before rehearsing complex, long-horizon tasks end-to-end, robot agents follow a similar cognitive progression. Here we depart from the conventional per-task training and evaluation protocol in robotics and frame robot learning as a continual process, where it generates and refines skills through practice and gradually moves on to more complex tasks by reusing the skills in the repository. Taking household tasks as an example, we let the agent practice individual subtasks, such as picking and placing (PnP) boxes and buckets or sorting tabletop items; each practice session is an exploration session on a single sub-task. During this process, the agent develops autonomous programs, rehearses them end-to-end to improve robustness, and eventually commits them to a multi-task skill library. During deployment on long-horizon compositional tasks, the agent reuses these subtask skills directly while verifying execution evidence and preconditions to handle environmental variations beyond those encountered during practice. Such skill acquisition and practice can also occur in parallel across multiple robots and tasks, enabling collective learning across multi-robot systems.

3.5 Parallel Multi-Agent and Multi-Environment Exploration

Online exploration can be time-consuming due to the latency of agent reasoning and environment execution. To accelerate this process, we parallelize exploration across multiple agents and environments, enabling agents to evaluate distinct strategies concurrently rather than sequentially within a single environment. In our simulation experiments, leveraging parallel environments and agents reduces the wall-clock time required to reach target performance by more than $2\times$, highlighting the scalability of parallelized exploration for robotic systems. Section 5.3.1 measures how parallel exploration changes the time to consume the exploration budget and the score of the resulting program.

4 The RAX System

RAX supports a wide range of embodiments in simulation and in the real world, including single-arm and bimanual manipulators, dexterous hands, mobile manipulators, and humanoids, with one agent architecture. To do so, it adopts a separation familiar from software engineering, in which a frontend is kept apart from the backend that serves it. In RAX, the backend is the robot's physical stack: the hardware drivers or the simulator, the low-level controllers, the sensor pipelines, and the safeguards around them. The frontend is the agent's workspace, where the agent observes, decides, and acts, and where it writes the programs that encode its decision logic. The two meet only at a tool interface, and the same frontend connects to the backend of every platform, from the real Franka and Rainbow RB-Y1 robots to each simulated benchmark. The agent never touches the backend: it cannot modify controllers, drivers, or robot infrastructure. Because the agent acts only through these tools, its programs contain the decision logic of a task, namely perception, planning, and the checks between stages, while low-level control, from joint-level PD control to whole-body motion tracking on humanoids, and the enforcement of safety limits remain in the backend. The remainder of this section describes the separation and what each side provides (Section 4.1), and the control interface through which one frontend drives arms, dexterous hands, mobile manipulators, and humanoids (Section 4.2).

4.1 Frontend–Backend Separation

On a real robot, the backend owns everything that keeps the hardware calibrated and safe. It runs the vendor's drivers and real-time controllers, serves calibrated RGB-D images together with the intrinsics and poses of their cameras, and records the robot's sensors and state continuously, so that the agent can review the outcome of an action after the fact. It also enforces the safeguards. Motion is enabled only while an operator has armed the robot, and disarming halts it immediately; every motion command is validated before execution against workspace bounds, joint limits, and self-collision, and on the RB-Y1 mobile manipulator also against joint load. A command that violates a limit is refused before the robot moves, together with the limit it violated, so that the agent can change its approach.

The frontend receives exactly two things from the system, tools and skills. The *tools* expose as much of the backend as possible: observation, raw control, and model-based helpers in the form of forward and inverse kinematics, motion planning where the platform offers it, and a preview that executes a candidate motion on the robot model alone, without the scene. They also provide the geometric primitives on which perception is built, namely the back-projection of pixels to metric 3D points, the poses of the cameras in the robot's frame, and the kinematic chain from each camera to the end-effectors. The *skills* are documents that describe how to work with these tools: the robot's frames, action spaces, rates, limits, and failure modes, the conventions of perception, and the workflow of exploration and memory. They are organized along four axes, namely task strategy, coordination, perception, and robot-specific control, and are composed into a single workspace for each session according to its embodiment and observation mode. Everything that solves a task, from object models and perception pipelines to grasps and control code, is written by the agent, which works in an isolated container with a persistent workspace and standard programming tools.

This division has three consequences. First, programs written by the agent call exactly the tools that the agent itself calls, so code developed interactively runs unchanged as a fully automated program. Second, the interface returns compact results, consisting of structured state and the final image of each motion, while the complete images and actions of every motion are stored as episodic evidence that the agent retrieves on demand, which keeps the agent's context small over sessions that last many hours. Third, on real robots, the skills that agents verify are kept in a versioned repository for each robot, in which every session works on its own branch, and a contribution is merged only after its structure, its entry point, and the evidence of its unattended success have been validated.

Most importantly, the separation makes RAX multi-embodiment by construction. Bringing up a new robot requires only a backend that exposes it through the tool interface, together with documentation of its control and sensing interfaces, while the frontend, the organization of memory, and the evaluation protocol are shared across all platforms. No expert teleoperation data is collected and no policy is trained to bring a robot into RAX. The single-arm, bimanual, dexterous, mobile, and humanoid embodiments of our experiments, in simulation and in the real world, differ only in their backends and in the documentation of their interfaces, and the agent learns to control each of them through exploration.

4.2 Multi-Embodiment Control Interface

RAX adopts a layered control design. Low-level controllers, from joint position and impedance controllers to whole-body motion trackers, run in the backend and are responsible for high-rate tracking and stability, while the agent makes decisions at the level of short action chunks and motion goals. Each platform exposes raw control for full expressiveness, together with

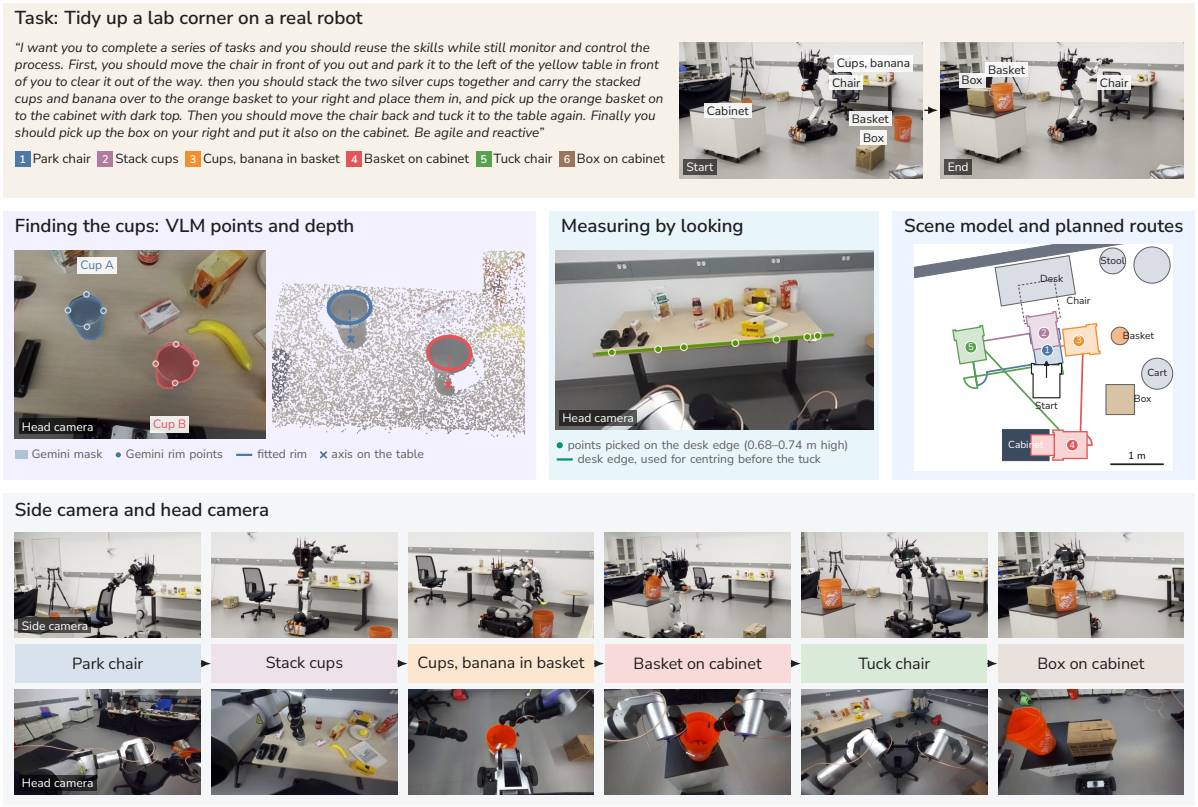


Figure 4 | **A long-horizon task on the real RB-Y1.** The agent composes six sub-tasks from skills learned in earlier sessions. It finds the cups from Gemini's rim points and depth, measures the desk edge by picking points in the head image, and plans the base routes on a model of the scene.

model-based helpers in the form of forward and inverse kinematics, collision-aware motion planning (Sundaralingam et al., 2023), and a robot-only preview that executes a candidate motion on the robot model without the scene. The agent thus chooses its own level of abstraction, from individual joint targets to planned free-space motions. Table 1 summarizes this construction for each platform.

On the real robots, the backend wraps each vendor's own control stack. The Franka FR3 arm runs joint impedance control, and the agent commands straight-line Cartesian moves of the tool center point, joint-space moves whose paths are checked against the workspace and the table before the arm moves, and force-controlled grasps that close the gripper to a target width with a set force. The Rainbow RB-Y1, a wheeled mobile manipulator with a torso, two arms, and a two-joint head that carries an RGB-D camera, runs joint position control on its robot controller, which also accepts Cartesian IK commands for the torso and arms. The agent commands end-effector poses, joint-space postures of the torso, arms, and head, coordinated motions of the torso and arms, motions of the differential-drive base, and the gaze of the head camera. It can query the vendor's kinematic model for the collisions, joint load, and center of mass of any posture, and dry-run any motion command to learn whether it would be accepted. Neither robot offers a motion planner, so the agent composes long motions from short checked moves through clear intermediate postures.

On the HumanoidArena benchmark (Wang et al., 2026c), the agent commands the Unitree G1 through whole-body kinematic references rather than joint torques: each reference specifies a desired heading, a full joint configuration, and hand commands, and a whole-body motion

Table 1 | **Low-level control on each platform.** EEF: end effector; OSC: operational-space control; PD: proportional-derivative.

Platform	Low-level control	Agent action space	Model-based helpers
LIBERO, LIBERO-Pro single-arm Franka Panda	EEF pose control (OSC)	EEF position and rotation deltas; gripper command	Pixel-to-3D projection from depth
RoboDojo dual-arm ARX X5	Joint position control (PD)	Joint-target chunks for both arms and grippers; bounded EEF steps	Forward kinematics, differential IK, pose algebra, collision-aware motion planning, robot-only preview
DexJoCo Franka Panda + Allegro hands	Arm: EEF pose control (OSC); hand: joint position control	Flange pose and sixteen hand joint targets per arm, single- or dual-arm	Pixel-to-3D projection from depth, robot-only preview
BEHAVIOR-1K Galaxea R1Pro	Base velocity control; trunk and arm joint position control	Base velocity, trunk and arm joint targets, grippers; closed-loop base moves from odometry	Closed-loop arm IK, arm and trunk motion planning, robot-only preview
HumanoidArena Unitree G1 + Dex3	Whole-body motion tracker	Whole-body reference frames: heading, 29 joint targets, hand open or close	Arm and waist motion planning to hand poses; kinematic model for forward kinematics and image projection
Franka FR3 real	Joint impedance control	Straight-line EEF moves, joint moves, force-controlled grasps	Inverse kinematics; path checks against workspace and table
Rainbow RB-Y1 real, mobile manipulator	Joint position control; Cartesian IK commands for torso and arms; wheel velocity control	EEF poses; torso, arm, and head joint targets; coordinated torso and arm motions; base motion; head gaze	Forward and inverse kinematics; dry-run checks of collision, joint load, and center of mass

tracker converts these references into balanced whole-body motion. Because the robot moves through space only by stepping, locomotion is not provided as a primitive but must be constructed by the agent from leg references, and agents do so by measuring their own gait, calibrating step length, and composing walking, turning, and stepping over obstacles from these measurements.

On DexJoCo (Wang et al., 2026a), the official operational-space controller tracks the flange pose of each arm while position actuators drive the sixteen joints of each Allegro hand, and no motion planner is provided, so the agent plans the motion of each arm and the configuration of each hand itself. The remaining simulated platforms span the same range of abstraction. The single arm of LIBERO and LIBERO-Pro (Liu et al., 2023; Zhou et al., 2025) is driven through the benchmark's own operational-space controller by relative end-effector commands alone, whereas on RoboDojo (Chen et al., 2026a) and BEHAVIOR-1K (Li et al., 2023) the native controllers execute joint targets under each benchmark's own dynamics, and inverse kinematics and motion planning are offered as helpers alongside raw control rather than as replacements for it. Motion planning accounts for the robot and static structures but never for task objects, which the agent must perceive itself.

5 Experiments

In this section we report our simulation benchmark results as well as case studies on both simulation and real robot experiments.

5.1 Setup

We comprehensively evaluate RAX in both simulated environments and on real-world robotic hardware. In simulation, the evaluation spans six benchmarks covering single-arm, bimanual, dexterous, mobile, and humanoid embodiments: RoboDojo, BEHAVIOR-1K, HumanoidArena, DexJoCo, LIBERO, and LIBERO-Pro. In each session, a coding agent starts in a localized workspace containing only the platform's baseline skills. Guided by a task description and RGB-D observations, the agent explores training scenes within a strictly defined episode budget. The submitted code is then executed as a fully automated program under each benchmark's official evaluation protocol (Appendix C).

On physical robots, RAX is deployed on a Franka FR3 manipulator to master five distinct tabletop tasks: (1) *memory*, tracking and finding a hidden cube randomly shuffled under four cups (Appendix F.1); (2) *articulated objects*, opening an air fryer drawer, inserting a banana, and closing it (Appendix F.2); (3) *deformable objects*, wiping a stain off a table using a wet towel (Appendix F.3); (4) *precision*, inserting pegs into a ManipulationNet board with a stringent 0.1,mm clearance (Appendix F.4); and (5) *precision and long horizon*, dismantling and rebuilding a Bandu tower of irregular pieces (Appendix F.5).

Additionally, we evaluate RAX on a Rainbow RB-Y1 mobile manipulator to solve a long-horizon logistical task. This involves pulling out a chair, stacking cups, picking up a banana, loading the items into a basket, carrying the basket to a cabinet, tucking the chair back, and subsequently transferring a box to the cabinet (Appendix F.6). Crucially, the agent receives only natural-language instructions and zero teleoperated demonstrations. It is designed to autonomously evaluate its own success criteria, only prompting a human operator to reset the physical scene between trials.

5.2 Simulation Results

Here we report the simulation benchmark results across RoboDojo, BEHAVIOR-1K, HumanoidArena, DexJoco, and LIBERO and LIBERO-Pro.

5.2.1 RoboDojo

Setup. We measure exploration scaling on seven RoboDojo tasks with the dual-arm ARX X5 (Appendix C.3), spanning domain-randomized variants (*make toast random*, *hang mugs random*, *sweep blocks random*), as well as precision (*plug in charger*), memory (*swap blocks*), long-horizon (*make kong*), and open-instruction (*pour by language*) tasks. For each task, a single agent explores a single environment from an empty workspace, given the task's public scoring rubric with a short clarification of its conditions, RGB-D observations, and one image of the task's completed state. Sessions differ only in the exploration budget N (i.e., the number of exploration episodes), with each episode conducted in a new exploration scene. At the end of exploration, the agent submits its rehearsed bundle. The resulting fully automated program is then evaluated without the agent on 30 to 50 unseen scenes, scored by the native evaluator with partial credit.

Baselines. We compare our approach with the five strongest learned policies on the RoboDojo-Sim leaderboard (Chen et al., 2026a), which are trained on demonstrations, and with PhysicalRSI, an agent-based system listed separately on the same leaderboard (Table 2).

Results. With its largest budget on each task, RAX exceeds every baseline on all seven tasks, averaging 78.3% compared to 24.9% for VPP2, the strongest policy on the leaderboard (Table 2). The margins are largest where learned policies make little progress: on *sweep blocks random*, RAX reaches 71.7% where the best baseline achieves only 8.0%; and on *pour by language*, which every learned policy fails, it reaches 100.0%. Performance consistently grows with the exploration budget. *Plug in charger* rises from 26.0% with 4 exploration episodes to 94.0% with 8; *swap blocks* improves from 34.0% to 100.0% with 16; and *make kong*—for which the agent submitted no program within 8 episodes—increases from 64.0% with 16 episodes to 96.0% with 32. The domain-randomized tasks require the largest budgets: *make toast random* rises from 2.5% with 16 episodes to 31.7% with 32 and 47.2% with 64, and *sweep blocks random* improves from 53.3% to 71.7%, while *hang mugs random* stays between 33.3% and 39.2% across all budgets.

Table 2 | **Scaling exploration on RoboDojo.** Score (%) on unseen layouts; baselines from the top five policies of the RoboDojo-Sim leaderboard (Chen et al., 2026a). Cells with more than one run report the mean. –: not run. †No program submitted. ‡Agent + VLA; listed among agents and unranked on the leaderboard.

Method	**Budget**	Make toast random	Hang mugs random	Sweep blocks random	Plug charger	Swap blocks	Make kong	Pour by language
VPP2	–	38.0	13.5	1.3	16.7	26.0	78.7	0.0
Awomo-0.5	–	7.0	9.6	0.0	7.3	4.7	30.0	0.0
Simate-beta	–	7.7	11.4	0.0	29.3	38.7	76.0	0.0
Rex-M1 Preview	–	14.3	20.2	8.0	3.3	13.3	33.3	0.0
VPP2-Preview	–	13.7	14.5	2.7	6.0	22.7	34.7	0.0
PhysicalRSI‡	–	4.7	2.0	0.0	30.7	0.0	82.0	10.1
RAX	4	–	–	–	26.0	34.0	0.0	94.4
	8	–	–	–	**94.0**	74.0	0.0†	**100.0**
	16	2.5	36.9	53.3	–	**100.0**	64.0	–
	32	31.7	33.3	68.3	–	–	**96.0**	–
	64	**47.2**	**39.2**	**71.7**	–	–	–	–

5.2.2 BEHAVIOR-1K

Setup. We evaluate RAX on six long-horizon household tasks from the 2025 BEHAVIOR Challenge (Li et al., 2023), where previous challenge entries have struggled to make progress. For each task, cooperating agents explore parallel environments using the Galaxea R1Pro for 128 exploration episodes, given the task description, RGB-D observations, and an image sequence of a successful completion (keyframes of official demonstrations, without actions or states). We evaluate the agent-in-the-loop program: a fresh copy of the agent starts from the frozen workspace and attempts each of the ten public test instances of the challenge—the same instances used to report public results.

Review reports. Throughout exploration, after each run of the agent's code, a separate review model provides detailed feedback to the agent regarding the outcome. This report outlines any missed goal conditions and their likely causes, suggests the most valuable code changes, and includes the run's partial score alongside key camera frames (Appendix C.4).

Scoring and baselines. Episodes are evaluated using the challenge's Q-score, which measures the fraction of satisfied goal conditions and awards full credit for a complete success. We compare our approach against the five highest-ranked entries from the challenge (Table 3). Figure 5 illustrates the program RAX generates for the *boxing books* task.

Table 3 | **BEHAVIOR-1K.** Q-score (%) on the ten public test instances; baselines from the five highest-ranked entries of the 2025 BEHAVIOR Challenge (Li et al., 2023). †Fully automated program. ‡Privileged track.

Method	Boxing books	Getting organized	Sorting items	Clearing food	Clean up desk	Loading the car	**Avg.**
Robot Learning Collective	0.0	2.0	6.2	8.0	12.7	20.0	8.2
Comet	0.0	0.0	0.0	0.0	0.0	0.0	0.0
SimpleAI Robot	0.0	0.0	0.0	0.0	0.0	0.0	0.0
The North Star	0.0	0.0	0.0	0.0	0.0	20.0	3.3
Embodied Intelligence‡	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RAX	**26.7**†	**9.0**	**75.0**	**40.0**	**28.2**	**100.0**	**46.5**

5.2.3 HumanoidArena

Setup. We evaluate RAX on all seven tasks of HumanoidArena (Wang et al., 2026c) that have official test episodes: three human–object interaction tasks (Football, DoubleDesk, and P&PBox) and four human–scene interaction tasks (OpenDoor, SitSofa, Boxing, and VisNavi), on a Unitree G1 with Dex3 hands (Appendix C.5). The agent commands whole-body reference poses, which the SONIC motion tracker (Luo et al., 2026) turns into balanced whole-body motion. The agent perceives the scene through the egocentric RGB-D camera and receives the task description together with three teleoperated demonstrations, which are available only to the exploring agent. It explores with a budget of up to 128 episodes on DoubleDesk and P&PBox and up to 32 episodes on each of the other tasks, on scenes drawn from the training distribution and never on a test episode, and the resulting fully automated program is evaluated alone, without the agent, on unseen scenes, the 60 official test episodes of each task.

Baselines. We compare with the baselines of the benchmark, ACT (Zhao et al., 2023), Diffusion Policy (Chi et al., 2025), a flow-matching policy (Boston Dynamics and Toyota Research Institute, 2025), and the VLA $\pi_{0.5}$ (Intelligence et al., 2025), each trained on 100 teleoperated demonstrations per task, and with WB-WAM (Qin et al., 2026), a world action model pretrained on 1,880 hours of video and motion data and fine-tuned on the same 100 demonstrations per task, all with the same tracker (Table 4).

Results. RAX reaches an average success rate of 89.8% (Table 4), 7.9 points above WB-WAM and 29.8 points above the strongest baseline of the benchmark, from three demonstrations instead of one hundred. It succeeds in every test episode of OpenDoor and Boxing, and on Football and VisNavi it improves on WB-WAM by 28.3 and 16.6 points. Performance grows with the exploration budget: averaged over the seven tasks, the success rate rises from 52.1% with 2 exploration episodes to 80.7% with 32. Tasks governed by whole-body posture and locomotion, such as OpenDoor and SitSofa, are solved within a few episodes, Football and VisNavi rise from 40.0% and 71.7% to 98.3% and 93.3%, and the two whole-body loco-manipulation tasks keep improving with further exploration, P&PBox from 0.0% to 83.3% and DoubleDesk from 0.0% to 56.7% at 128 episodes. We also test RAX with the three demonstrations without their actions (RAX *, Section 5.3.4): without any action data, it still matches the 81.9% average of WB-WAM, which is fine-tuned on 100 action demonstrations per task.

5.2.4 DexJoCo

Setup. DexJoCo (Wang et al., 2026a) evaluates task-oriented dexterous manipulation with Franka arms that carry 16-degree-of-freedom Allegro hands, on six single-arm and five bimanual tasks. The agent commands the end-effector pose of each arm and the joint positions of each

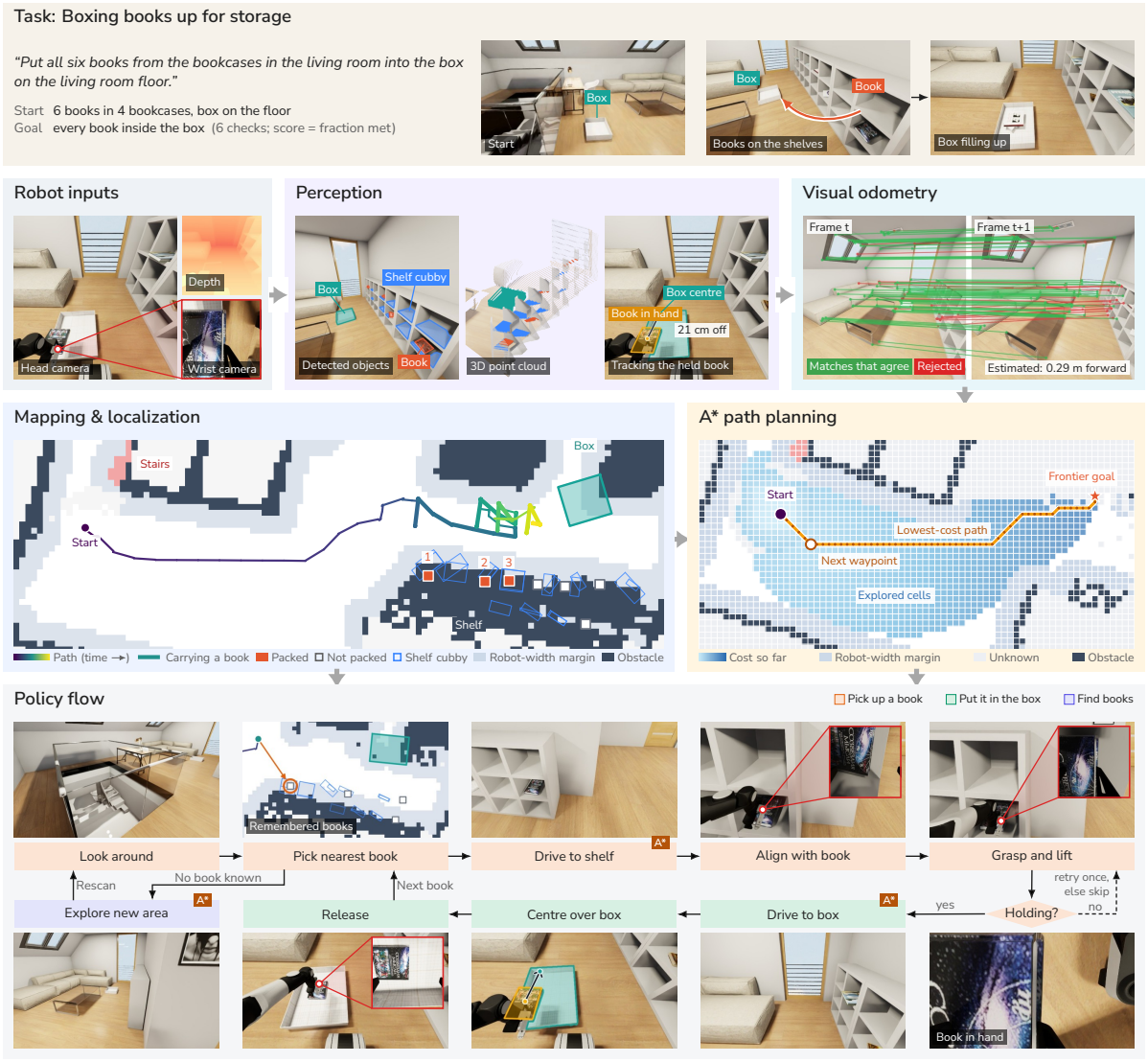


Figure 5 | **The program RAX writes for boxing books on BEHAVIOR-1K** (Table 3). It detects the books, shelves and box in the head camera, tracks the robot's motion by visual odometry, maps the scene, and plans paths with A*; the policy flow shows the camera view at each step.

hand. It perceives the scene through RGB-D observations and receives the task description and three completion images of expert episodes as a goal reference, but no action demonstrations or expert videos. For each task, a single agent explores a single environment from an empty workspace, with the exploration episodes drawn from the training distribution of the rand-obj setting and never on an evaluation layout. Sessions differ only in the exploration budget, from 2 to 32 exploration episodes, and the resulting fully automated program is evaluated alone, without the agent, on unseen scenes, the 150 official evaluation episodes of each task (Appendix C.6).

Baselines. We compare with the baselines of the benchmark, Diffusion Policy with transformer and CNN backbones (Chi et al., 2025), ACT (Zhao et al., 2023), and the VLAs $\pi_{0.5}$ (Intelligence et al., 2025) and GR00T N1.5 (Bjorck et al., 2025), each trained on 100 teleoperated demonstrations per task, and with subsequent results on the same benchmark: GR00T N1.6, the world action models Fast-WAM (Yuan et al., 2026), PointWAM (Park et al., 2026), which pretrains on

Table 4 | **HumanoidArena.** Success rate (%) on the 60 official test episodes of each task, all methods with the SONIC tracker. Baselines from Wang et al. (2026c) and Qin et al. (2026). Max: each task at its largest budget. *Three demonstrations without actions. –: not run.

			Human–object interaction			Human–scene interaction				
Method	**Demos**	**Budget**	Football	DoubleDesk	P&PBox	OpenDoor	SitSofa	Boxing	VisNavi	**Avg.**
ACT	100	–	16.7	18.3	56.7	78.3	73.3	56.7	33.3	47.6
Diffusion Policy	100	–	45.0	36.7	75.0	85.0	78.3	76.7	23.3	60.0
Flow matching	100	–	13.3	38.3	73.3	70.0	15.0	70.0	38.3	45.5
$\pi_{0.5}$	100	–	10.0	43.3	71.7	66.7	73.3	70.0	23.3	51.2
WB-WAM	100	–	70.0	**65.0**	**86.7**	98.3	95.0	81.7	76.7	81.9
RAX	3	2	40.0	0.0	0.0	98.3	96.7	58.3	71.7	52.1
		4	51.7	1.7	16.7	93.3	98.3	95.0	80.0	62.4
		8	76.7	5.0	23.3	93.3	**100.0**	80.0	80.0	65.5
		16	96.7	28.3	20.0	93.3	96.7	80.0	83.3	71.2
		32	**98.3**	25.0	51.7	**100.0**	96.7	**100.0**	**93.3**	80.7
		64	–	25.0	46.7	–	–	–	–	–
		128	–	56.7	83.3	–	–	–	–	–
		Max	**98.3**	56.7	83.3	**100.0**	96.7	**100.0**	**93.3**	**89.8**
RAX *	3	Max	**98.3**	46.7	70.0	95.0	**100.0**	86.7	76.7	81.9

1.15 million human videos, and ME-Dex (Zhang et al., 2026), and the diffusion transformer DECO (Li et al., 2026) (Table 5).

Results. Performance grows rapidly with exploration (Table 5): averaged over the eleven tasks, the success rate rises from 31.9% with 2 exploration episodes to 86.9% with 8 and 96.7% with 32, and with 32 exploration episodes RAX exceeds the best published result on every task. Hammer, Mouse, and iPad are solved from two exploration episodes, most other tasks within four to eight, and the bimanual Assembly within sixteen. Taking the best session of each task, RAX succeeds in every evaluation episode of eight of the eleven tasks and reaches an average success rate of 99.5%, 45.4 points above the strongest baseline of the benchmark and 34.2 points above the best published average, without a single action demonstration. The margins are largest on the bimanual tasks that existing methods rarely complete: RAX solves Hanoi (Appendix D.2) and iPad in every episode, where the best published results are 42.0 and 52.0, and reaches 96.0 on Assembly, where the best published result is 28.0.

5.2.5 LIBERO and LIBERO-Pro

Setup. LIBERO (Liu et al., 2023) evaluates a single Franka Panda arm with a parallel-jaw gripper on four ten-task suites, Spatial, Object, Goal, and Long (Appendix C.7), and LIBERO-Pro (Zhou et al., 2025) perturbs the layouts of the same tasks. The agent commands delta end-effector poses and the gripper. It perceives the scene through the agent-view and wrist RGB-D cameras and receives the task description and three images of the completed task, but no action demonstrations. It explores with a budget of up to 20 episodes on fresh layouts drawn from the official placement distribution and may continue with up to 5 episodes of adaptation on layouts drawn from the position-perturbed distribution of LIBERO-Pro, in which objects exchange their initial regions (Appendix D.4), never on an evaluation state. The resulting fully automated program is evaluated alone, without the agent, on unseen scenes, the 50 official initial states of each task and the 50 position-perturbed states of LIBERO-Pro.

Baselines. We compare with the VLAs OpenVLA (Kim et al., 2024), OpenVLA-OFT (Kim et al., 2025), π_0 (Black et al., 2024), $\pi_{0.5}$ (Intelligence et al., 2025), and X-VLA (Zheng et al.,

Table 5 | **DexJoCo, rand-obj setting.** Success rate (%) on the 150 official evaluation episodes of each task. Baselines from Wang et al. (2026a), Park et al. (2026) (GR00T N1.6, Fast-WAM, PointWAM), Zhang et al. (2026), and Li et al. (2026). DP-T and DP-C: Diffusion Policy with transformer and CNN backbones. †One policy for all tasks. ‡Mean over the ten tasks other than iPad. §With simulated tactile input. –: not reported.

		Single-arm						Bimanual					
Method	**Budget**	Hammer	Mouse	Bucket	Tongs	Glasses	Water	iPad	Hanoi	Assembly	Microwave	Photo	**Avg.**
DP-T	–	81.3	62.0	83.3	22.7	53.3	84.0	8.0	24.7	4.7	73.3	56.7	50.4
DP-C	–	58.7	74.0	70.0	57.3	54.0	63.3	52.0	12.7	3.3	54.0	24.0	47.6
ACT	–	50.0	61.3	64.0	31.3	47.3	47.3	9.3	6.0	0.0	66.0	7.3	35.5
$\pi_{0.5}$	–	84.7	64.7	84.0	24.0	72.0	88.7	36.0	23.3	2.7	60.0	54.7	54.1
GR00T N1.5	–	67.3	85.3	72.0	12.7	27.3	72.7	12.7	0.7	0.7	50.7	40.7	40.2
GR00T N1.6	–	58.0	66.0	94.0	33.3	51.3	77.3	–	24.7	4.0	76.7	30.0	51.5‡
DECO	–	68.0	90.0	80.0	46.0	50.0	88.0	40.0	18.0	0.0	64.0	76.0	56.4
Fast-WAM†	–	43.3	77.3	81.3	47.3	62.7	42.7	–	42.0	2.7	88.7	40.7	52.9‡
PointWAM	–	82.0	89.3	96.7	50.0	62.0	94.7	–	36.0	11.3	72.0	24.0	61.8‡
ME-Dex†§	–	86.0	76.0	92.0	92.0	56.0	72.0	48.0	32.0	28.0	88.0	48.0	65.3
RAX	2	98.7	**100.0**	0.0	36.7	0.7	0.0	**100.0**	0.0	14.7	0.0	0.0	31.9
	4	98.7	97.3	93.3	2.7	0.0	0.0	**100.0**	**100.0**	0.7	0.0	0.0	44.8
	8	94.0	**100.0**	**100.0**	**100.0**	**100.0**	91.3	**100.0**	**100.0**	0.0	84.0	86.7	86.9
	16	98.0	**100.0**	96.7	99.3	55.3	98.7	**100.0**	98.7	**96.0**	93.3	68.7	91.3
	32	**100.0**	**100.0**	**100.0**	**100.0**	93.3	**99.3**	**100.0**	**100.0**	72.0	**99.3**	**100.0**	**96.7**

2026), and with the world action models Cosmos Policy (Kim et al., 2026), Fast-WAM (Yuan et al., 2026), PatchWAM (Wang et al., 2026b), and VPP2 (Guo et al., 2026), all of which fine-tune on the benchmark's demonstrations (Table 6).

Results. Without a single action demonstration, RAX achieves one of the highest average success rates among these models on the original suites, 98.5% (Table 6). Under position perturbation the learned policies collapse: OpenVLA, π_0, and X-VLA succeed in at most 2% of the episodes of any suite, $\pi_{0.5}$ in at most 38%, and the world action models PatchWAM and VPP2 in 8.8% and 42.3% on average. Without a single episode on perturbed layouts, the programs of RAX, which locate every object anew in each episode, already retain 76.0% on average and exceed every learned policy on every suite, 33.7 points above VPP2; they also exceed by 12.8 points the coding agent ASPIRE (Lu et al., 2026), which develops a program for each task and perturbation with up to 75 rollouts on instances from the evaluation distribution (Appendix E). With at most 5 episodes of adaptation on perturbed layouts, RAX raises its average by a further 8.4 points to 84.4%, from 71.4% on Long to 100.0% on Object, and exceeds ASPIRE on every suite. Exploration thus yields programs that transfer to new layouts, and a few further episodes of exploration repair those that do not.

5.3 Ablations

In this section we ablate the key design choices of our system, including parallel exploration with multi-agent and multiple environments, the effect of depth sensing over RGB-only observations, and the evolve of inference tokens and inference time over exploration.

5.3.1 Parallelization

Exploration can also be parallelized (Section 3.5): the same budget of exploration episodes is distributed across K parallel environments, explored by a team of cooperating agents that share a common workspace. In the same way that multiple agents scale test-time compute in parallel rather than purely serially, trading some efficiency for wall-clock time, parallel environments let a team of agents consume an exploration budget concurrently. We parallelize

Table 6 | **LIBERO and LIBERO-Pro.** Success rate (%) on the 50 official initial states of each task and on the 50 position-perturbed states of LIBERO-Pro. Baselines from their papers, openpi (π_0, $\pi_{0.5}$), and the LIBERO-Pro leaderboard (Zhou et al., 2025); PatchWAM reports only the mean, and VPP2 uses 10 initial states per task. †Developed on the perturbed setting under evaluation (Appendix E). ‡Zero-shot with a skill library built on LIBERO-90; included in the average. –: not reported.

		LIBERO					LIBERO-Pro, position perturbation				
Method	**Demos**	Spatial	Object	Goal	Long	**Avg.**	Spatial	Object	Goal	Long	**Avg.**
OpenVLA	50	84.7	88.4	79.2	53.7	76.5	0.0	0.0	0.0	0.0	0.0
OpenVLA-OFT	50	97.6	98.4	97.9	94.5	97.1	–	–	–	–	–
π_0	50	96.8	98.8	95.8	85.2	94.2	0.0	0.0	0.0	0.0	0.0
$\pi_{0.5}$	50	98.8	98.2	98.0	92.4	96.9	20.0	17.0	38.0	8.0	20.8
X-VLA	50	98.2	98.6	97.8	97.6	98.1	0.0	2.0	1.0	0.0	0.8
Cosmos Policy	50	98.1	**100.0**	98.2	97.6	98.5	–	–	–	–	–
Fast-WAM	50	98.2	**100.0**	97.0	95.2	97.6	–	–	–	–	–
PatchWAM	50	96.5	**100.0**	**99.0**	**98.0**	98.4	–	–	–	–	8.8
VPP2	50	**99.2**	99.8	98.2	**98.0**	**98.8**	54.0	58.0	37.0	20.0	42.3
ASPIRE†	0	–	–	–	–	–	51.0	98.0	81.0	22.6‡	63.2‡
RAX, no adaptation	0	**99.2**	**100.0**	98.6	96.2	98.5	**79.2**	**100.0**	73.4	51.2	76.0
RAX, adaptation	0	**99.2**	**100.0**	98.6	96.2	98.5	**79.2**	**100.0**	**87.0**	**71.4**	**84.4**

exploration on *make toast*. All sessions use RGB-D observations, an image sequence of one successful completion, and instructions that summarize the official task code, including its numeric success thresholds; the fully automated program is evaluated on the same 32 unseen scenes. Parallelization reduces the time needed to consume the exploration budget for both agents (Table 7). With four agents in four environments, Claude Code reaches a score of 92.2 in 4.9 hours, against 63.3 in 13.7 hours for a single agent with a larger budget; with two agents in two environments, Codex matches its single-agent score in 9.2 hours instead of 14.2.

Table 7 | **Parallel exploration on make toast.** Agent time until submission, and success rate and score (%) of the fully automated program on 32 unseen scenes.

Agent	**Agents**	**Envs**	**Budget**	**Time (h)**	**Success**	**Score**
Codex	1	1	32	14.2	40.6	66.4
Codex	2	2	32	9.2	56.3	65.6
Claude Code	1	1	48	13.7	50.0	63.3
Claude Code	4	4	32	4.9	87.5	92.2

5.3.2 Depth versus RGB-Only Perception

To measure the contribution of depth, we compare RGB-D and RGB-only sessions on the three domain-randomized RoboDojo tasks. In the RGB-only profile, depth and the geometric primitives built on it are removed, and the agent must perceive through motion, relating image coordinates to robot motion by moving the robot by known amounts. Each pair shares the task, the agent, and the exploration budget N, and its fully automated program is evaluated on the same unseen scenes as in Section 5.2.1. With depth, the agent outperforms the RGB-only agent in every pair, by 17.3 points on average (Table 8).

Case study: perception with and without depth. To understand where this gap comes from, we inspect the fully automated programs submitted by the RGB-D and RGB-only agents on make toast random with a budget of 32 episodes. Both agents use the vision-language model

Table 8 | **Depth versus RGB-only perception.** Score (%) on unseen layouts at matched exploration budgets. Cells with more than one run report the mean.

Task	Budget	RGB-D	RGB-only	Δ
Make toast random	16	**2.5**	0.0	+2.5
Make toast random	32	**31.7**	7.8	+23.9
Hang mugs random	16	**36.9**	13.9	+23.0
Sweep blocks random	16	**53.3**	33.3	+20.0
Average		**31.1**	13.8	+17.3

in the same role, to point at semantic image locations such as the toaster slots, the top crust of a bread slice, and the lever. They differ in how they turn these image points into motion.

With depth. The RGB-D agent lifts every image point into a metric 3D point and builds the task's geometry from these points. It first refines each slot opening by fitting a line to its visible contour, then measures both rim endpoints of each slot in 3D, from which it derives the slot centers and their common heading. On the bread, it measures two points along the top crust and one on the near side, and recovers the slice's full 3D orientation from the plane through these points. After the grasp, it measures the slice again in the wrist camera to obtain its pose in the hand, and repeats this measurement above the slot to correct any slip before release. The agent also uses metric geometry to verify its perception: a measurement is rejected when a rim point does not lie within a narrow band at the toaster's roof height, when the two rim endpoints are not 9 to 20 cm apart, or when the two slots are not 2.6 to 7.5 cm apart, and a press of the lever counts only when the lever is measured to have travelled 2.5 to 13 cm downward.

Without depth. The RGB-only agent does not give up on 3D reasoning; it reconstructs a coarse camera model of its own. It maps overhead-camera pixels to table coordinates through a linear calibration at an assumed height, about 1.8 mm per pixel across the image and 1.4 mm per pixel along it, and then refines every target by visual servoing in the wrist camera: it moves the gripper until the target is centered, with hand-tuned gains of 0.4 to 0.8 mm per pixel, and rotates it until the slot or the bread is aligned in the image. The quantities that depth would provide are replaced by constants, including the grasp and release heights, the 3.8 cm spacing between the slots, and the height to which the lever is pressed, and success is judged by asking the vision-language model whether a slice is held or seated. These approximations hold only where the scene matches the agent's assumptions. In evaluation, 36 of its 45 episodes ended in the lever stage without success, the step at which a fixed press height replaces a measured one, against 8 successes for the RGB-D agent.

5.3.3 Inference Time and Tokens

Here we demonstrate the decrease of inference wall time and output tokens with exploration as more and more decisions are being automated with programs.

Simulation As the agent compiles its experience into autonomous code programs, the computational overhead of action generation decreases substantially. We evaluate this efficiency gain across two benchmark tasks: *loading the car* in BEHAVIOR-1K (128 exploration episodes across two parallel environments) and *make toast random* in RoboDojo (64 exploration episodes). We define episode *inference time* as the cumulative wall-clock latency dedicated to action generation—excluding simulator and hardware step times—regardless of whether actions stem from direct LLM deliberation, code program execution, or a hybrid mode. Figure 6 tracks inference latency and output token consumption across all successful exploration episodes. On *loading the car*, the initial successful episode required 123 minutes of inference time and

254k output tokens. By the final exploration episode, inference overhead dropped to 4 minutes and 4k tokens, while environment execution time remained over 1 hour per episode. Similarly, on *make toast random*, initial task completion required 17 minutes of inference latency. Once execution transitioned to the compiled program, inference latency for the final agent-monitored successes fell to 6 seconds per episode—matching the 3–7 seconds observed during standalone program rehearsals (which accounted for 7 of the 15 total successes).

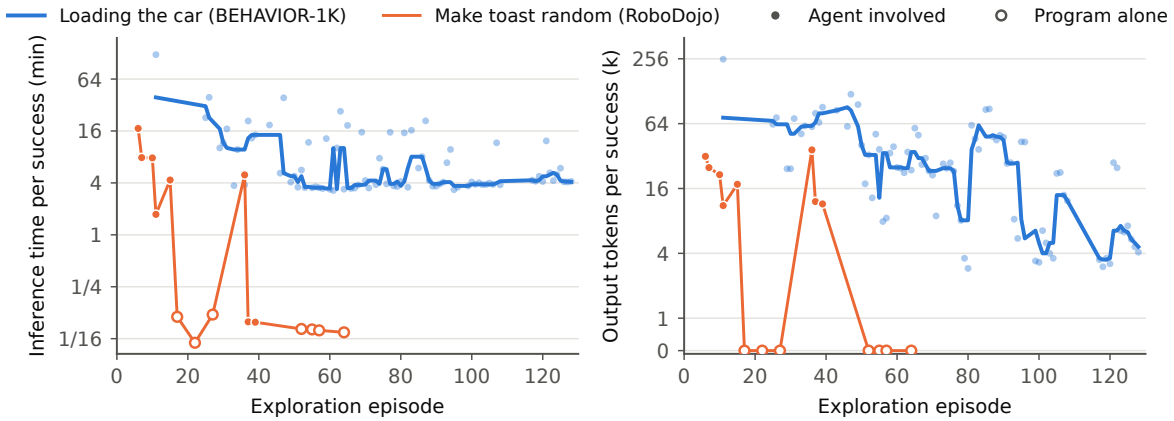


Figure 6 | **From exploration to code.** Inference time (left) and output tokens (right) of each successful exploration episode, on base-2 logarithmic vertical axes; for rehearsals, inference time is the program's computation time as measured by the harness, without the overhead of running in isolation. The token axis is linear below one thousand tokens so that zero can be shown. Filled dots mark episodes in which the agent takes part, alone or together with its program; hollow circles mark rehearsals, in which the program runs alone. On loading the car, raw points are faint and the line is a running median over five successes; output tokens of parallel episodes are attributed to the environment an agent was assigned, or split evenly across the episodes active at that moment.

Real robots Similar efficiency trends hold on real robots using the Franka manipulator (Table 9). We evaluate air fryer manipulation (Appendix F.2) and Bandu tower assembly (Appendix F.5), comparing an initial trial in which the agent generates each action via active deliberation against a subsequent execution of the compiled autonomous program. In both paradigms, the system actuates the robot through identical tool interfaces, with robot time defined as the duration spent executing arm, gripper, and visual perception calls. For the air fryer task, initial task completion (including online exploration) required 72 minutes of inference latency and 262k output tokens, against 3 minutes of robot time. Upon compilation, the initial autonomous run required only 18 seconds of inference latency (attributable entirely to local programmatic execution), with four subsequent unattended trials consistently yielding latencies between 18 and 23 seconds. Similarly, on the Bandu tower, step-by-step LLM deliberation to assemble six blocks required 17 minutes of inference latency; once compiled, autonomous assembly of seven blocks completed with 10 seconds of program overhead alongside 11 minutes of continuous robot execution. During these autonomous runs, control proceeded without human intervention or online LLM action synthesis, while background monitoring generated at most 2.2k tokens per run in parallel without imposing latency on robot execution.

5.3.4 Demonstrations Without Actions

On HumanoidArena, the exploring agent reads three teleoperated demonstrations of each task, and its programs may reuse motion segments cut from them, such as a gait cycle (Appendix D.3). We remove the actions from these demonstrations and keep everything else

Table 9 | **From the agent's actions to a fully automated program on the real Franka.** The first success with actions generated by the agent and a later unattended success of its program. Inference time: wall-clock time outside robot calls. The agent's Bandu rebuild places six pieces, the program seven.

Task	Actions from	Inference	Robot time	Output tokens
Air fryer	Agent	72.3 min	2.9 min	262.4 k
	Program	18 s	2.5 min	2.0 k
Bandu tower	Agent	17.2 min	9.3 min	88.4 k
	Program	10 s	10.9 min	2.2 k

unchanged (RAX ⋆), with a budget of 256 exploration episodes on DoubleDesk, 128 on P&PBox, and 64 on each of the other five tasks, and evaluate the program on the same 60 official test episodes. Without any action data, RAX ⋆ reaches 81.9% on average and matches WB-WAM, which is fine-tuned on 100 action demonstrations per task (Table 10). On Football, OpenDoor, and SitSofa, it stays within 5 points of RAX with action demonstrations. The more dynamic and contact-rich whole-body tasks, which require sustained locomotion or whole-body contact, VisNavi, Boxing, P&PBox, and DoubleDesk, remain 10.0 to 16.6 points lower despite larger budgets. When the agent has to discover a suitable gait and whole-body control on its own, whole-body loco-manipulation requires more exploration.

Table 10 | **HumanoidArena without action demonstrations.** Success rate (%) on the 60 official test episodes of each task. RAX: largest budgets of Table 4. ⋆Three demonstrations without actions.

Method	Demos	Budget	Football	DoubleDesk	P&PBox	OpenDoor	SitSofa	Boxing	VisNavi	Avg.
WB-WAM	100	–	70.0	**65.0**	**86.7**	98.3	95.0	81.7	76.7	81.9
RAX	3	32–128	**98.3**	56.7	83.3	**100.0**	96.7	**100.0**	**93.3**	**89.8**
RAX ⋆	3	64–256	**98.3**	46.7	70.0	95.0	**100.0**	86.7	76.7	81.9

5.4 Intriguing Emerging Agentic Behaviors with Scaling Exploration

In this section, we analyze agentic behaviors that emerge as exploration scales, in simulation and on real robots; each task is detailed in Appendices D and F.

5.4.1 Active Perception

Active perception (Xiong et al., 2025) emerges naturally as the agent does not treat perception as a single look at the scene (Figure 7). It moves its cameras and its hand to obtain the measurements that a step needs, measures again before it acts, and checks the outcome of each action with a new measurement.

Simulation. On the bimanual Tower of Hanoi of DexJoCo (Appendix D.2), the agent received no model of its Allegro hands. It spent its first episode on its own hand: it moved single joints, lifted the wrist cameras' depth into points to see what each joint moves, segmented the white fingertip caps, and fitted the fingertip offsets on which every grasp of its program is built; its thumb offset lies within 1.1 mm of the official model it never saw. On the LIBERO-Long task of putting two cans into a basket (Appendix D.1), the program perceives the scene anew in every round; before each grasp, it hovers 7 cm above the can, refits the lid from the wrist camera, and rejects a top that is not can-shaped. Because the agent-view camera cannot see the floor of the basket, the program maps the basket from the wrist camera while it holds the first can, and maps it again after the first release. On the bimanual Tower of Hanoi of DexJoCo (Appendix D.2), the program verifies every lift by measuring the held disk from three cameras and pinches again when the disk rose less than half of the lift; after placing the red disk, it

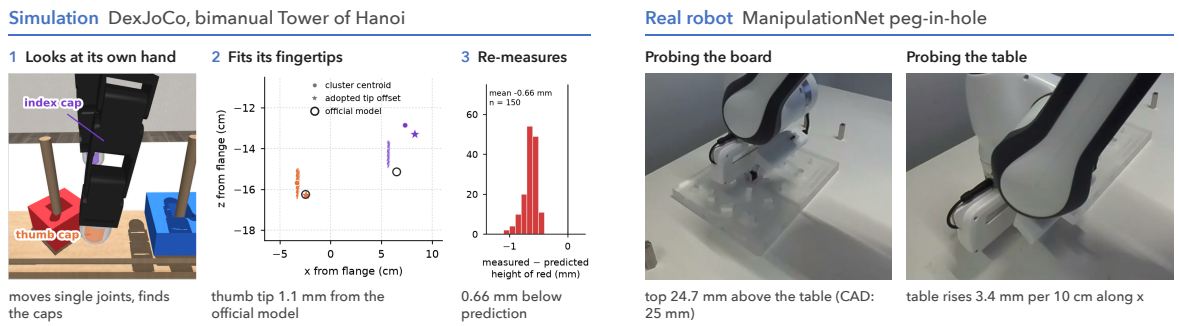


Figure 7 | **Active perception.** Simulation, DexJoCo bimanual Tower of Hanoi (Appendix D.2): (1) in its first episode, the agent moves single joints of the Allegro hand and finds the white fingertip caps in the wrist camera; (2) from the caps' points it fits the fingertip offsets relative to the flange, and, without a model of the hand, its thumb tip lies 1.1 mm from the official model; (3) after each placement, the program measures the disk again, on average 0.66 mm below the height it predicted over 150 evaluation episodes. Real robot, ManipulationNet peg-in-hole (Appendix F.4): the agent probes with closed fingers to measure the height of the transparent board and the tilt of the table.

measures the disk's top again and finds it on average 0.66 mm below the height it predicted. On the humanoid that carries a hammer over a bar (Appendix D.3), the program perceives anew at every stage and before every step at the bar, and it confirms a grasp by measuring the height of the handle again. Without depth, the RGB-only agent recovers metric relations by moving the robot by known amounts and servoing on the wrist camera (Section 5.3.2).

Real robot. On the ManipulationNet peg-in-hole challenge (Chen et al., 2026b) (Appendix F.4), the board is transparent acrylic, and the task text warned that the wrist camera's depth is inaccurate at close range. The agent turned this warning into a measurement procedure that uses the arm itself as a probe: it closes the gripper, descends in 1.5 mm steps, and records the height at which the vertical force drops by 3 to 4 N. Touching the table at three points, it found that the table rises by about 3.4 mm per 10 cm along the robot's x axis and that the top of the board lies 24.7 mm above the table, against 25 mm in the board's CAD model, and it models the height of the board's top as a plane that rises along x. Before each insertion into the row with 0.1 mm clearance, it touches the board next to the hole with the held peg, and these reference heights agree within 0.5 mm. Probing also calibrates the camera: from a hole whose centre it had located by touch, the agent found that the wrist camera's absolute positions were offset by 6.4 and -0.8 mm, corrected every hole by this offset, and the next peg dropped into its hole on the first probe.

5.4.2 Error Recovery and Safety

On real robots, the agent's mistakes have physical consequences, and it turns each one into a rule that its later programs follow (Figure 8). On the RB-Y1, an in-place turn that the agent planned by hand swung the rear of the base into a cabinet; from then on, the agent checked every base move against the area the base sweeps. Lifting a bucket while squatting overloaded a wrist and stopped the robot; standing up before the lift, a correction from the operator, resolved it, and the agent kept it as a rule in its skill. On the Franka, an unusual arm posture drove the elbow into a camera; the agent then restricted itself to natural arm postures, kept the arm away from the cameras, and recalibrated the bumped camera itself (Appendix F.2.1). In simulation, the humanoid crosses a bar only from foot placements that its own falls showed to be safe (Appendix D.3).

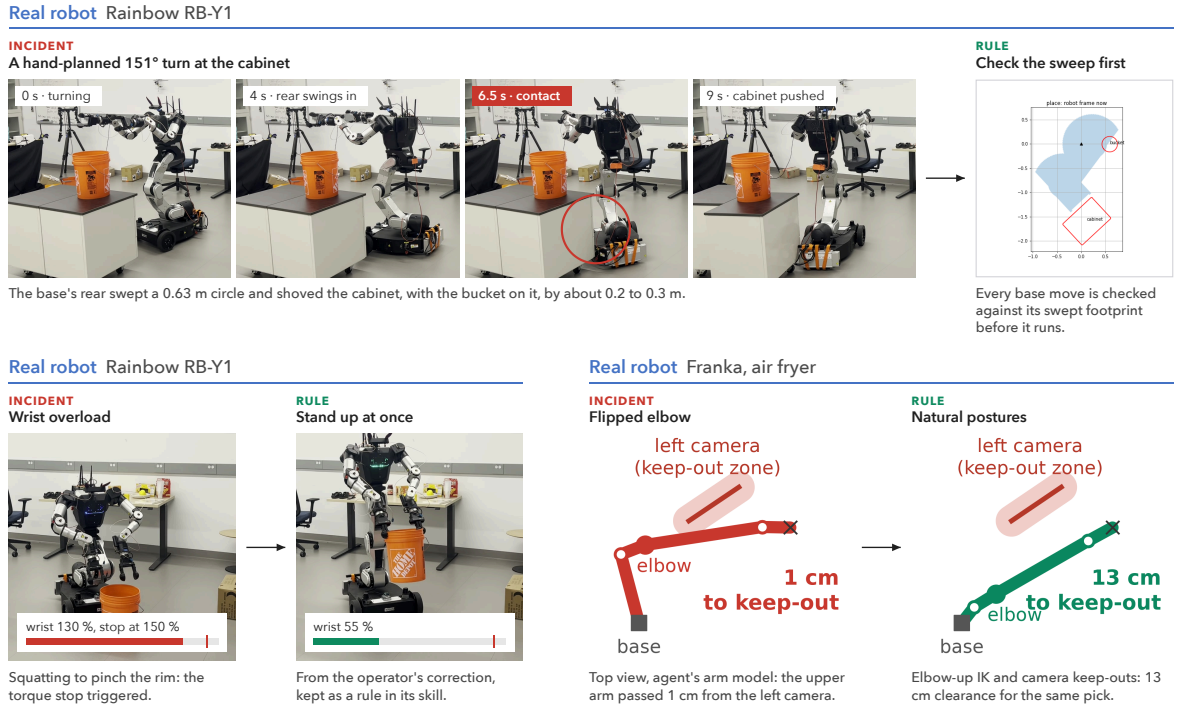


Figure 8 | **Error recovery and safety on real robots.** Each incident and the rule the agent derived from it. RB-Y1: a turn planned by hand pushes a cabinet, after which every base move is checked against the area the base sweeps; squatting to lift a bucket overloads a wrist, and standing up first resolves it. Franka: an unusual arm posture hits a camera, after which the agent plans only natural postures that keep the arm away from the cameras.

5.4.3 Human–Robot Interaction

The agent reaches the operator through a request tool and uses it with purpose (Figure 9). On the ManipulationNet board, after the board shifted during its attempts at tight fits, it asked the operator to tape the board to the table. On the air fryer (Appendix F.2), it refused to pick a banana placed in front of the camera it had hit earlier, because its forearm would pass within about 1 cm of the camera, and asked for the banana to be moved. After unattended successes, it asked the operator to move and rotate the fryer, in effect designing its own tests of generalization: across runs 6 to 10, the fryer moved by up to 7 cm and its handle turned by up to 18°, and the program succeeded unattended in every new placement but one, where it needed help and which exposed two errors in its program that the agent then fixed.

5.5 Continual Learning

In this section, we demonstrate how RAX achieves continual learning with robot programs: by reusing skills learned in earlier exploration sessions, and by rapidly adapting its programs when the environment or the task changes.

5.5.1 Reusing Skills

On the RB-Y1 mobile manipulator, RAX acquires skills through isolated exploration sessions and reuses them to solve long-horizon compositional tasks (Figure 10). In initial sessions, the agent independently learned to park and tuck an office chair, stack nested cups, transport a bucket by its rim onto a cabinet, and pick and place a box. Each skill was committed to the repository only after achieving successful end-to-end execution, stored as a versioned program

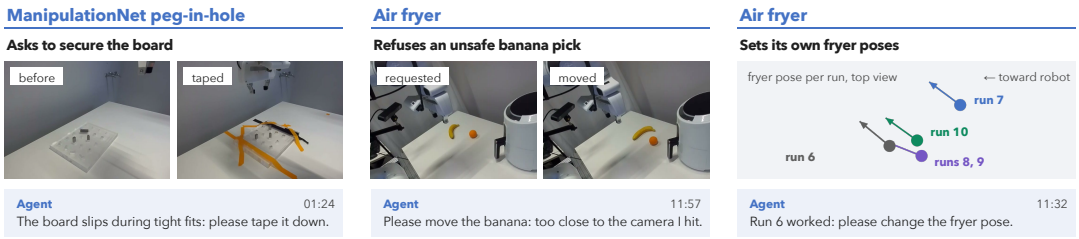


Figure 9 | **Human–robot interaction.** Requests that the agent sent to the operator, summarized, with the scene before and after. Right: the fryer poses that the agent requested across runs 6 to 10, from its own measurements of the fryer's axis and handle direction.

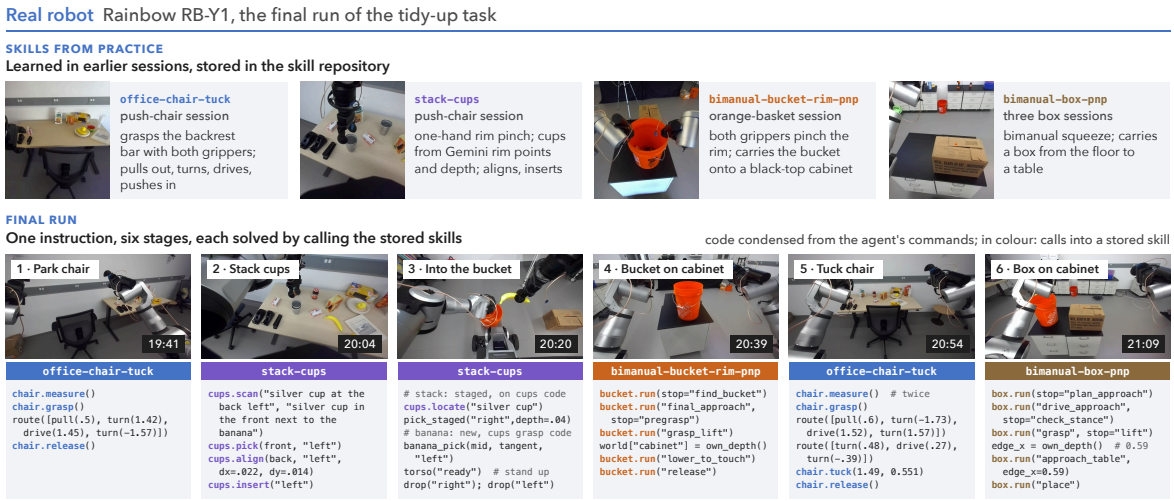


Figure 10 | **Continual learning on the real-world RB-Y1 mobile manipulator. Top:** Modular skills acquired during prior exploration sessions and stored in the skill repository, complete with descriptions from their respective skill files. **Middle:** The final execution of a complex tidy-up task—a single high-level instruction carried out across six sequential stages, annotated with completion timestamps and invoked skills. **Bottom:** The agent's generated code for each stage (condensed from execution logs with passed arguments). Code invoking pre-stored skills is color-coded by skill (`chair`, `cups`, `bucket`, and `box` correspond to each skill's `run.py`), while ad-hoc control logic written directly by the agent is shown in black.

bound to explicit precondition checks. The multi-stage tidy-up task (Figure 4) evaluates the composition of these skills under a single high-level goal: park the chair, stack the cups, place the cups and a banana inside the bucket, transport the bucket onto the cabinet, tuck the chair back under the desk, and finally pick the box onto the other empty cabinet.

Skills are called, not relearned In the final execution of the tidy-up task, every stage is resolved by composing stored skills from the repository (Figure 10). The agent executes the chair skill sequentially: it estimates the chair geometry, plans a collision-free motion path, grasps the backrest, navigates along the path, and releases the target. It invokes the cup-stacking skill using semantic visual grounding from the current scene, applying fine-grained closed-loop offsets prior to insertion. For the bucket and box skills, the agent parameterizes execution using real-time perception data, such as the estimated cabinet boundary derived from depth sensing. To bridge gaps between stored skills, the agent dynamically synthesizes ad-hoc code—such as extending the grasp planner to handle a banana, commanding torso elevation prior to transit, and re-estimating desk pose before final chair insertion.

Skills are refined in use Across three iterations of the tidy-up task, the agent iteratively updated program versions rather than rewriting entire control loops. The bucket and box skills required minor programmatic adjustments (∼25 out of more than 1200 lines) to integrate dynamic cabinet-edge measurements from onboard depth perception. Skills encountering greater environmental domain shift underwent larger revisions: following two failed grasp attempts, the agent updated the chair skill to grasp 4 cm deeper, while extending the cup-stacking skill to support picking additional object classes.

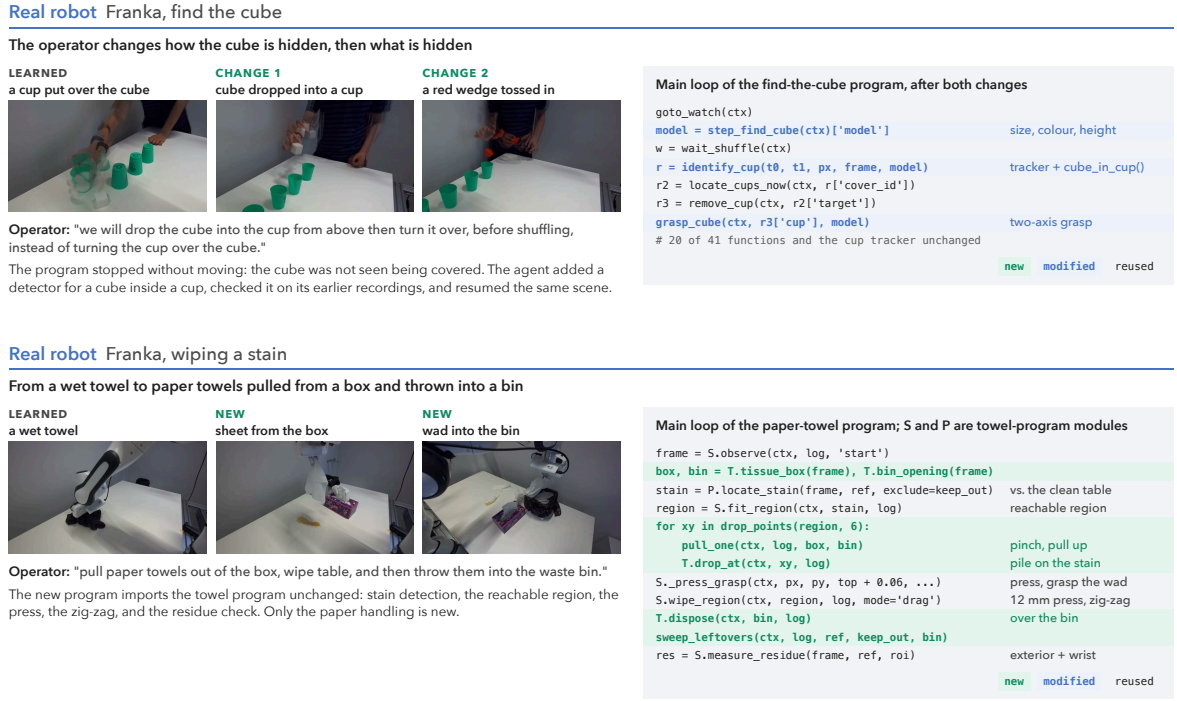


Figure 11 | **Adapting skills and programs on the real Franka.** Each row is one program that already worked, with the task it learned (left image) and the operator's change (green). Right: the program's entry point after the change, condensed from the agent's code, with new lines in green, modified lines in blue, and reused lines in black. Top: find the cube, where each image overlays frames from the left camera a quarter of a second apart, earlier ones fainter, to show how the object is hidden; the cube is first dropped into a cup and then replaced by a red wedge. Bottom: wiping with paper towels from a box; `S` and `P` are the towel program's modules, imported unchanged.

Repository expansion Intermediary control code synthesized by the agent to bridge task stages is modularized and saved as new skills. For instance, a synthesized `hand-reach` primitive was automatically abstracted, committed to the repository, and subsequently reused across both the chair-manipulation and cup-stacking routines.

5.5.2 Rapid Adaptation of Programs

When task conditions vary post-deployment, RAX preserves valid program subroutines and selectively updates only the modules invalidated by environmental shifts (Figure 11). On the Franka manipulator, two end-to-end autonomous programs were adapted to novel task variants within the same session.

Find the cube In a hidden-object task, the physical occlusion sequence was modified: rather than directly inverting a cup over the target cube, the cube was placed inside an upright cup prior to inversion. The original program detected no covering cup and safely halted

execution via exception handling without initiating ungrounded motion. The agent extended its perception pipeline with a detector for cubes contained within upright cups, validated the new logic against current and historical sensor logs, and resumed execution to complete the task (Figure 21). Crucially, the cup-tracking, execution-monitoring, and removal routines were reused without modification. When substituting the cube with a red wedge, the agent generalized its geometric object representation from a fixed cube primitive to an object parameterized by estimated dimensions, color, and height, adjusting grasp synthesis accordingly while preserving 20 of the original 41 program functions (including the tracking pipeline; Figure 22).

Wiping with paper towels Following the acquisition of a reliable surface-wiping skill using a wet cloth, the task was extended to require pulling disposable paper towels from a dispenser box, executing the wiping motion, and discarding used towels into a receptacle. The agent synthesized a composite program that imported the original wiping module as a dependency. The new program reused existing subroutines for stain segmentation against the table background, reachable workspace estimation, pressing by position control (12 mm below the measured surface contact, along a serpentine trajectory), and post-wipe residue verification. Only paper-handling subroutines were added: locating the dispenser box and disposal bin, extracting individual sheets onto the stain, discarding the crumpled towel, and clearing loose paper from the workspace. The stain was removed in both wet runs, although leftover sheets still had to be cleared with supervised scripts (Appendix F.3, Figure 28).

6 Discussion

Progress in robot learning has largely been driven by collecting more demonstrations and training larger models. Our results show that a capable agent, when allowed to explore, continues to improve with the experience it accumulates, without teleoperation and without updates to its parameters. Because this form of learning relies on general-purpose agents whose capabilities are advancing rapidly in other domains, every improvement in those agents translates directly into improved robot learning.

The knowledge acquired by RAX agents takes the form of perception and control programs, skills, and notes. In this form, knowledge executes at the rate of the robot, can be inspected and verified, composes into longer behaviors, and transfers across tasks, and we expect code to play the same role for learning from experience that network parameters play for learning from data. Since the instructions, skills, and memory structures of RAX are themselves code and text, a natural next step is to let agents improve the architecture itself from experience, and to share verified skills across fleets of robots, so that a skill learned by one robot becomes available to all.

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Appendix

A Interface Contracts

On the real robots the agent acts only through tools that wrap each vendor's control SDK, and its own programs call the same tools. Every command is checked before the robot moves; workspace, joint limit, or IK violations are refused, and motion works only while the operator has armed the robot. Table 11 summarizes the tools; both robots also offer `status` and `request_operator`.

Table 11 | **Tools on the real robots.**

	Franka FR3	**Rainbow RB-Y1**
Perception	RGB-D images from two exterior and one wrist camera; pixel to 3D point	Head RGB-D images and gaze control; pixel to 3D point; a vision-language model
Motion	Straight-line end-effector moves; joint moves with path checks	Joint, end-effector, and torso targets; IK over torso and both arms; base drive
Gripper	Grasp to a width with a set force	Close with a current limit, or set an opening
Model	None (the agent writes its own kinematics)	Forward kinematics; dry runs for collision, joint load, and center of mass

B Agent Instructions and Skills

Every session starts from the same workspace template: one instruction file and a folder of skills, written once and shared by all platforms (`auto_research_agent/` and `agent_skills/` in our code). Robot-specific facts, such as tool names, cameras, and control rates, are filled in when the workspace is built, so the agent reads the same instructions on every robot.

Instructions. The instruction file fixes the order of work rather than how to solve any task. The agent first reads the task text and its own notes from earlier sessions, then (1) solves the task by controlling the robot directly through tools until it confirms success, (2) turns what it did into a program whose steps check their preconditions, verify their results, and report states they were not built for, (3) runs that program on every further exploration episode, stepping in where it falls short and fixing the code, and (4) keeps its notes, code, and a hand-off note current, since only those carry over to later sessions. In benchmark sessions it then freezes the program, rehearses it once end to end without help, and submits it. The file also states the budget rules: new layouts in every episode, so scene geometry is measured again rather than assumed, and a blocker is reported rather than bypassed.

Skills. Skills are short documents that the agent opens when it reaches the step they cover. They are grouped by axis, task strategy, coordination, perception, and robot control, and a session receives only the layers that match its robot and observation mode. The task and coordination skills are robot-agnostic (Table 12); the perception and control layers give each robot's interface under the same names, so a task skill can point to, for example, the robot's interface or its free-space motion without knowing which robot it runs on.

Table 12 | **Robot-agnostic skills** given to every session.

Skill	**What it asks of the agent**
Manual robot control	Look first, explore with small moves, choose each action from the current observation, and confirm every condition of the task before handing off.
In-context action learning	When images of a completed task are supplied, infer the goal and useful strategy from them and adapt it to the current layout, without copying actions.
Experience memory	Record what worked right away, with the observable conditions under which it worked; retrieve it for similar situations and distill it as evidence grows.
Autonomous control	Start from a confirmed manual success, turn it into observation-driven code that can be supervised and resumed, and test it across layouts.
Transfer hand-off	Before exploration ends, write how each subtask and each object is handled, for a later session on new layouts.
Multi-environment exploration	Run parallel environments, track their scenes separately, and share one episode budget.
Multi-agent exploration	Decide whether to delegate; give each subagent its own environment and a complementary goal, and combine their findings and code.

C Benchmarks and Tasks

This section gives the protocol that all simulation experiments share and then the setting of each benchmark and its tasks. The real-robot tasks are described in Appendix F.

C.1 Sessions

Agents. Every experiment is a session of one coding agent in its own command-line harness: Claude Code with Claude Opus 5.5 or, on four RoboDojo tasks and one BEHAVIOR-1K task, Codex with a GPT-6 model, at the xhigh reasoning effort unless stated otherwise. A session provides one or more simulated environments; with more than one, the agent may start subagents of the same model, which share its workspace and its budget. The weights of the model are never updated.

Workspace. The agent starts from an empty workspace. Its folders for code, notes, and outputs are empty, and it holds only the instructions of the harness, the task description, the contract of the tool interface, and the skills of the platform, composed for the embodiment and the observation mode of the session (Appendix B); the agent keeps no memory from earlier sessions. It runs in a container without network access, with 16 GB of memory and four CPUs, and reaches the simulator only through its tools. Every program it registers runs in a fresh container with 8 GB of memory, four CPUs, and no network access.

Task information. The task description is generated from the public definition of the task; its content differs between benchmarks and is given below. Images of the completed task are frames of successful official demonstrations, either the final frame of each or a sequence of keyframes along each. They contain no actions and no states, and only the exploring agent sees them, never its programs. During an interactive episode, the agent may ask the benchmark's evaluator whether the task is complete.

Budget and scenes. The exploration budget counts exploration episodes. A single ledger charges every episode start and every rehearsal, across all environments and subagents, and the agent sees how many episodes remain. The budget is an upper bound, since submission ends exploration. Exploration scenes come in a fixed order, which the agent cannot choose, from the training distribution of the benchmark, and evaluation scenes from its official evaluation set, which exploration never uses.

Limits. Each episode has the native step limit of the benchmark and a wall-clock limit, which during an interactive episode also bounds the agent's own deliberation. The agent and its programs may call a vision-language model, Gemini 3.8 Flash, with a budget of $10 per session that exploration and evaluation share.

C.2 Rehearsal and Evaluation

Rehearsal. When the agent considers its program ready, it freezes a copy of its code with a registered entry point into a bundle and requests a rehearsal. The bundle is executed alone in a fresh isolated environment on a new exploration scene while the agent is paused, and the rehearsal consumes one exploration episode, like an episode in which the agent acts itself. Afterwards, the agent reads the outcome and the logs of the rehearsal and may revise the bundle and rehearse again. Only a bundle that has completed a rehearsal without error can be submitted, and submission ends exploration.

Fully automated program. Evaluation executes the submitted bundle on each unseen scene in a fresh isolated environment, without the agent and without edits or retries. The results are never revealed to the agent. Because the bundle calls the same tools that the agent called during exploration, the program that is evaluated is the program that was rehearsed.

Agent-in-the-loop program. At the end of exploration, the workspace is frozen, including the code, the notes, the handoff, and a compressed version of the conversation. For each evaluation scene, a fresh copy of the agent starts from this frozen workspace and attempts the scene, running its code, observing, and acting through the same tools as during exploration.

Scoring. Both forms are scored by the benchmark's native evaluator, with partial credit wherever the benchmark defines it. An error, a timeout, or an early exit of the program counts as a failure, and scores are averaged over all scheduled evaluation episodes.

C.3 RoboDojo

Tasks and interface. We use seven tasks of RoboDojo-Sim (Chen et al., 2026a) with the dual-arm ARX X5 (Table 13). The agent commands absolute joint targets at 25 Hz, six arm joints and a gripper opening for each arm, with between 1 and 50 rows per call. It may also command bounded end-effector steps, plan collision-free motions of one arm against the robot and the table, lift pixels to 3D points, and preview a motion on the robot alone. It observes the scene through a high camera and a camera on each wrist at 640×480; the benchmark renders RGB, and the harness adds metric depth, the intrinsics, and the pose of every camera.

Task information. The task description is the public scoring rubric of the task, which for the randomized variants is that of the base task, with its score table. For make kong, pour by language, and swap blocks it adds a short clarification of the conditions, and for plug in charger and pour by language a summary of the official task code without its numeric thresholds. The agent also receives one image of the completed task, the last frame of the official video of the task.

Exploration and evaluation. A single agent explores a single environment; on plug in charger and pour by language it may start one subagent. Exploration scenes are the layouts of one official collection in order, which repeat once the collection is exhausted, and the program is evaluated on the first layouts of another collection (Table 13); on hang mugs random, both come from a pooled collection, with evaluation on layouts 0 to 49 and exploration on layouts 50 to 89. An episode is limited to 6 s of wall-clock time per allowed step. A score is 100 on native success and the native partial score otherwise. The parallel experiment (Table 7) uses make toast with a description that includes the numeric thresholds, keyframes of a completed run, and 32 evaluation layouts from a third collection; the RGB-only sessions of Table 8 receive no depth, calibration, or pixel lifting.

Table 13 | **RoboDojo tasks.** Step limit at 25 Hz; partial credit: native score tiers; evaluation: number of official layouts.

Task	Benchmark id	Step limit	Partial credit	Evaluation	Agent
Make toast random	`make_toast_random`	1400	25/50/100	45	Codex
Hang mugs random	`hang_mugs_random`	800	15/40/100	50	Claude Code
Sweep blocks random	`sweep_blocks_random`	1000	–	30	Codex
Plug in charger	`plug_in_charger`	400	–	50	Claude Code
Swap blocks	`swap_blocks`	700	–	50	Codex
Make kong	`make_kong`	600	–	50	Codex
Pour by language	`pour_by_language`	800	20/50/100	50	Claude Code

C.4 BEHAVIOR-1K

Tasks and interface. We use six tasks of the 2025 BEHAVIOR Challenge (Li et al., 2023) with the Galaxea R1Pro, in the official evaluation environment of BEHAVIOR-1K (Table 14). The agent commands the native action at 30 Hz, base velocity together with absolute joint targets of the four trunk joints and the two seven-joint arms and a command for each gripper, with between 1 and 150 rows per call. It may also move the base in closed loop, step an end effector through inverse kinematics, plan collision-free arm motions, lift pixels to 3D points, and preview a motion on the robot alone. It observes the scene through the head camera at 720×720 and a camera on each wrist at 480×480, all RGB-D with intrinsics and poses relative to the base, together with the official proprioception and an odometry estimate integrated from the base velocity. The step limit of an episode is 1.5 times the mean length of the human demonstrations of the task, as in the 2026 evaluation, whereas the 2025 challenge allowed twice the mean.

Task information. The task description contains the official instruction, the goal and initial conditions of the task's BDDL definition, the scene and its rooms, and the limits of an episode. Except on boxing books, the agent also receives keyframes of successful official demonstrations from the head camera: the first frame, the end of each annotated skill, the final frame, and enough frames in between that no gap exceeds 10 s.

Exploration and evaluation. Exploration scenes are the 300 training instances of each task, and evaluation uses the test instances 301 to 310 (Table 14). An episode is limited to 900 s plus 0.5 s per allowed step. Boxing books is evaluated as a fully automated program and the other five tasks as agent-in-the-loop programs, with one fresh copy of the agent per test instance. On sorting items, clearing food, and clean up desk, each copy also receives the reports that a review model, Gemini 3.8 Flash, wrote on the fully automated program's runs on the other nine test instances, built from the final status of each goal condition, the partial score, the program's log, and camera frames, and referring to objects only by their task names. Episodes are scored by the Q-score of the challenge: 1 on success and otherwise the fraction of goal conditions that become satisfied during the episode, maximized over the alternative groundings of the goal and averaged over the ten instances.

Table 14 | **BEHAVIOR-1K tasks and sessions.** Step limit at 30 Hz; budget: exploration episodes allowed; envs: parallel environments; program: the program evaluated in Table 3. *Maximum reasoning effort. †Without demonstration keyframes or the vision-language model; 32 episodes used.

Task	Benchmark id	Agent	Step limit	Budget	Envs	Program
Boxing books	`boxing_books_up_for_storage`	Codex†	36,341	128	6	fully automated
Getting organized	`getting_organized_for_work`	Claude Code*	23,506	128	2	agent-in-the-loop
Sorting items	`sorting_household_items`	Claude Code	23,711	128	8	agent-in-the-loop
Clearing food	`clearing_food_from_table_into_fridge`	Claude Code	19,602	128	4	agent-in-the-loop
Clean up desk	`clean_up_your_desk`	Claude Code	32,126	128	4	agent-in-the-loop
Loading the car	`loading_the_car`	Claude Code*	28,839	128	2	agent-in-the-loop

C.5 HumanoidArena

Tasks and interface. We use the seven tasks of HumanoidArena (Wang et al., 2026c) that have official test episodes, with a Unitree G1 that has 29 body joints and Dex3 hands (Table 15). The agent commands the official 40-dimensional whole-body reference at 50 Hz, with between 1 and 100 frames per call: the root translation and height, which the SONIC tracker (Luo et al., 2026) does not follow, since the robot translates only by stepping, the root orientation relative to the initial heading, 29 joint targets, and an open or close command for each hand. It may

also plan collision-free motions of the arms, optionally with the waist, and lift pixels to 3D points in the pelvis frame and in a gravity-aligned frame. It observes the scene through the head camera at 640×480 in RGB-D, with the intrinsics and the camera pose in the pelvis frame, together with the root orientation, the joint positions and velocities, and the state of the hands. An episode ends when the robot falls, by the rule of the benchmark.

Task information. The agent can read three teleoperated demonstrations of the task from the official raw dataset. Each is a complete episode with the 40-dimensional references, joint states, root poses, and the head camera's RGB video and depth. The demonstrations are mounted read-only for the exploring agent only; its programs may contain motion segments that the agent cut from them (Appendix D.3).

Exploration and evaluation. We use Claude Opus 5.5 as our agent. Exploration scenes use seeds derived from a hash that excludes every official seed. Sessions with a budget above 32 explore two environments and may start two subagents. An episode is limited to 600 s plus 1 s per allowed step. The program is evaluated on the 60 official test episodes of each task, seeds 0, 1, and 2 with 20 episodes each, derived as in the official evaluation; every episode runs in a fresh simulator and container, four in parallel, and the simulator advances only when the program sends actions. Success is binary, and a fall or an error counts as a failure.

Table 15 | **HumanoidArena tasks.** Step limit at 50 Hz; budget: largest exploration budget in Table 4. HOI and HSI: human–object and human–scene interaction.

Task	Type	Step limit	Success condition	Budget
Football	HOI	2000	the ball passes the goal line between the posts	32
DoubleDesk	HOI	2000	the hammer rests inside the basket on the second desk	128
P&PBox	HOI	1450	the box rests on the shelf, inside its outline	128
OpenDoor	HSI	1800	the latch is released and the door is open by at least 60°, robot standing	32
SitSofa	HSI	2000	the pelvis is over the seat and below 0.66 m	32
Boxing	HSI	900	a palm comes within 0.18 m of the target on the bag	32
VisNavi	HSI	1800	both ankles are inside the destination, robot standing	32

C.6 DexJoCo

Tasks and interface. We use the eleven tasks of DexJoCo (Wang et al., 2026a) in its rand-obj setting, in which the placement of the objects and the height of the table are randomized at every reset (Table 16). Single-arm tasks use one Franka Panda arm with a 16-degree-of-freedom Allegro hand, and bimanual tasks use two. The agent commands the official policy action of the benchmark: for each arm, an absolute target for the flange position and orientation in the world frame, the orientation as a rotation vector, together with sixteen hand joint targets, which gives a 22-dimensional row for one arm and a 44-dimensional row for two. The official operational-space controller pulls the flange toward its target and position actuators drive the hand joints, without a motion planner. Each row advances the simulation by 20 ms, and one call to the interface executes between 1 and 150 rows. Observations are 640×640 RGB-D images from the official cameras of each task, with depth in millimeters and the intrinsics and world pose of every camera, together with the official proprioceptive state, the flange pose of each arm and the joint positions of each hand. No force, torque, contact, or tactile signal is exposed, and the object poses and table height in the simulator's state never reach the agent. The interface also offers a dry run of a motion on the robot alone, without the table and objects, that does not advance the episode, and helpers for rotation arithmetic and for lifting a pixel with its depth to a world position.

Table 16 | **DexJoCo tasks, rand-obj setting.** All cameras are RGB-D. *Including 30 steps before control is handed over.

Task	Benchmark id	Arms	Cameras	Step limit
Hammer	`hammer_nail`	one	front, wrist	1000
Mouse	`click_mouse`	one	ego_right, wrist	1200*
Bucket	`pick_bucket`	one	front, wrist	1200
Tongs	`pinch_tongs`	one	front, wrist	1000
Glasses	`fold_glasses`	one	front, wrist	1200
Water	`water_plant`	one	front, wrist	1000
iPad	`bimanual_unlock_ipad`	two	ego, wrist_left, wrist_right	1200
Hanoi	`bimanual_hanoi`	two	ego, wrist_left, wrist_right	1500
Assembly	`bimanual_assembly`	two	ego, wrist_left, wrist_right	1500
Microwave	`bimanual_microwave_cook`	two	ego, wrist_left, wrist_right	1100
Photo	`bimanual_photograph`	two	ego, wrist_left, wrist_right	1000

Task information. The agent receives a task description generated from the task definition: the official instruction, the robot and its cameras, the step limit and wall-clock limit of an episode, the fact that object placement and table height change between episodes, and the binary success criterion without partial credit. It contains no detail of the success check. The agent also receives three images of the completed task, the final frames of three successful human demonstrations from the official dataset, each in a different layout, taken from the task's overview camera. They contain no actions and no states. The three demonstrations are the first that replay successfully in the official environment from initial layouts that differ from all 150 evaluation layouts.

Exploration. A session gives one agent, Claude Code with Claude Opus 5.5 at the xhigh effort setting and without subagents, one simulated environment, and a budget of $N \in \{2, 4, 8, 16, 32\}$ exploration episodes. The agent starts from an empty workspace (Appendix C.1). A exploration episode is either interactive, in which the agent drives the robot itself through the interface or through its own scripts, or a rehearsal, in which its registered program runs alone in a fresh container on a fresh episode while the agent is paused; both consume the budget. The agent ends exploration by submitting a program that has completed a rehearsal without error, so the budget is an upper bound, and over the 55 sessions the agents used 592 of the 682 budgeted episodes (Table 17). Exploration layout i is the first reset of the official rand-obj environment with seed $1000 + i$, the same randomization as in evaluation, whose layouts come from seeds 0 to 2. An episode is limited to the official number of steps and to 600 s plus 5 s per allowed step of wall-clock time, a limit that no episode reached. The agent and its programs may call a vision-language model, Gemini 3.8 Flash, with a budget of $10 per session.

Evaluation. Once the agent has exited, its workspace is frozen, and the submitted program is evaluated alone on 150 episodes that reproduce the official evaluation of DexJoCo layout for layout: seeds 0, 1, and 2, with 50 consecutive resets each. Every episode runs in a fresh simulator process and a fresh container for the program, four episodes in parallel, under the official step limit and success check, and an error or an early exit of the program counts as a failure. The simulator advances only when the program sends actions, so evaluation is synchronous and free of inference latency, whereas the official client of the benchmark is asynchronous.

C.7 LIBERO and LIBERO-Pro

Tasks and interface. We use the 40 tasks of the Spatial, Object, Goal, and Long suites of LIBERO (Liu et al., 2023) in the official environment, numbered from 0 in the official order, and

Table 17 | **DexJoCo sessions by exploration budget.** One session per task. Episodes are summed over the eleven sessions; agent time runs from the start of the agent to evaluation and is the median over the eleven sessions, with the range in parentheses.

Budget	Interactive	Rehearsal	Used / budgeted	Submitted early	Agent time (h)
2	11	11	22 / 22	0 of 11	0.70 (0.44–1.28)
4	26	18	44 / 44	0 of 11	1.05 (0.51–2.46)
8	57	29	86 / 88	2 of 11	1.34 (0.59–3.02)
16	93	67	160 / 176	5 of 11	1.52 (0.72–2.89)
32	179	101	280 / 352	8 of 11	1.76 (0.98–3.89)

the position-perturbed variants of LIBERO-Pro (Zhou et al., 2025), in which objects exchange their initial regions. The agent commands the official action at 20 Hz, increments of the end-effector position and rotation, scaled so that 1 corresponds to 5 cm or 0.5 rad, and a gripper command, with between 1 and 100 rows per call and no motion planner. It observes the scene through the agent-view and wrist cameras at 256×256 in RGB-D, with the intrinsics and the pose of each camera, together with the end-effector pose and the finger positions, and it may lift pixels to 3D points. The step limit of an episode is 220 on Spatial, 280 on Object, 300 on Goal, and 520 on Long, after ten initial steps without motion.

Task information. The task description contains the official instruction, the robot and its cameras, the limits of an episode, and the binary success criterion without the detail of its check. The agent also receives three images of the completed task, the final agent-view frames of three official human demonstrations that replay successfully.

Exploration and evaluation. We use Claude Opus 5.5 as our agent in two environments. The budget is up to 20 exploration episodes on the official placement distribution, followed by at most 5 optional episodes of adaptation on the perturbed distribution, which the agent may use when it judges that its program will not transfer (Appendix D.4). Exploration layout i is a fresh reset of the official environment with seed $1000 + i$, a sample of the official placement distribution and never a stored initial state. An episode is limited to 600 s plus 5 s per allowed step. The program is evaluated on the 50 official initial states of each task and on the 50 position-perturbed states of LIBERO-Pro, each episode in a fresh container, synchronously, and with binary success. Adaptation episodes are fresh resets of the perturbed task with seeds from 2000.

D Simulation Case Studies

This section follows single sessions on the simulation benchmarks in detail: what the agent explored, the program it wrote, and how that program behaves on unseen scenes.

D.1 Re-Perception Under Perturbation: Two Cans into a Basket

The first task of the LIBERO-Long suite asks the robot to put both the alphabet soup and the tomato sauce in the basket, on a table that also holds a ketchup bottle, a carton of orange juice, milk, butter, and cream cheese. The agent explored the task for 15 episodes in about an hour, ten of them interactive and five rehearsals of its program, and wrote a program of 1,362 lines that succeeds in all 50 official initial states and in 44 of the 50 perturbed states of LIBERO-Pro. The perturbation exchanges the soup with the ketchup and the tomato sauce with the orange juice, which moves the two cans by 15 and 32 cm (Figure 13a). This section shows what the program perceives, how it plans, and why its behavior transfers.

Perception. The program perceives the scene anew in every round, and it divides the work between a vision-language model and depth (Figure 12). Gemini only names the objects: it returns a pixel on each can and in the basket, together with a short description of each can's state. Every geometric quantity comes from depth. The program lifts the agent-view depth into the world frame, takes the table as the mode of the point heights, and, starting from each Gemini pixel, segments the object by a flood fill over 3D-connected points; the top height is a high percentile of the segment, and the axis is the centroid of its top 1.2 cm. A segment counts as a can only if its top is 4.5 to 8 cm wide and 5 to 30 cm above the table, bounds that the agent set after measuring every object on the table and that reject the taller and narrower ketchup bottle. A second detector, which uses depth alone, finds flat, round plateaus of the size of a lid and adds any can that Gemini missed. Before grasping, the program hovers 7 cm above the lid and refits the lid disk from the wrist camera; a top that is not can-shaped is blacklisted, and the grasp yaw is the first of four candidates whose finger zones are clear of other objects.

Placing by footprint reach. The agent-view camera cannot see the floor of the basket, which its near wall hides, and the agent found during exploration that this flat floor is only about 7 by 10 cm: drop spots chosen by clearance put the second can's footprint over the sloped walls, where the can stalled. The program therefore maps the interior from the wrist camera while it holds the first can, as a height map with 4 mm cells, and computes for every cell the highest point under the footprint of a can, its *footprint reach* (Figure 12c,d). It chooses the pair of spots, at least 6.8 cm apart, that minimizes the larger of their two reaches, releases each can just above its reach, and remaps the basket after the first release. During exploration, the agent also arranged perturbed layouts of its own, stacking the tomato sauce on the soup, standing the soup on the cream-cheese box, placing the cans side by side, and pushing the basket by 8 cm and rotating it by 20°, and it ran its program on each of them.

Why the program transfers. Nothing in the program refers to where the objects stood during exploration, so the perturbation changes its inputs but not its procedure. Under perturbation, its first estimate of each can's axis lies within a median of 3.1 mm of the ground truth, against 2.9 mm on the official layouts, and the wrist refit within a median of 1.0 mm, while the cans themselves moved by 15 and 32 cm (Figure 13b); a policy that reproduced its official motions would miss each can by that displacement. The checks also absorb labelling errors. When Gemini placed the soup's label on the ketchup bottle, which now stands where the soup stood in the official layouts, the wrist refit measured a top 3.7 cm across, and the program rejected it as not can-shaped, blacklisted its position, and found the soup in the next round. All six failures share one cause: the soup is not visible from the agent-view camera at the start, and

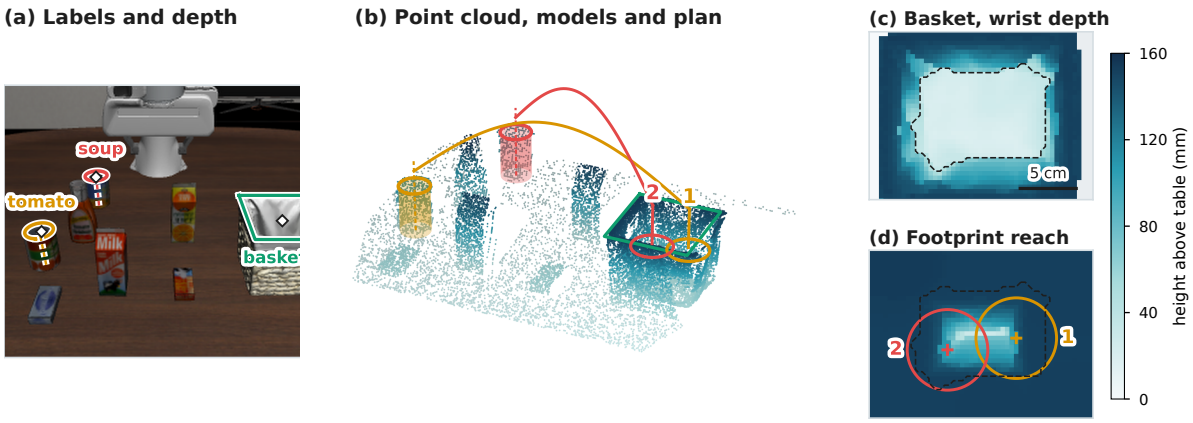


Figure 12 | **Perception and planning on a LIBERO-Pro layout,** re-run with the agent's code on the stored frames of one evaluation episode. (a) The agent-view image with Gemini's points (diamonds) and the geometry that the program derives from depth: the top and axis of each can and the basket rim. (b) The point cloud of the agent-view camera, completed over the basket by the wrist camera and coloured by height above the table on the scale of (c) and (d), with the fitted cans, the basket rim, and the two planned placements, numbered in the order of execution. (c) The basket measured by the wrist camera in 4 mm cells, with the interior outlined. (d) Footprint reach, the highest point under the footprint of a can, and the two spots that minimize it; the circles show the footprint.

the program, which re-perceives only from its home pose, stops with about 300 steps left. The listings below condense the program's main loop and give the agent's code for the footprint reach and the choice of drop spots, without docstrings and one log line.

Pseudocode · run(), condensed from controller.py

```
def run(robot):
    table_z = mode of the agent-view point heights
    repeat up to 3 rounds:                          # perceive anew in every round
        lids   = flat, round plateaus in depth      # depth alone
        labels = Gemini(agent view)                 # one pixel per can and the basket
        cans   = [flood_fill(p) for p in labels if is_can(top width, height)]
        cans  += lids that no label explains, minus blacklisted positions
        for can in order(cans):                     # stacked cans: the higher first
            hover 7 cm above the lid; refit the lid disk from wrist depth
            if not can_shaped(disk): blacklist(can); continue
            yaw = first of {0, 90, 45, -45} deg with clear finger zones
            descend to top - 3 cm; close; lift
            if not 3.5 cm < finger opening < 7.4 cm: continue
            if no basket map: map the basket from the wrist, holding the can
            spot = plan_spots(map) for the first can, else lowest_spot(remap)
            lower to the release height above the footprint reach; open; lift
            if task complete: return
```

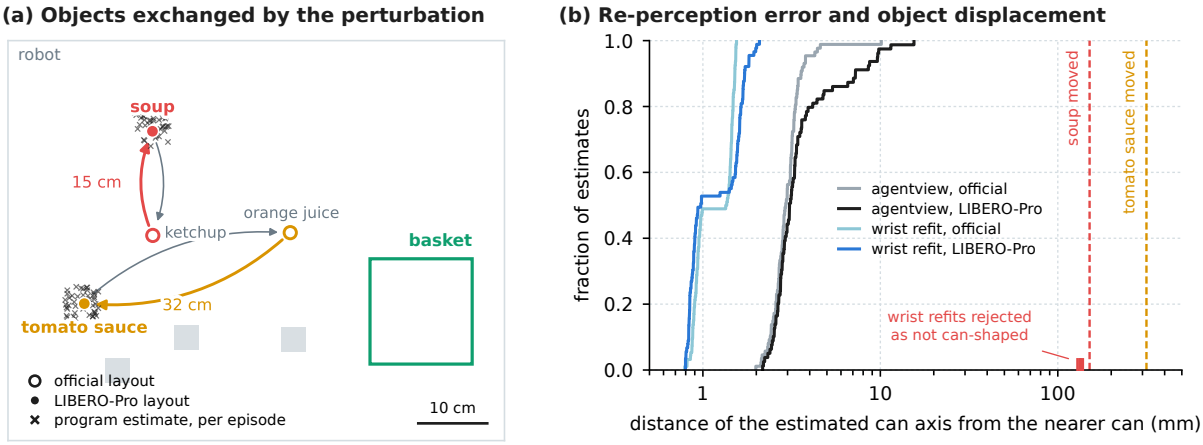


Figure 13 | **Why the program transfers to LIBERO-Pro.** (a) Mean object positions over the 50 official and the 50 perturbed layouts, seen from above with the robot at the top; the perturbation exchanges the soup with the ketchup and the tomato sauce with the orange juice. Crosses mark the program's first estimate of each can in every perturbed episode. (b) The distance between the program's estimate of a can's axis and the nearer can in the simulator, for the first agent-view estimate and for the wrist refit before each grasp, against the displacement of the two cans. The task takes the cans in either order, and in two perturbed episodes Gemini exchanged their labels; the rejected refits are those on the ketchup bottle.

basket.py · footprint reach and drop-spot planning

```python
def footprint_max(b, r=0.036):
    H = np.where(np.isfinite(b["H"]), b["H"], b["rim_z"])   # unobserved cells count as rim
    res = b["grid"][2]
    k = int(np.ceil(r / res))
    yy, xx = np.mgrid[-k:k + 1, -k:k + 1]
    fp = (xx ** 2 + yy ** 2) * res ** 2 <= r * r           # disk of the can's footprint
    return ndimage.maximum_filter(H, footprint=fp, mode="constant", cval=b["rim_z"])

def plan_spots(b, r=0.036, min_sep=0.068):
    F = footprint_max(b, r)
    ii, jj = np.nonzero(b["interior"])
    if len(ii) < 2:
        return None
    f = F[ii, jj]
    keep = np.argsort(f)[:400]
    ii, jj, f = ii[keep], jj[keep], f[keep]
    res = b["grid"][2]
    D = np.hypot(ii[:, None] - ii[None], jj[:, None] - jj[None]) * res
    cost = np.maximum(f[:, None], f[None]) + 0.5 * np.maximum(0.0, 0.08 - D)
    cost[D < min_sep] = np.inf
    a, c = np.unravel_index(np.argmin(cost), cost.shape)
    if not np.isfinite(cost[a, c]):
        return None
    s1, s2 = cell_xy(b["grid"], ii[a], jj[a]), cell_xy(b["grid"], ii[c], jj[c])
    return [(s1, float(f[a])), (s2, float(f[c]))]
```

D.2 Self-Calibration and Grasp Design: Tower of Hanoi with Two Dexterous Hands

The bimanual Tower of Hanoi task of DexJoCo asks for the final two moves of a three-level tower: the medium red disk goes from the middle peg onto the large blue disk on the right peg, and the small yellow disk from the left peg on top of it. Two Franka arms carry 16-degree-of-freedom Allegro hands, and the disks slide over pegs that stand 13 cm above them, so a grasp has to hold a disk level as it travels along its peg without touching the peg. The agent received no grasp and no model of the hand, only the joint semantics and a few generic postures in its skills. It wrote a program that completed all 150 formal episodes on unseen layouts, in which the board shifts by up to 4.9 cm and the table height varies by as much.

Measuring its own fingertips. The agent spent its first episode on its own hand, without touching a disk. It moved single joints of the right hand and lifted the wrist cameras' depth into world points to see what each joint moves, finding that the first thumb joint turns the thumb from pointing sideways to pointing straight down and that the last two flex its tip forward. It then held the posture it meant to grasp with, the index finger and the thumb both pointing down, segmented the white fingertip caps in the right wrist camera, lifted them to 3D, and split them into two clusters (Figure 14). Relative to the measured flange, the thumb cluster lies at $(-3.3, 4.1, -15.7)$ cm and the index cluster at $(7.3, 3.1, -12.9)$ cm; the agent adopted tip offsets of $(-2.4, 3.8, -16.3)$ and $(8.3, 3.6, -13.3)$ cm, about 1 cm toward the centre of each tip, and every grasp of its program is built on them, as the first listing at the end of this section shows. The agent never saw a model of the hand, yet its thumb offset lies within 1.1 mm of the fingertip of the official model. Its index offset lies 27 mm from the model's, because the camera sees only the upper part of that cap, but the error shifts the planned pinch line by only about 8 mm across a 7.7 cm disk. The same measurements placed the wrist camera at $(-14.0, -3.0, 4.1)$ cm from the flange, exactly where the official model puts it.

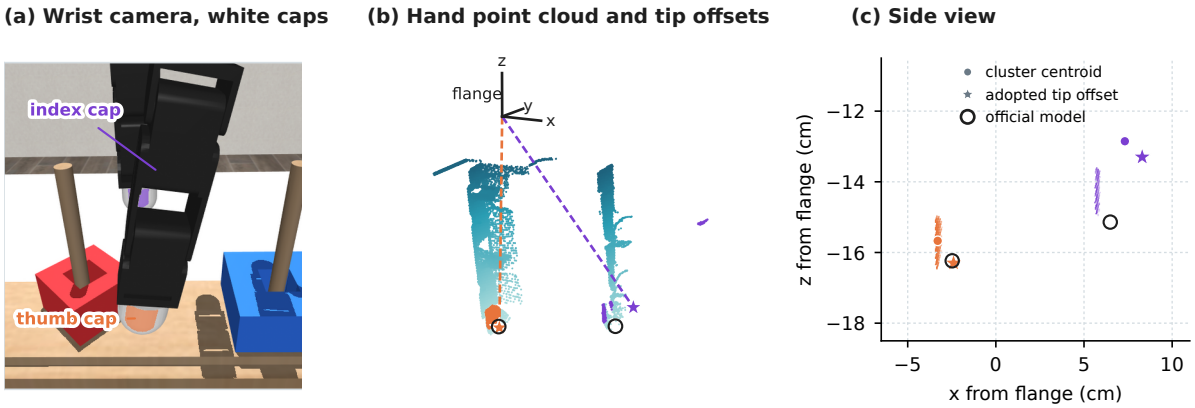


Figure 14 | **Measuring the fingertips with the wrist camera,** in the agent's first episode, re-run on its stored frames. (a) The right wrist camera with the hand in its grasp posture; the agent segments the white fingertip caps and splits them into two clusters. (b) The hand's point cloud in the frame of the measured flange, with the two clusters and the tip offsets the agent adopted (stars). (c) A side view of the clusters, their centroids and the adopted offsets, with the fingertips of the official hand model, which the agent never saw, for comparison.

Designing the grasp. The agent then designed its grasp by trying candidates on the red and yellow disks (Figure 15). A palm-down hook pressed the palm onto the disk, and flexing the thumb turned the disk by 45° about its peg. A palm-down pinch at the wall of the disk lifted it, but the disk swung by about 40° about the pinch line, which the agent measured to pass 2.8 cm from the disk's centre of mass, and a firmer squeeze dropped it. A side pinch lifted

the yellow disk by 10 cm before it slipped back onto its peg, and a cage grasp left only 4 mm between the hand's fixed thumb–index gap of 8.1 cm and the 7.7 cm red disk, so that only the thumb touched it. From these failures, the agent derived that rolling the hand by an angle θ moves the index edge of the palm $0.10 \sin\theta - 0.013 \cos\theta$ m away from the thumb–index line, so that a roll of about 25° clears the peg and lets the pinch line pass through the centre of the disk. The resulting rolled vertical pinch, with a roll of 0.44 rad, held both disks through a 22 cm lift and a 40 cm carry in the fourth episode.

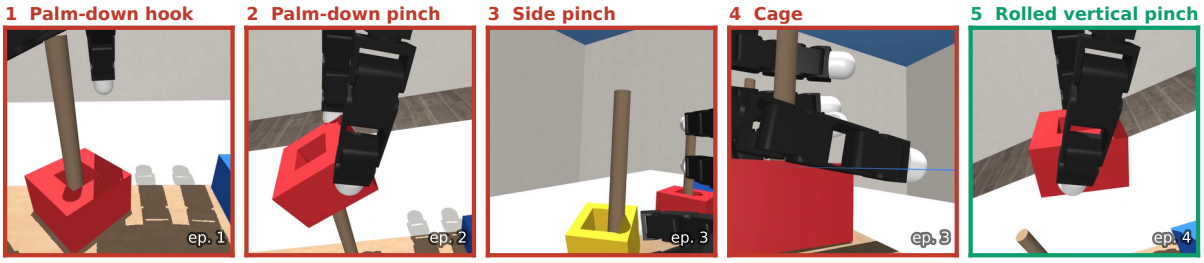


Figure 15 | **The grasps the agent tried,** seen by the wrist camera of the grasping hand in its exploration episodes, with panels 3 and 4 rotated so that the peg stands upright. The first four failed for the reasons given below them; the rolled vertical pinch carried both disks and became the grasp of the program.

The program. On every layout, the program measures the disks and pegs from the egocentric depth image, pinches both disks at once, and moves the red disk first while the left hand waits. It verifies every lift by measuring the held disk from three cameras, and it opens, backs off, and pinches again when the disk rose less than half of the lift. Because each arm keeps converging to its command after a move, the agent measured the tracking lag, which decays by a factor of 0.78 every five steps, and the program aligns the disk with the peg on its predicted final position rather than on its current one. Before submitting, the agent injected alignment errors of 8 and 15 mm and a missed pinch into its own runs, and the program recovered from all of them; its five versions completed all of their autonomous runs.

Precision on unseen layouts. The program's estimate of the target peg follows the hidden pose of the board across all 150 formal layouts, with residual standard deviations of 0.24, 0.05, and 0.18 mm in x, y, and height (Figure 16). After placing the red disk, the program measures its top again and finds it on average 0.66 mm, and at most 1.0 mm, below the height it predicted from the blue disk, the counterpart of the 6.5 mm check on the air fryer (Appendix F.2.1).

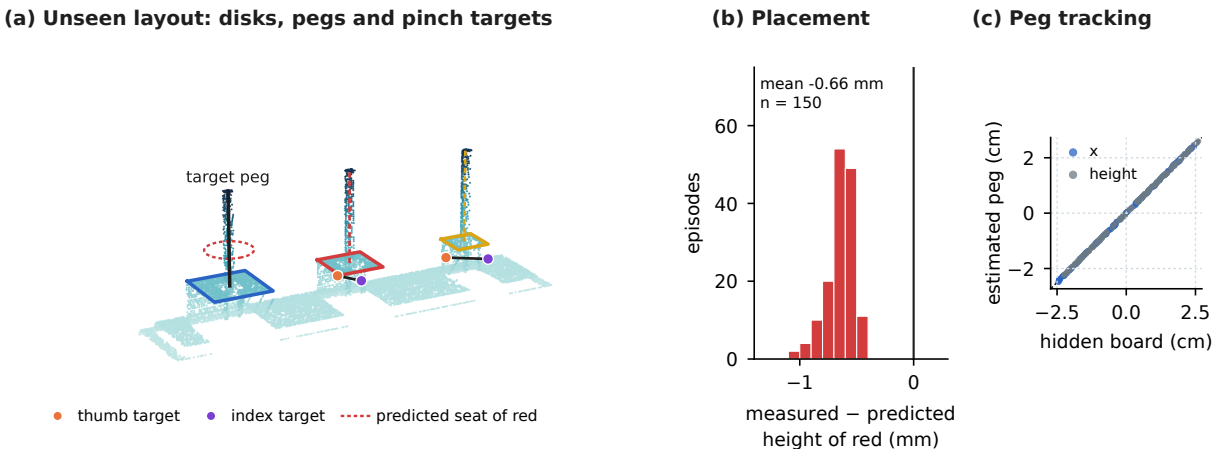


Figure 16 | **Perception on an unseen layout and precision over the 150 formal episodes,** re-run with the agent's code. (a) The egocentric point cloud, coloured by height, with the measured disks, the peg axes, the target peg, the thumb and index targets of both pinches, and the predicted seat of the red disk. (b) The height of the red disk measured after placement, minus the height the program predicted. (c) The program's estimate of the target peg against the hidden pose of the board, both relative to their means over the 150 layouts.

task.py · tip offsets and the rolled vertical pinch

```
TH_TIP = np.array([-0.024, 0.038, -0.163])   # default-frame right-hand thumb tip (th0=1.396, straight)
IX_TIP = np.array([0.083, 0.036, -0.133])    # index tip, x1=1.57 straight
R_T = 0.012
ROLL = 0.44
DISK_H = 0.05
PEG_ABOVE_TOP = 0.135                        # peg top above top of a disk resting on the board
PRESHAPE = dict(ff=(0, 1.3, 0, 0), th=(1.396, 0, 0, 0))
SQUEEZE = dict(ff=(0, 1.75, 0.5, 0.3), th=(1.396, 0, 0.7, 0.4))
RELEASE = dict(ff=(0, 1.2, 0, 0), th=(1.396, 0, 0, 0))
PARK = {"right": np.array([-0.45, -0.40, 1.42]), "left": np.array([-0.45, 0.40, 1.42])}


def pinch_pose(side, c, half, yaw, mid, gap_t=0.006, dz=-0.010):
    """Flange pose so thumb tip is gap_t outside the near face and the thumb-index line crosses the centre."""
    c = np.asarray(c, float)
    th_off = world_offset(side, TH_TIP, ROLL, yaw)
    ix_off = world_offset(side, IX_TIP, ROLL, yaw)
    u = np.array([np.cos(yaw), np.sin(yaw)]); v = np.array([-np.sin(yaw), np.cos(yaw)])
    lat_t = (th_off[:2] @ v - ix_off[:2] @ v) / 2
    t_xy = c + (-half - R_T - gap_t) * u + lat_t * v
    t = np.array([t_xy[0], t_xy[1], mid + dz])
    flange = t - th_off
    rv = rotvec_near(hand_rot(side, ROLL, yaw), np.array([-np.pi, 0, 0]))
    return flange, rv
```

Pseudocode · run(), condensed from controller.py

```
def run():
    disks  = colour and depth boxes from the egocentric camera   # measured on every layout
    pegs   = depth column above each disk; target = fuse(blue disk, blue peg)
    pinch(red, right hand) and pinch(yellow, left hand) at once     # rolled vertical pinch
    for hand, disk in [(right, red), (left, yellow)]:
        repeat up to 3 times:
            lift until the disk clears the peg top
            if the held disk rose less than half the lift:          # three cameras
                open, back off, measure again, pinch again          # GRASP FAILED
        carry: flange = target - (disk centre - measured flange)
        repeat up to 3 times:                       # settle, then predict the final pose
            lag = measured flange - command
            err = target - (disk centre - lag);  stop if |err| < 4 mm
        lower onto the support; check the bottom; open; lift by 15 cm
        if disk is red: measure its top, predicted as the blue top + 5 cm
```

D.3 From Exploration to Code: Carrying a Hammer over a Bar on a Humanoid

In the DoubleDesk task of HumanoidArena, a Unitree G1 humanoid has to take a hammer from one desk, carry it over a bar on the floor, and drop it into a basket on a second desk. The agent commands the robot through whole-body kinematic references that a learned tracker, SONIC, executes; it was given the tracker, three teleoperated demonstrations, a forward-kinematic model of the robot, and a tool that returns the level frame of the pelvis. Over 125 exploration episodes and 10.7 hours, it wrote a program of 3,639 lines in eleven files, of which the 637-line kinematic model is the one it was given. The program completed 34 of the 60 formal episodes; before submitting it, the agent predicted "roughly half to two-thirds".

Perception. Every quantity the program uses comes from the depth image of the head camera, lifted into a point cloud in the level frame of the pelvis (Figure 17). The rotation into that frame comes from the provided tool, and the agent's own fit of the floor plane by random sample consensus agrees with it to within 0.07°. On this cloud, the agent's code finds the floor as the mode of the point heights, the bar as the dominant line among points 5 to 11 cm above the floor, the desk tops and their near edges among upward-facing points, and the back wall, and it localizes the robot from the bar, the desks, and the wall against a map whose constants it measured itself. It finds the hammer by its orange handle, takes the handle's principal axis, decides which end carries the head from the dark points beyond the handle, and measures its length, using the 0.302 to 0.310 m it found during exploration to place the centre when part of the hammer is hidden; on the unseen scene of Figure 17, the length is 0.305 m. It finds the basket as the largest cluster of points above a desk and fits a rectangle of 0.60 by 0.40 m with its rim 0.90 m above the floor, and it measures the distance of each sole from the bar by combining the fitted bar with the kinematic model. The program perceives anew at every stage and before every step at the bar, and after a grasp it confirms that the hammer is held by measuring the height of the handle again.

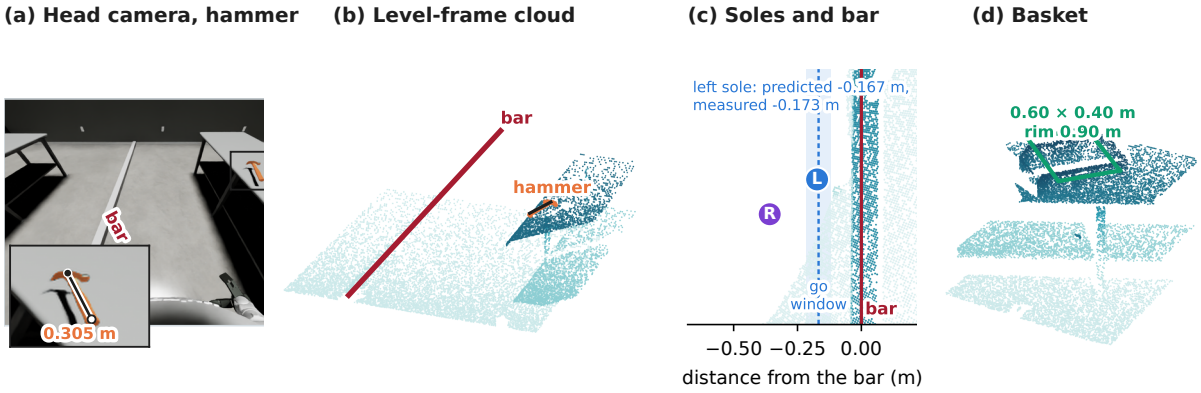


Figure 17 | **Perception on an unseen DoubleDesk scene,** re-run with the agent's code on the stored frames of one formal episode. (a) The head camera at the start, with the hammer the program finds and, enlarged, its axis and length. (b) The point cloud in the level frame, coloured by height, with the hammer and the bar fitted to it. (c) The last stance before the crossing, seen from above in the frame of the bar: the soles, the window from which the program starts a crossing, and the position of the left sole that the stance model predicted. (d) The basket found at the placing stance.

Locomotion and the bar. The agent's skills recommended walking by replaying a cycle cut from a teleoperated recording. The agent cut a symmetric 66-step cycle from one demonstration, looped it with a ramp in heading, and measured that its amplitude does not control the length of a single step, which varied by ±10 cm; it therefore calibrated multi-step walks against its

own localization and used them as a table of distances. Sidestepping moved the robot by 1 to 2 cm per cycle, so lateral placement turns and then walks. The crossing replays the step over the bar from one demonstration: with the recorded pitch, the pelvis did not travel and the lead foot landed on the bar, so the agent scaled the pitch to zero; perpendicular approaches landed the lead foot on the edge of the bar, so it approaches at 64°, where the swinging foot also reaches sideways; and it adds a trailing step when the robot ends astride the bar, after learning never to turn in that position. Whether a crossing succeeds depends on the stance from which it starts, and the agent built a model of how each step moves the two soles relative to the bar, from its own measurements of 102 transitions (Figure 18a). The model predicts that a left step places the left sole 14.5 cm ahead of the right one up to an amplitude of 0.35, and up to 25 cm at 0.6; a planner searches three actions ahead to bring the left sole into the window from which the program crosses. The bounds of that window are measured too: crossings started with the left sole within 21 cm of the bar succeeded in 17 of 17 trials, which set the far edge of the window at 21.5 cm, and the trailing sole is kept within 37 cm of it, since the agent found that crossings with the trailing sole closer than 35.5 cm had succeeded 27 times out of 27, and those with it between 35.5 and 42 cm had fallen 4 times out of 13. The agent then wrote its own Monte Carlo simulators of the approach, which replay the stance model with its measured errors, and used them to choose that margin, which lowered the simulated risk of a fall from 9.3 to about 6% at a cost of 0.3 actions. Over all 137 crossings in its logs, every crossing started with the trailing sole within 37 cm of the bar succeeded, and in the formal evaluation the robot crossed the bar in all 59 logged episodes without a fall (Figure 18b). In the episode of Figure 17c, the model predicted that the left step would bring the left sole to 16.7 cm before the bar, and the point cloud measured it at 17.3 cm.

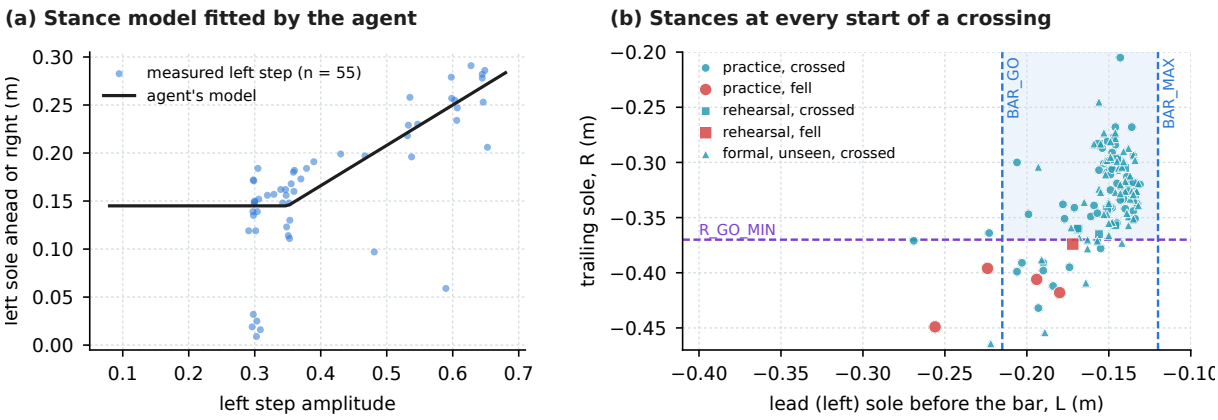


Figure 18 | **The stance model and the crossing window.** (a) How far a left step from a squared stance places the left sole ahead of the right one, against the step amplitude: the agent's measurements and the piecewise-linear model it wrote. (b) The positions of the two soles before the bar at every start of a crossing in exploration, rehearsal, and formal episodes, with the window that the program uses.

The program and its versions. The program runs seven stages, each with its own checks: it approaches the hammer until the handle lies at about 60° to its heading, which the agent found to give grasps that hold; it grasps with up to two attempts, treating a closure without torque as a miss; it carries the hammer to the bar while correcting its odometry from the bar and the basket; it plans its stance at the bar and crosses only from the window; it verifies after the crossing that both soles are past the bar; and it finds the basket and places the hammer inside its fitted rectangle. Each stage has a deadline in steps, measured from the durations of earlier runs. The agent kept its own table of every stage in every episode, linked each failure to the

change it made, and grew the program over more than thirty versions (Table 18 and Figure 19). Its interactive and rehearsal runs completed 41 of 82 episodes, rising to 6 of 9 for the final versions, and in the formal evaluation most failures occurred at the grasp, which held in 40 of the 59 logged episodes.

Table 18 | **Versions of the DoubleDesk program.** From the agent's own records. Completed counts interactive and rehearsal runs of each group of versions.

Versions	Main changes	Failures that motivated them	Completed
up to v8	First full chain; localization from the bar and the basket; crossing window and stance planner; grasp retries; basket approach by turning, then walking	Odometry off by 0.55 m; a walk onto the bar; a fall from 25.6 cm; a false miss that dropped the hammer	4/13
v9–v9g	Left step aimed by the stance model; handle at 60°; deadlines; a subagent's grasp measured at 3/13 against 13/22 and reverted	Late grasps; sweeps of the desk	5/10
v9h–v9k	Looser carrying posture; limited fine steps and backing off; window back to 21.5 cm	The arm over the head; the hammer lost; a fall from 22.4 cm	3/7
v9l–v9o	Limits on the trailing sole; stance model refitted on 60 transitions with a calibration episode; correction of the handle axis	A fall with the trailing sole 41.8 cm back; a handle at 85°; a flipped hammer	8/12
v9p–v9t	Trailing-sole margin from the 27/27 against 4/13 record; search turn; stall guard; placing in a partly seen basket	A fall in rehearsal; stalled steps; a basket not seen	6/13
v9u–v9u5b	Re-approach; correction of lateral drift (9.5 cm, $n = 11$); cycle guard; a faster place near the step limit	Three to six actions at the bar; a rehearsal cut off with 49 steps left	9/18
v9v–v9w	Turning toward the basket and looking before walking; position tolerance of 9.5 cm	Walks back at the basket; an overshoot of 35 cm	6/9

phases.py · the crossing window and its evidence

```
BAR_GO = -0.215         # crossings from L >= -0.21 succeeded 17/17 (ep26-57); -0.232 (x1), -0.224 (x2) and -0.223
                        # (ep43) crossed but -0.224 with R -0.396 fell (ep59), as did -0.256/-0.257
BAR_MAX = -0.12
R_GO_MIN = -0.37        # trailing (right) sole at the go: R > -0.355 crossed 27/27; R -0.42..-0.355 fell 4/13
                        # (ep72 -0.364, R5 -0.374, ep59 -0.396, ep64 -0.418)
R_PLAN_MIN = -0.355     # planned stances keep the right sole at least here (sim: fall risk 9% -> 6%, +0.3 actions)
```

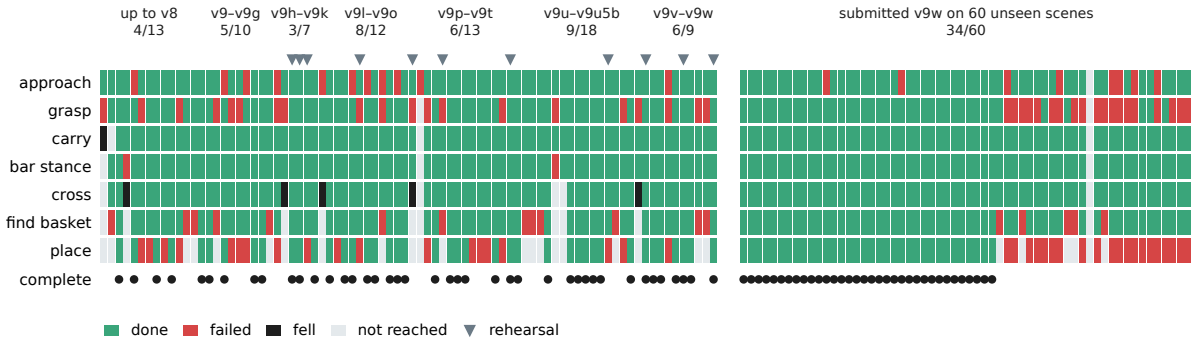


Figure 19 | **The outcome of every stage in every program run,** in interactive and rehearsal episodes grouped by version (left), and for the submitted version on the 60 formal episodes, ordered by outcome (right). Each column is one episode; the counts give completed episodes.

phases.py · crossing with verification

```
252 def cross_bar(r, log, f=None):
253     """Two high steps over the bar (demo01 replay, expert roll, no pitch). Returns True when both feet are
        past."""
254     if f is None:
255         f = C.measure_feet(r)
256     lead = 'L'   # diagonal approach (heading = normal - 26 deg): the left foot leads
257     r.step(C.cross_frames(r.base_joints(), r.base_yaw(), r.hands, mirror=(lead == 'R'), rp=1.0, pitch=0.0))
258     r.hold(20)
259     f2 = C.measure_feet(r)
260     log(f"[bar] crossed lead {lead}: after {None if f2 is None else (round(f2['L'], 3), round(f2['R'], 3))}")
261     if f2 is not None and min(f2['L'], f2['R']) < 0.0 < max(f2['L'], f2['R']):
262         trail = 'L' if f2['L'] < f2['R'] else 'R'
263         log(f"[bar] straddle; trail step {trail}")
264         r.step(C.cross_frames(r.base_joints(), r.base_yaw(), r.hands, 905, 965, mirror=(trail == 'L'), rp=1.0,
265                               pitch=0.0))
266         r.hold(20)
267         f2 = C.measure_feet(r)
268     return f2 is None or min(f2['L'], f2['R']) > 0.0
```

Pseudocode · run() and deliver(), condensed from controller.py

```
1  def run():
2      approach the hammer until the handle lies at about 60 deg   # search turns if unseen
3      grasp with up to 2 attempts                  # a closure without torque is a miss
4      while not held and step < 1000:              # deadlines measured from earlier runs
5          re-approach if the hammer moved out of reach; grasp again
6      deliver()
7
8  def deliver():
9      carry: locate from the bar and the basket; turn; walk a calibrated distance; face 64 deg
10     repeat up to 3 times:
11         repeat up to 8 times:                    # stance planning at the bar
12             measure L, R from depth and kinematics
13             action = bar_action(L, R)            # go, left step, right step, back, or walk
14             if action is go: break
15         if cross_bar(): break                    # both soles past the bar
16     raise the hand; find the basket (up to 6 looks); place inside the fitted rectangle
```

D.4 Adaptation Under Perturbation on LIBERO-Pro

After its exploration on unperturbed layouts, RAX may adapt its program to the perturbed distribution of LIBERO-Pro. A subagent reads the program, the description of the perturbed task, and the first frame of a perturbed layout, and judges whether the program will transfer. When it judges that the program will not, the agent resumes from its workspace, with the notes and code of its exploration, and continues for at most 5 episodes on perturbed layouts before the program is evaluated again on its own. In this mode, RAX runs as an agent-in-the-loop program across the change of distribution: the program acts alone, and the agent steps in only to revise it, so that adaptation exercises the continual learning of Section 3.4. The agent adapted its program on five of the 40 tasks, tasks 3, 8, and 9 of the Long suite and tasks 1 and 5 of the Goal suite (Table 19).

Table 19 | **Tasks adapted to LIBERO-Pro.** Success rate (%) on the 50 position-perturbed states before and after adaptation, with the exploration episodes it used.

Task	Instruction	Episodes	Before $\rightarrow$ after
Long 3	put the black bowl in the bottom drawer of the cabinet and close it	5	$0 \rightarrow 100$
Long 8	put both moka pots on the stove	5	$0 \rightarrow 100$
Long 9	put the yellow and white mug in the microwave and close it	4	$0 \rightarrow 2$
Goal 1	put the bowl on the stove	5	$2 \rightarrow 96$
Goal 5	push the plate to the front of the stove	4	$2 \rightarrow 44$

A mirrored cabinet. The third task of the Long suite asks the robot to put the black bowl in the bottom drawer of the cabinet and close it. The program solved all 50 official states, but LIBERO-Pro places the cabinet on the other side of the table, facing the other way (Figure 20a,b), and the program, whose code assumed a drawer that opens toward $-y$, failed in all 50 perturbed states. Its checks kept it from acting blindly: in 38 of them, it found no drawer panel where it expected one and stopped at the first step. Resuming from its workspace, the agent identified the cause in its first episode on the perturbed layouts and, rather than rewriting the program, added a 36-line layer that mirrors the frame in which it plans: points, orientations, and action rows are reflected across the plane $y = 0$ whenever the cabinet stands at $-y$ (the listing below). The program decides the side itself, from the flat top of the cabinet in the agent-view point cloud (Figure 20c). In a stress test, the agent moved the wine bottle in front of the drawer and found that its detector of the drawer panel took the bottle for the panel; it narrowed the detector's window, and with its five episodes spent, the adapted program completed all 50 perturbed states (Figure 20d).

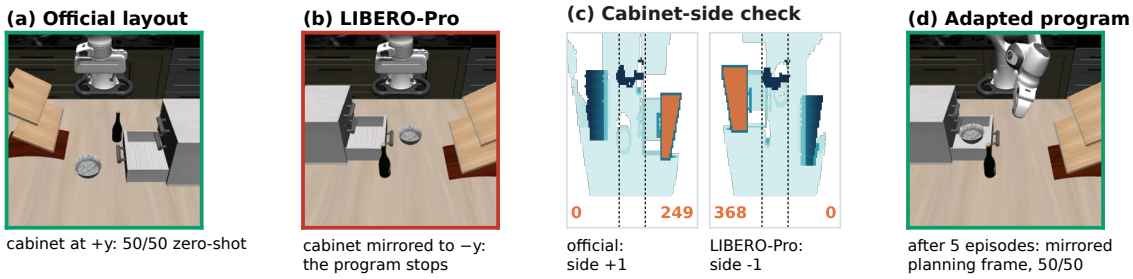


Figure 20 | **Adapting to a mirrored cabinet.** (a) An official layout of the third Long task and (b) the same layout under the LIBERO-Pro perturbation, which moves the cabinet to the other side of the table. (c) The adapted program's check of the side, re-run on both first frames: level cells of the cabinet top above 1.08 m (orange) on either side of the robot, counted on the height map of the agent-view point cloud. (d) The adapted program on the perturbed layout, with the bowl in the drawer.

frame.py · the mirror layer added during adaptation (excerpt)

```python
"""World <-> planning frame. The controller plans in a frame where the cabinet stands at +y and its drawer opens
toward -y (the layouts it was developed on). A cabinet at -y facing +y (swapped layouts) is handled by mirroring
y: points (x, -y, z), rotation vectors (-wx, wy, -wz) (pseudovector under the reflection), action rows likewise.
The mirror is an involution, so the same functions map both ways."""
MIRROR = False

def pts(P):
    if not MIRROR:
        return P
    P = np.array(P, dtype=float, copy=True)
    P[..., 1] *= -1
    return P

# state(s) mirrors the 8-D state in the same way

def row(r):
    """7-D action row [dx, dy, dz, drx, dry, drz, grip]."""
    r = list(r)
    if MIRROR:
        r[1], r[3], r[5] = -r[1], -r[3], -r[5]
    return r
```

E Exploration Budget of ASPIRE on LIBERO-Pro

ASPIRE (Lu et al., 2026) writes robot programs in CaP-X (Fu et al., 2026) with Claude Code and Claude Opus 4.6. Its programs call the CaP-X interfaces for perception, including text-prompted segmentation and vision-language pointing, for grasp planning, including a top-down grasp selector, for inverse kinematics, and for joint-space motion, and they may not read the simulator's ground-truth state. On LIBERO-Pro, the unit of development is a task under one perturbation: each of the 30 tasks of the Spatial, Object, and Goal suites is developed separately for the position and the task perturbation, giving 60 programs, each written for its own perturbed suite.

For each task and perturbation, ASPIRE executes an initial program on a debugging split of 15 environment seeds (51–65), where a seed fixes one instance of the task, including object poses, distractors, and the initial state of the robot. After each execution, the coding agent reads a trace of every primitive call, with its inputs, outputs, and return status, the camera frames before and after the call, grasp candidates, object poses, and motion-planning results. It diagnoses the failure, patches the program, and replays the debugging seeds, with at most three replays per seed. On some tasks, an evolutionary search additionally proposes K candidate programs per iteration, conditioned on the three best programs and their failure traces, and executes each candidate on the debugging split; the reported searches run for up to four iterations, and K is not reported. The final program is executed once on a validation split of 15 further seeds (66–80) and then evaluated on seeds 1–50. A coordinator promotes validated repairs into a shared library of skills, each stored with its failure signature, its condition of use, its repair strategy, and a code sketch, which later sessions read as in-context guidance. With the same loop on Claude Opus 5, Kang et al. (2026) report a median of 0.99 hours and 91 thousand output tokens per task and perturbation.

The debugging and validation seeds index instances of the same perturbed suite as the evaluation seeds, so every rollout before the final evaluation is drawn from the distribution on which the program is evaluated. This is also required by the task perturbation, which changes the goal of the task: a program developed on the original task would pursue the original goal. ASPIRE's result on the Long suite comes from a separate setting, in which a skill library built on the 90 tasks of LIBERO-90 is applied to the Long tasks without any debugging on them. ASPIRE does not report the total number of rollouts per task; Table 20 gives the budget that its protocol implies, alongside that of RAX. With at most 5 rollouts from the evaluation distribution on the five tasks that adapted to the perturbation (Appendix D.4), and none on the other 35, RAX exceeds ASPIRE on every suite under position perturbation, by 28.2 points on Spatial, 2.0 on Object, 6.0 on Goal, and 48.8 on Long (Table 6).

Table 20 | **Exploration before evaluation under position perturbation.** ASPIRE's counts are its debugging and validation splits for one task and perturbation; evolutionary search adds K rollouts per debugging instance and iteration.

	ASPIRE	RAX
Programs per task	one per perturbation	one, adapted on 5 of 40 tasks
Instances from the evaluation distribution	30	0, or up to 5 when adapted
Rollouts on these instances	up to 75, plus evolutionary search	0, or up to 5 when adapted
Exploration on unperturbed layouts	–	up to 20 episodes
Evaluation under perturbation	after development on it	zero-shot, or after adaptation

F Real-Robot Tasks

This section describes the setup of every real-robot experiment: the task, the robot and its sensors, what the agent was given, what the operator did, how many trials the agent ran, how success was judged, and the final outcome. Five tasks use a Franka FR3 arm with a parallel gripper, two exterior RGB-D cameras (left and right of the table), and an RGB-D camera on the wrist; the sixth uses a Rainbow RB-Y1 mobile manipulator with an RGB-D camera on its head. The control interfaces are those of Section 4.2. In every task the agent received only the task text quoted below, typed by the operator, and the robot's tools; it received no demonstrations, no object models, and no trained policy. Each session ran Claude Code with maximum reasoning effort. The agent could send requests to the operator through its tools, and the operator could reply in a chat; we report every reset and intervention recorded in the agent's notes. Times are local wall-clock times taken from the agent's notes and the camera recordings, which are timestamped frame by frame. The key-frame figures show raw camera frames without overlays.

F.1 Find the Cube

Task. The operator hides a white cube of about 3 cm under one of four inverted green cups (about 9.3 cm tall) and shuffles the cups by hand; the robot must then lift the cup that holds the cube away and grasp and lift the cube. The task text was: *"First wait for the operator to place a cube under a green cup and watch it permute the positions of the cups. after it finishes, you should identify the cup with the cube insides and lift the cup away, then grasp the cube and lift it in the gripper, as success."* The agent had to watch the shuffle while it happened: it held the arm still at a watch pose above the cups and recorded the wrist camera's stream, then identified the cup from that recording.

Operator. The operator hid the cube, shuffled the cups, and reset the scene between trials when the agent asked. During the session the operator made the game harder in three steps. At first the cube was covered by a cup and the cups were slid around. In later trials the operator also lifted the cube and moved it, or set a decoy cup on its old spot. From 21:35 the operator dropped the cube into an upright cup, not necessarily the nearest, turned that cup over, and then shuffled. From 21:50 the cube was replaced by a red wedge of about 4 cm.

Trials and outcome. The agent ran 12 trials between 20:08 and 22:35, and recorded each in its notes; trials that stopped partway were resumed from the step that failed. In the first trial the shuffle started before the arm reached the watch pose. In the next two the agent finished parts by hand-issued commands, and in the third its tracker assumed a static cube and chose the decoy. Trials 4, 5, and 6 were three consecutive end-to-end successes of the agent's program without intervention, in 166, 231, and 194 s. Trial 7 missed the moment the cup covered the cube; trial 8 identified the cup correctly but pinched the cube through the cup wall and carried both, after which the agent grasped cups higher and with less force. Trial 9 introduced a new way of hiding the cube: dropped into an upright cup that was then turned over. The program recognized no hiding event it knew and stopped without moving the robot. The agent adapted on the spot: it added a detector for a cube lying inside an upright cup, checked it on the recorded stream and on all seven recordings so far, and resumed the same scene one minute later, when the robot lifted the predicted cup and grasped the cube in 98 s (Figure 21). Figure 23 shows how its program tracks the cups. With the red wedge, the cup was identified correctly in trials 10 and 11; the wedge was grasped after several fixes and manual help in trial 10, and on the first attempt in the resumed trial 11 (Figure 22). For the wedge the agent kept the cup tracker and 20 of the program's 41 functions unchanged, generalized its model of the hidden object to a measured size, color, and height, and changed the grasp to close on the

wedge along two axes. Success was judged from the gripper width after the grasp and from the wrist camera's view of the object in the gripper.



Figure 21 | **Find the cube, trial 9: adapting to a new way of hiding the cube.** (a)–(d) Left exterior camera: the operator drops the cube into an upright cup, turns the cup over, and shuffles the four cups. (e)–(f) Right exterior camera: after the agent added a detector for this case, the robot sets the identified cup aside and grasps the cube. The identification comes from the updated detector run on the recorded wrist stream, and the run resumed from locating the cups. Time is from the first frame.

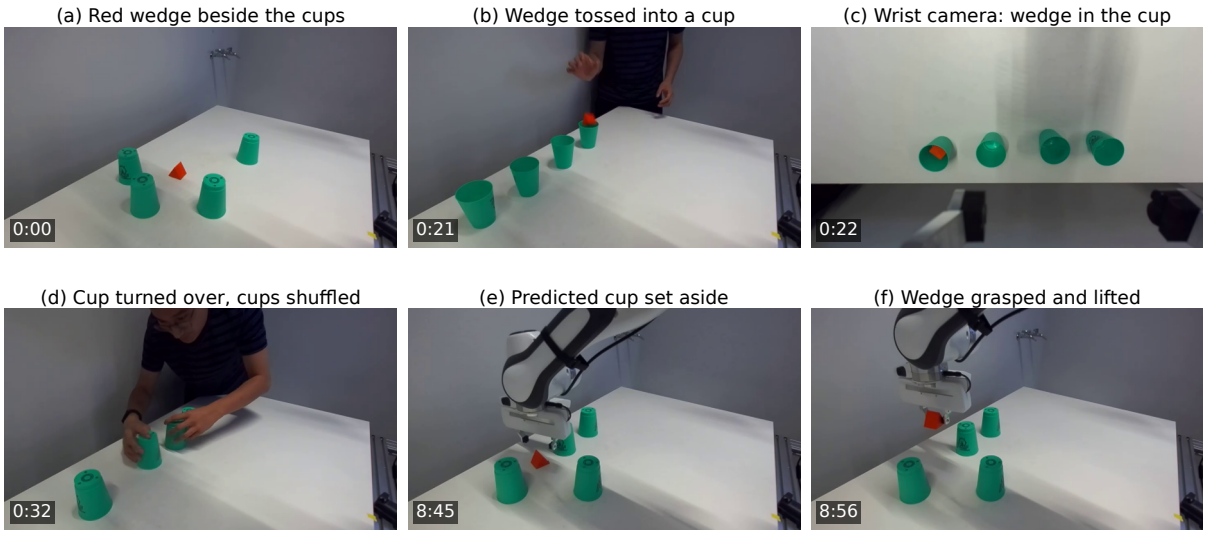


Figure 22 | **Find the cube, trial 11: a red wedge instead of the cube.** (a)–(d) The operator tosses the wedge into an upright cup, turns the cup over, and shuffles; (c) is the wrist camera, which the agent's detector watches. (e)–(f) On the resumed run, the robot sets the predicted cup aside and grasps the wedge on the first attempt. Left exterior camera except (c); time is from the first frame.

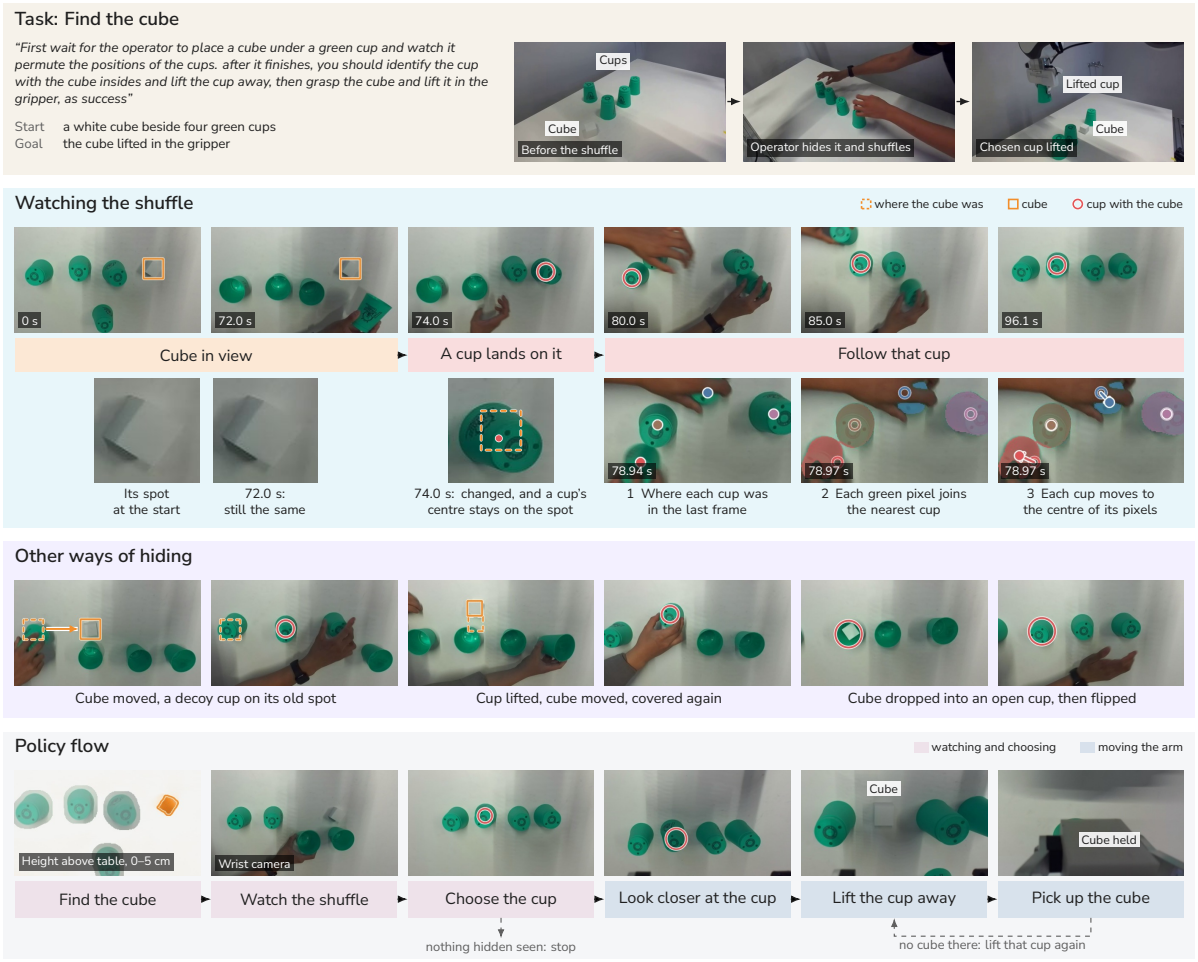


Figure 23 | **The find-the-cube program.** During the shuffle it follows every cup on the wrist camera and remembers the cup that covered the cube (top); it also catches a cube that is moved or dropped into a cup (middle). It then lifts the cup that hides the cube and picks the cube up (bottom).

F.2 Air Fryer

Task. The robot must open the drawer of an air fryer by its handle, put a banana into the basket, and close the drawer. The task text was: *"You should open the air fry pan, put the banana inside, and close it. Reminder: the plate is really heavy and you should figure out where to grasp and the force applied. Also if the position of the air fry pan is out of range you may ask for operator reset. Be careful about the forces and maybe not pull out the plate entirely so we do not have to lift it in everytime."* The agent's perception, its articulated model of the fryer, and its verification are described in Appendix F.2.1.

Operator. The operator reset the scene between runs; after the first run, for example, the operator took the banana out of the basket, which the robot cannot reach into. Between runs the fryer was moved (by about 6 cm and 4 cm in the two table directions before run 7) and rotated it (so that the handle faced the robot's base, before run 8). Before the final run the operator moved the banana away from the left camera at the agent's request and added an orange ball as a second object.

Runs and outcome. The session lasted from 09:18 to 12:05. The agent first completed the task once by direct tool calls (run 1, about 75 minutes including exploration), then wrote a program and ran it nine times (runs 2–10). Run 2 ended with a safety stop when the elbow hit the left camera; the collision moved the right camera, which the agent stopped using. Runs 4, 5, and 8 needed resumes after a false-negative check, a descent timeout, and refused long moves, and run 8 also needed recovery commands issued by the agent. Runs 6, 7, 9, and 10 were clean unattended successes, including run 7 on the moved fryer and run 9 on the rotated one. In the final run the program opened the drawer by 13.1 cm, placed the banana and then the ball in the basket, and closed the drawer, in 210 s, leaving the handle 3.5 mm from its initial position; Figure 26 shows it. Figure 24 shows how its program handles the fryer's layout, the camera bump, and the release. Success was judged by the agent's own measurements: the drawer opening, the objects' points inside the basket rim, and the handle's final position.

F.2.1 Agent-Written 3D Perception

This section details the perception code behind Figure 3. The task is to open the drawer of an air fryer by its handle, put a banana into the basket, and close the drawer, on a real Franka FR3 with a left exterior and a wrist RGB-D camera. The agent wrote the perception and the program itself, in about 1,500 lines of Python; they use no learned detector and no vision-language model. All numbers below are from the agent's final run.

Lifting to a metric frame. Every depth pixel is back-projected with the camera intrinsics and the camera-to-base pose into the robot's base frame, and the views of the exterior and wrist cameras are fused in that frame. All later computations operate on these points; estimates are projected back into the images only to check them.

Segmentation and geometric primitives. Each part of the scene is selected by a colour threshold combined with a height band and a workspace region: the dark lid on top of the fryer, its white body, the dark handle outside the body, the yellow banana, and an orange ball just above the table. Geometric primitives are then fitted to the selected points, each with a plausibility check on its size:

- the fryer body, by a robust circle fit to its side (radius 8 to 16 cm accepted), and the lid, seen from the wrist camera above, by a second circle (4.5 to 8 cm), whose centre gives the fryer's vertical axis c;

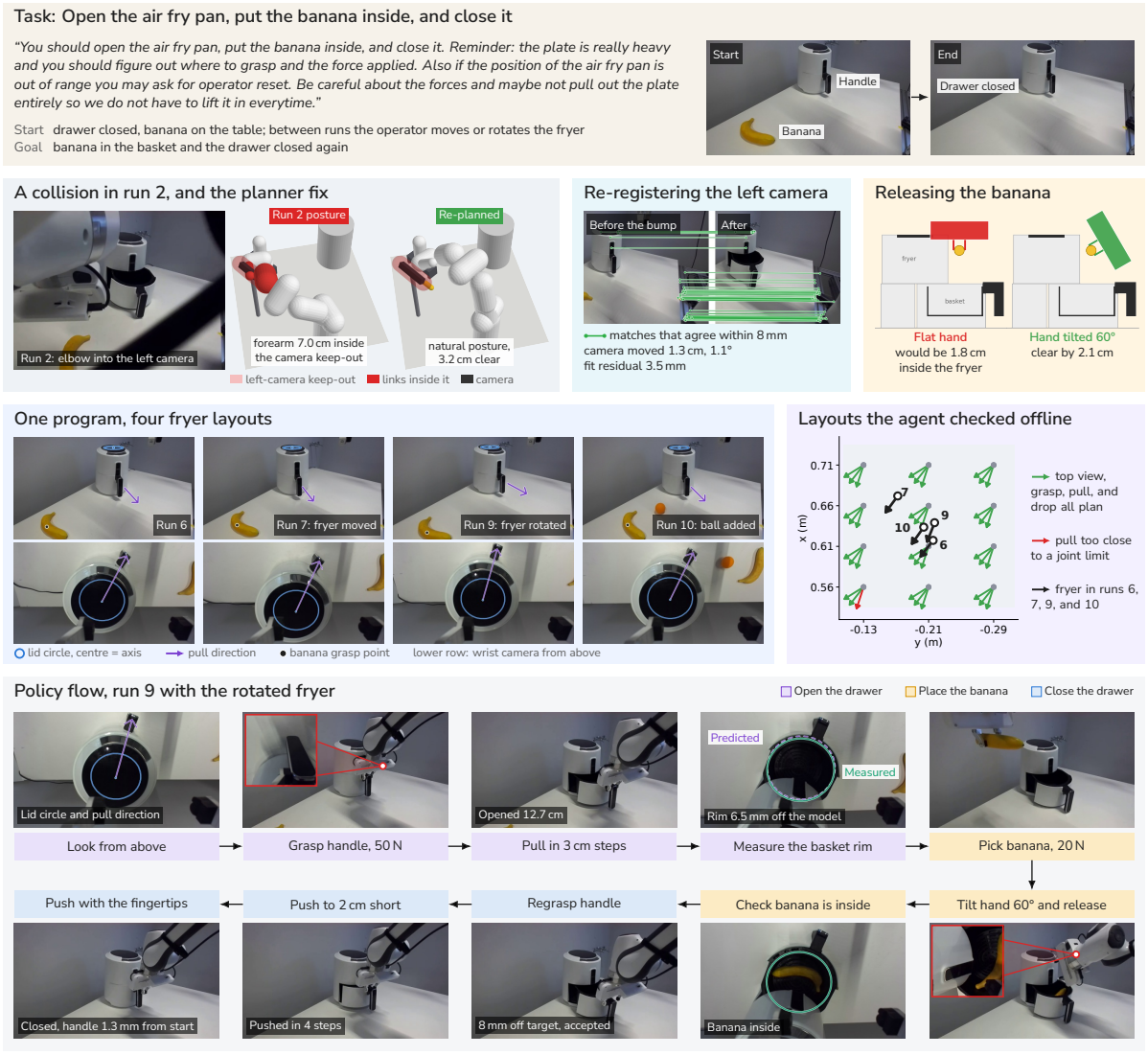


Figure 24 | **The air-fryer program.** After its elbow hit the left camera in run 2, the agent re-registered the camera and planned only postures clear of both cameras; it releases the banana with the hand tilted 60°. One program handles four fryer layouts; the bottom row follows the run with the fryer rotated.

- the handle, by the direction from the axis to its horizontal arm, which gives the pull direction d, the outer face of its grip in 1.5 cm height slices, the height of its arm, and its lateral offset;
- the banana, by its principal axis and a quadratic centre line, which give its arc middle, the normal on its convex side, its chord (13 to 21 cm accepted; 18.0 cm here), and its height;
- the basket, once the drawer is open, by a circle fitted to its rim.

The pull direction from the top view must agree with the one from the side view within 12°; the side view alone was biased by about 4°, which is why the agent added the top view.

Task frame and articulated model. All targets are expressed in a frame anchored on the fryer: the axis c, the pull direction d, and $t = z \times d$. The agent also wrote an articulated model of the fryer: a fixed upper cylinder and the back of the lower body, and a drawer cylinder with box models of the handle's grip and arm, which translate together by the drawer's opening along d. The model provides a signed distance from any point, and the program instantiates it at the drawer's current opening to check every planned motion for clearance between the arm, hand, fingers, wrist camera, and held object, and the fryer, the table, and keep-out regions around the exterior cameras.

How the agent built the model. No model of the fryer was provided. The agent first completed the task once by hand, controlling the robot step by step, and recorded what it measured in its notes: the fryer's top at a height of 0.415 m and the seam between body and drawer at 0.245 m, a handle with a horizontal arm and a hanging grip, and the basket centre relative to the handle once the drawer was out. It also noted that the drawer slides freely along the pull direction and tips slightly when released. About 90 minutes into the session it turned these observations into the model above, in the first version of its program, and revised it in later versions. Table 21 separates what the agent estimates in each new scene from what it fixed from these measurements.

Table 21 | **What the agent's fryer model takes from each scene.**

Part of the model	In each scene	Source
Position and orientation (axis c, pull direction d)	estimated	lid circle and handle direction in the point cloud
Handle face, arm height, lateral offset	estimated	handle points in the point cloud
Drawer opening (joint state)	measured	shift of the handle face after the pull
Radii, heights, handle dimensions	fixed	the agent's measurements from its first manual run

Estimators. The agent does not register the whole model to the point cloud at once. Instead, each parameter of the model comes from a small, robust estimate on the points of one part of the scene, illustrated on its final run in Figure 25:

1. **Axis** c (Figure 25a). From the wrist camera above the fryer, the dark points at the height of the lid are fitted with a circle by least squares: writing the circle as $x^2 + y^2 = 2ax + 2by + k$ turns the fit into a linear system for the centre (a, b) and k, with radius $\sqrt{k + a^2 + b^2}$. The fit is repeated six times, each time discarding the points that lie farthest from the current circle, so that the dark inner part of the lid and stray points do not pull the centre away. The centre is the fryer's axis; the radius, 6.3 cm here, must lie between 4.5 and 8 cm.
2. **Pull direction** d (Figure 25b). Dark points at the height of the handle and just outside the lid are the handle's arm seen from above. Points more than 0.25 rad from their median

bearing around the axis are dropped, and d points from the axis to the median of the rest; a median ignores the few points of other dark objects.

3. **Handle face and arm height** (Figure 25c). In the plane through the axis along d, the handle's points are cut into 1.5 cm height slices. In each slice, the 97th percentile of the distance along d marks the handle's outer face, the surface the fingers and the closing push rest on, while ignoring a few noisy points beyond it; the 95th percentile of height gives the top of the arm, which sets the grasp height.
4. **Banana** (Figure 25d). The banana's yellow points are expressed along their principal axis, found by singular value decomposition, and a quadratic curve fitted by least squares gives its centre line. The middle of that curve is the grasp point, its bulge gives the convex side that sets the hand's orientation, and its ends give the chord, 18.0 cm here, which must lie between 13 and 21 cm.
5. **Basket rim** (Figure 25e). After opening, the dark rim points are searched for around the rim predicted by the model and fitted with a circle in the same way as the lid. The measured centre lies 6.5 mm from the prediction, and the release poses use it.

The drawer's opening is the shift of the handle face between the measurements before and after the pull, and the camera pose after the bump is a rigid transform fitted by least squares to feature matches that agree with each other within 8 mm.

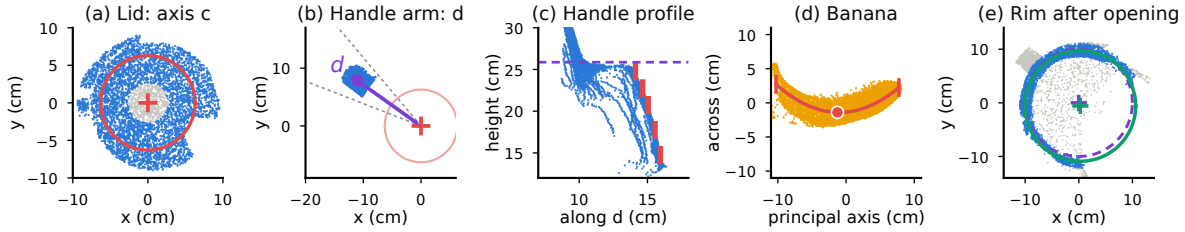


Figure 25 | **The agent's estimators on its final run**, computed by its own code. (a) Lid points from the wrist camera above the fryer; blue points are kept and grey points discarded by the trimmed least-squares circle fit (red), whose centre is the axis c. (b) Handle-arm points seen from above, around the lid circle; points outside the ± 0.25 rad window (dashed) are discarded, and d points to the median of the rest. (c) Handle points in the plane along d: the outer face in each 1.5 cm height slice (red bars) and the top of the arm (dashed). (d) Banana points along their principal axis, with the fitted centre line, its ends, and its middle (the grasp point). (e) Rim points after opening around the model's prediction (dashed), and the fitted rim (green), 6.5 mm apart. Distances are in centimetres.

Planning on the model. Figure 26 shows each planned action on the real robot, numbered as in Figure 3(b). The handle grasp lies 1.2 cm inside the measured grip face along d and 9 mm below the top of the arm, with the hand tilted 5° toward the fryer. The drawer is pulled 12.5 cm along d in absolute 3 cm steps at a fixed height, since relative steps sagged under the drawer's weight. The basket centre is predicted from the handle face and the opening, with an offset measured in the first manual run; the banana is released with its middle 1.5 cm beyond the basket centre and the hand tilted 60° about the banana's axis, and the ball slightly further along d. The drawer is closed by pushing along $-d$, finishing with the fingertips on the grip face.

Verification. Each step is checked by a new measurement. The opening is the shift of the handle face (13.1 cm here), and the step fails if it falls more than 3 cm short of the command. The rim is searched for around the model's prediction and found 6.5 mm from it, and the release poses use the measured rim. The banana counts as placed when its points lie inside

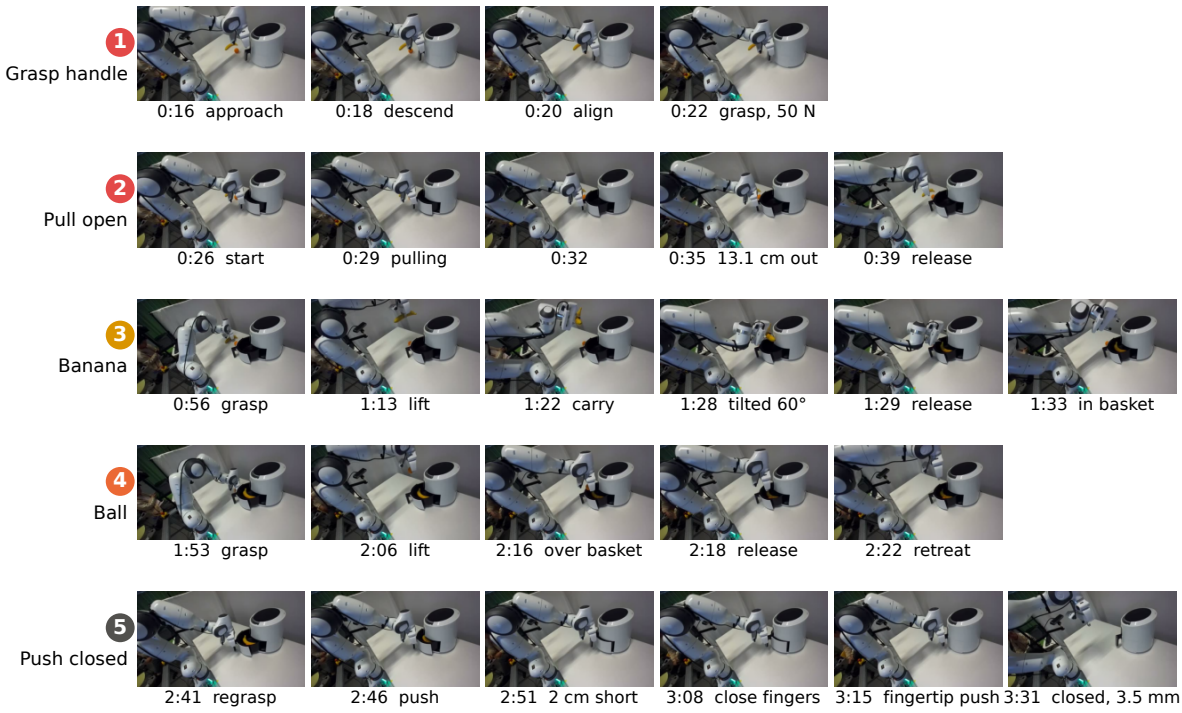


Figure 26 | **The air fryer task on the real robot.** The agent's final run, from the continuous recording of an exterior camera, with the time since the start of the run under each frame. Each row is one planned action, numbered as in Figure 3(b): grasping the handle, pulling the drawer open by 13.1 cm in 3 cm steps, carrying the banana and releasing it with the hand tilted 60° over the basket, releasing the ball, and closing the drawer by a push that stops about 2 cm short followed by a push with closed fingers, which leaves the handle within 3.5 mm of its initial position. This camera was moved by a collision earlier in the session and was not used for perception.

the rim and below its height: 99.99% of them, with its top at 17.9 cm against a rim at 23.5 cm. Before closing, the handle is measured again, since the released drawer tips by about 2.5 cm and shifts sideways by about 1.5 cm; after closing, its face lies 3.5 mm from its initial position.

Sensor limits and calibration. The agent recorded two limits of its sensors and coded around them: the wrist camera's depth is unreliable closer than about 30 cm, so metric targets come from the exterior camera; and side-view pixels on the edge of the banana take the depth of the rim behind it, so heights are measured from the top-down wrist view. When the arm bumped the left camera, whose reported pose stayed unchanged, the agent re-estimated that pose by matching image features against frames captured before the bump, lifting the matches to 3D with both depth maps, and fitting a rigid transform, with a residual of 3.5 mm and fixed landmarks within 2 to 9 mm of their references. It applies the correction only to frames captured after the bump, and it stopped using the right camera, which had moved too far to correct.

F.3 Wiping a Stain

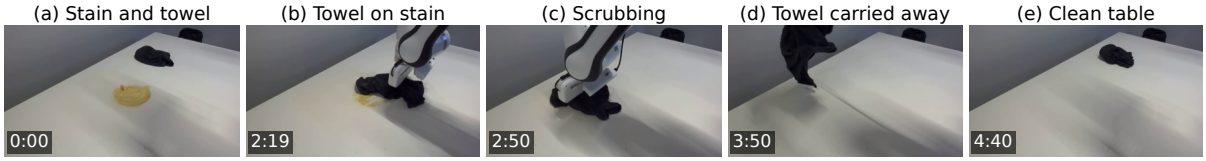


Figure 27 | **Wiping a stain, trial 7b,** from the left exterior camera. (a) The poured stain and the towel heap; (b) the grasped wad pressed on the stain; (c) scrubbing; (d) the towel is carried back; (e) the table after one pass. Time is from the first frame.

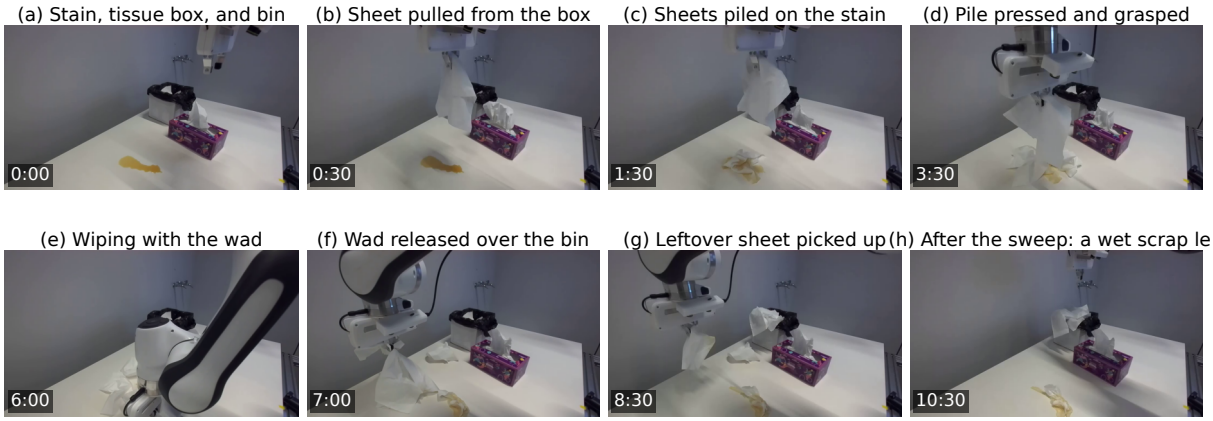


Figure 28 | **Wiping with paper towels, second wet run,** from the left exterior camera. (a) The poured stain, the tissue box, and the bin behind it; (b) a sheet pulled from the box; (c) six sheets piled on the stain; (d) the pile pressed and grasped as a wad; (e) wiping, with the towel program's zig-zag; (f) the wad released over the bin; (g) a leftover sheet picked up by the sweep; (h) the table afterwards, with a flat wet scrap the sweep missed. Time is from the first frame.

Task. The robot must grasp a wet towel and wipe a soy-sauce stain off the table. The task text was: *"You are given a wet towel, and I will ask you to wipe the soy sauce stain on the table using it. First get familiar with the robot and environment. When you are ready, please call for operator to pour the soy sauce on the table for you to wipe. Also, if you need help such as adding some soap to the towel also ask for operator help."* The towel is deformable and has no fixed shape on the table. The agent did not use force sensing: it pressed the towel onto the table by commanding a height 12 mm below the contact height it had measured.

Operator. The operator poured soy sauce when the agent asked, washed the towel and added soap to it between trials, and returned it to the table as a loose heap. The stains differed from trial to trial in size (the agent measured 10 to 311 cm^2), shape, and position, including one next to the left camera at the edge of the reachable workspace.

Trials and outcome. The session lasted from 09:21 to 12:46. The agent first wiped a stain by direct tool calls (two passes; the first left a smeared film, and the second, after the operator added soap, cleaned it), then wrote a program that locates the towel heap, grasps a wad of it, locates the stain against a clean reference image, scrubs it in a back-and-forth pattern, parks the towel, and checks for residue. It ran the program in 13 recorded trials and resumes (trials 1 to 7b). Failures were a refused far reach, an operator reply the program did not expect, one false "clean" verdict on thin streaks (the operator asked for a second wipe; the agent then added a

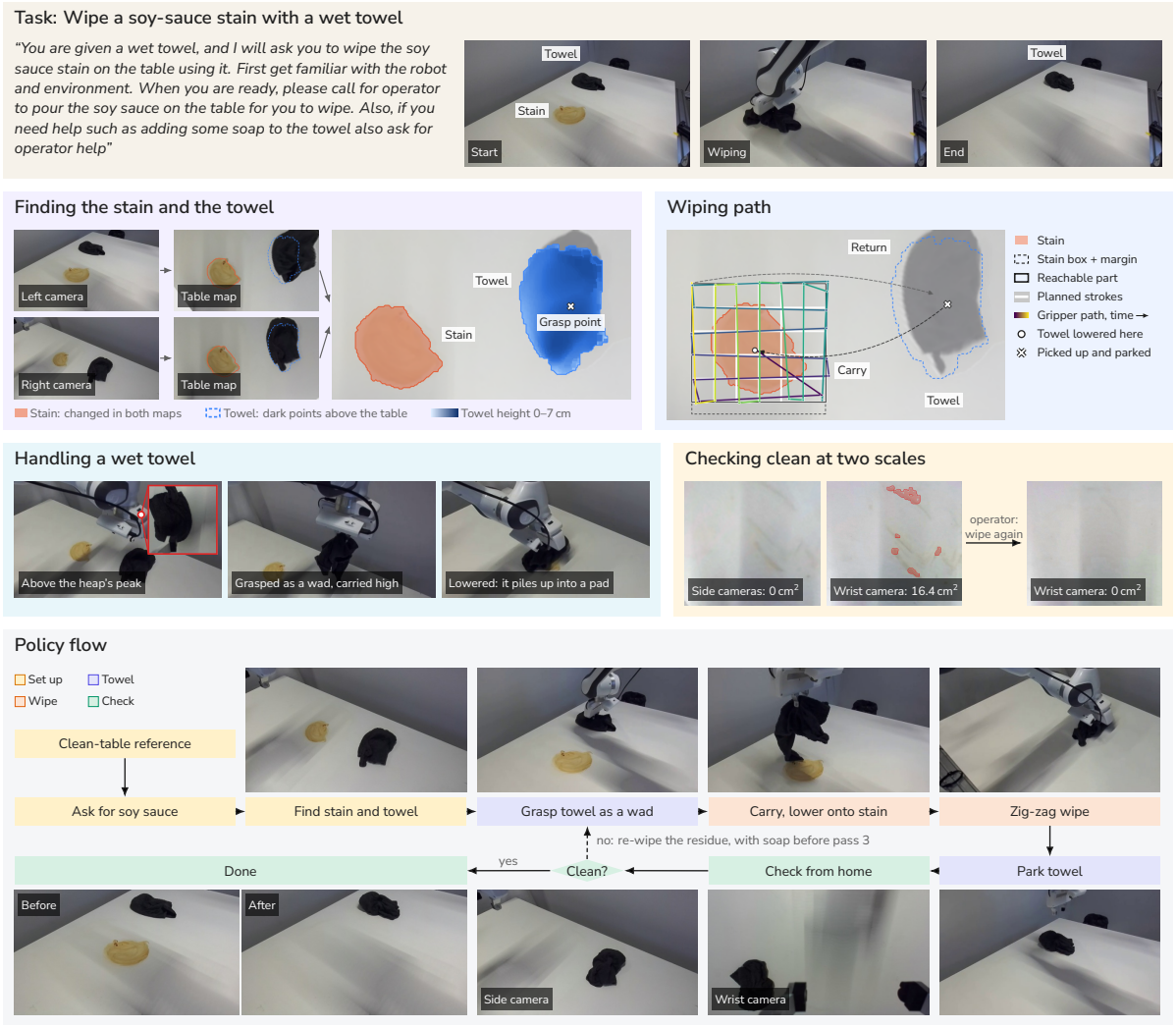


Figure 29 | **The wiping program.** It finds the stain by comparing both exterior cameras with a clean reference and the towel heap in depth, grasps the heap as a wad, and wipes a zig-zag fitted to the reachable area. The wrist camera catches thin streaks that the exterior cameras miss, so the table is checked at both scales.

check with the wrist camera at 1 mm resolution), stains next to the left camera that required fitting the wiping region to the reachable area, and a refused approach move. The last two trials, 6 and 7b, cleaned the stain in one pass without intervention; trial 7b reused the scene of trial 7 after the agent fixed the approach move (Figure 27). Figure 29 shows how its program finds, wipes, and checks the stain. Success was judged by the agent's residue measurement against the clean reference in the exterior and wrist cameras (0 cm^2 in both for trials 6 and 7b) and by eye in the three camera views.

Adaptation to paper towels. Afterwards the operator asked for a variant: *"pull paper towels out of the box, wipe table, and then throw them into the waste bin."* The agent wrote a new program that imports the towel program unchanged and calls its parts: locating the stain against the clean reference, fitting the wiping region to the reachable area, the press-then-grasp of a wad, the zig-zag wipe 12 mm below the measured contact, and the residue check. The new code handles the paper: it finds the box and the bin, pinches the top of a protruding sheet and pulls it out, piles six sheets on the stain, releases the wad over the bin and scrapes paper stuck to the fingers, and sweeps up sheets left on the table. After three dry runs, the stain was removed in both wet runs (Figure 28), but leftover sheets, and in the second run a flat wet scrap that the sweep missed, were cleaned up with supervised scripts.

F.4 Peg Insertion: the ManipulationNet Peg-in-Hole Challenge



Figure 30 | **The ManipulationNet peg-in-hole challenge,** final run, from a crop of the left exterior camera. (a) A peg inserted into L1 (3 mm clearance); (b)–(d) the rectangle peg entering L3 (0.1 mm clearance) corner first, turned upright, and seated; (e) the three pegs seated in L3. Time is from the first frame.

Task. This is the peg-in-hole challenge of ManipulationNet (Chen et al., 2026b). Its board is a transparent acrylic plate of 320 × 250 × 25 mm with blind holes 21.8 mm deep for five peg shapes, each at four clearance levels, rows L1 to L4 with 3, 1, 0.1, and 0.02 mm clearance. Neither the steel pegs, 50 mm long, nor the holes are chamfered. We used three pegs, the rectangle, the triangle, and the hexagon. The task text was: *"There is a white transparent board in front and below you. There are openings on top that match the shape of the silver metal pegs (3 of them. rectangle. triangle and hexagon). You need to grasp them up and insert into the holes upright. Reminder: the depth of the wrist is not accurate at close distance, so you can locate approximately and probe the position and yaw based on force feedback, to succeed in the fine insertion. If you need help or if the position is of bad IK you can ask for operator to move it."* The agent converted the public CAD files of the board and the pegs into a model of the hole positions, which it registered to the wrist camera's image. During the session the user asked it to work toward the tighter rows, from L2 to L3. L4 was not attempted.

Operator. The board rests on the table unfixed. After the agent dragged it while lifting a jammed peg with a closed gripper, the operator taped the board to the table (at 01:30). The operator re-seated a hexagon peg that had toppled in L2 after a failed L3 attempt. The L3 fits are tight enough that the robot could not pull a seated peg out (15 N with rocking); before the final run the operator pulled the three pegs out of L3 and stood them upright on the table. The user also gave advice in the chat, for example to lower the peg further before turning it upright.

Exploration and outcome. The agent worked by direct tool calls and its own scripts, supervised by the user; it never ran an unattended program in this task. Between 23:13 and 01:46 it inserted each peg into L1, L2, and L3 in turn (the triangle and the hexagon each failed once in L3 before succeeding). The user then asked for a full run, and between 02:28 and 03:22 the agent took each peg from the table and inserted it into L1, then L2, then L3, showing it upright in each row (Figure 30). Figure 31 shows how it registers the board, finds the holes by touch, and tilts the peg in. L3 needed a tilted entry with a corner of the peg first, followed by small rocking pushes while the peg was turned upright. All three pegs were seated in L3 at the end, at depths of 20.8 mm (hexagon), 19.7 mm (triangle), and 21.0 mm (rectangle), out of 21.8 mm. The full run took about 55 minutes including the operator's extraction of the pegs from the earlier L3 attempts. Success was judged by the insertion depth the agent measured with the arm and by images of the upright pegs.

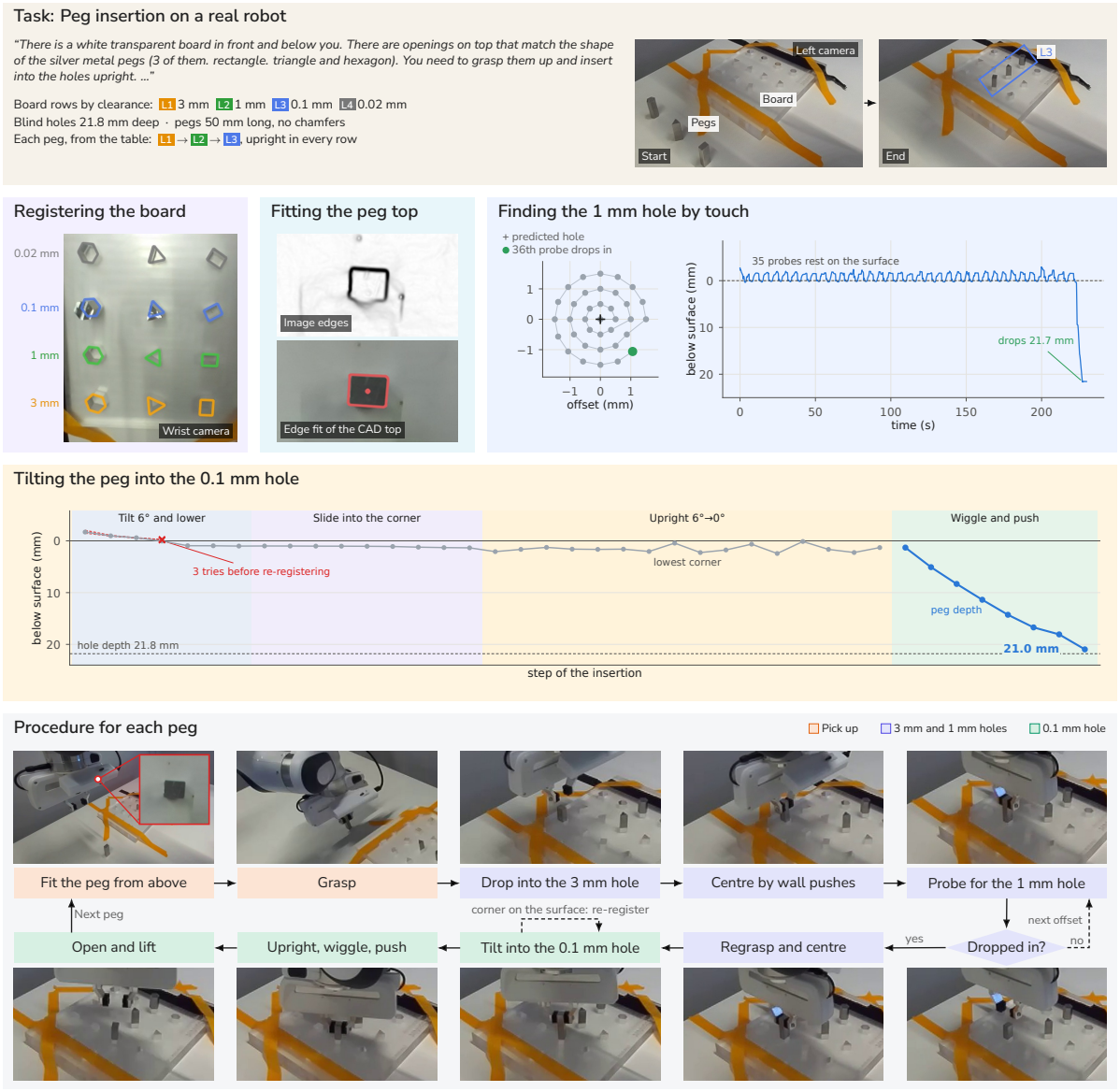


Figure 31 | **The peg-insertion procedure.** The agent fits the board's CAD model and each peg's top outline to the wrist camera's image edges. It drops the peg into the 3 mm row, probes by touch for the 1 mm hole, and tilts a corner into the 0.1 mm hole before turning the peg upright and pushing it down.

F.5 Bandu Tower



Figure 32 | **Bandu tower, final fully automatic assembly,** from the left exterior camera. (a) The pieces after disassembly; (b)–(c) pieces are placed on the base; (d) the knob is held over the urn for visual alignment; (e) the completed tower. Time is from the first frame.

Task. Bandu is a stacking game with irregular wooden pieces. The task text was a single sentence: *"Stack the bandu blocks as the demonstrated tower."* The "demonstration" was a tower standing on the table when the session began; the agent was not shown how it was built. The tower stands on a base of four rectangular blocks laid in a 2×2 pattern and consists of seven pieces: a ring with a dowel standing in it, a barrel, a goblet with an urn seated in its cup and a knob seated in the urn, and an oblique cylinder that the user added at 13:42, asking for it on top of the barrel. Several of these placements leave little play, and the knob's opening clearance was about 1 mm by the agent's estimate. The user told the agent to take the tower apart and rebuild it, and to repair its own disturbances rather than rely on human resets.

Operator. The operator reset the pieces once after the robot's hand swept the tower (at 15:06, during a sideways pick the user had asked for), restarted or re-homed the gripper twice, and moved one piece the arm could not reach. The user asked for visual alignment instead of a contact search for the knob ("align with at most 1 mm opening error").

Exploration and outcome. The session lasted from 11:13 to 17:20. The agent first took the tower apart and rebuilt it by direct tool calls, then wrote a program with a disassembly and an assembly mode and alternated them over runs 1 to 15, fixing a failure after each stop. In the final exploration cycle the program disassembled the tower (7/7 pieces, 459 s) and then rebuilt it fully automatically, without intervention: 7/7 pieces in 663 s, the knob placed by visual alignment from both exterior cameras to an offset of 0.19 mm before insertion (Figure 32). Figure 33 shows how its program perceives, places, and checks each piece. Each placement was checked by the measured seat height and, for the urn, an uprightness check in both cameras; the final tower was compared with the demonstrated one in both exterior views.

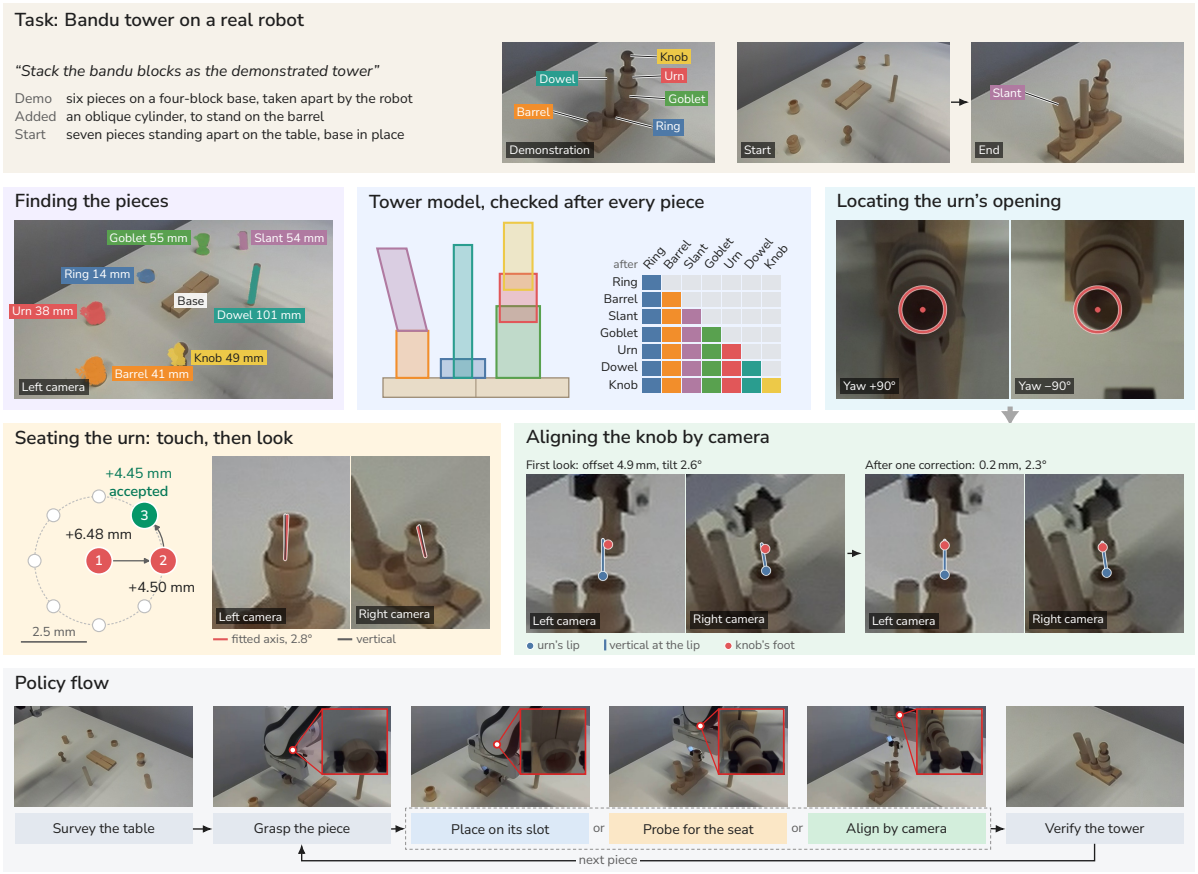


Figure 33 | **The Bandu assembly program,** running fully automatically. It finds the pieces with both exterior cameras and Gemini, keeps a slot model of the demonstrated tower, and places each piece by a guarded place, a tactile search for the seat (with a tilt check for the urn), or a camera alignment of the knob over the urn's opening.

F.6 Long-Horizon Tidying on the RB-Y1

Task. The RB-Y1 must tidy a lab corner in six chained subtasks given in one instruction: *"I want you to complete a series of tasks and you should reuse the skills while still monitor and control the process. First, you should move the chair in front of you out and park it to the left of the yellow table in front of you to clear it out of the way. then you should stack the two silver cups together and carry the stacked cups and banana over to the orange basket to your right and place them in, and pick up the orange basket on to the cabinet with dark top. Then you should move the chair back and tuck it to the table again. Finally you should pick up the box on your right and put it also on the cabinet. Be agile and reactive."* The robot perceives the scene only through its head camera and must move between the table, the basket, and the cabinet with its base.

Skills from earlier sessions. The agent started from a shared skill library written in five earlier RB-Y1 sessions: carrying a box between the floor and a table (`bimanual-box-pnp`, from three sessions), lifting the basket onto the cabinet by its rim with both grippers (`bimanual-bucket-rim-pnp`), and moving and tucking an office chair (`office-chair-tuck`) and nesting two cups (`stack-cups`), both from one chair session. Carrying the cups and the banana into the basket had no skill. During the session the agent added a skill for reaching hand moves (`oc-hand-reach`) and a composition skill for the whole task, and updated the library after each trial.

Operator. The operator reset the scene before each trial, stayed in the loop throughout, and gave advice in the chat, such as "straighten the arms", "lift first", and "check the torso and height against previous skills". In trial 1 the robot's base hit the cabinet during a hand-planned turn, after which the agent checked the base's swept footprint before every base move. In trial 3 the operator re-armed the robot after controller faults while the cups were being nested.

Trials and outcome. The agent ran the whole sequence three times; all six subtasks were completed in each trial (trials 1 and 2 overnight on the first day, trial 3 on the next evening). Trial 3, shown in Figure 4, ran from 19:31 to 21:09 (98 minutes), with the agent at every checkpoint of the skills. Success of each subtask was judged by the agent from head-camera images (for example, two rims visible for nested cups, both objects inside the basket, the basket and the box on the cabinet).

G Compute and Cost

Across all experiments in this paper, simulated and real, the agents' model calls consumed over 100k US dollars of API credits.